HL7 FHIR R4 — Chapter 6: Security and Privacy Module (sections 6 – 6.4.12)

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6 Security and Privacy Module

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6.0 Security and Privacy Module

6.0.1 Introduction

The Security and Privacy Module describes how to protect a FHIR server (through access control and authorization), how to document what permissions a user has granted (consent), and how to keep records about what events have been performed (audit logging and provenance). FHIR does not mandate a single technical approach to security and privacy; rather, the specification provides a set of building blocks that can be applied to create secure, private systems.

6.0.2 Index

The Security and Privacy module includes the following materials:

Resources | Datatypes | Implementation Guidance and Principles
Consent Provenance Audit Event | Signature | Security Principles Security Labels Signatures

The following common use-cases are elaborated below:

• Security and Privacy
• Authorization and Access Control
• Authorization considerations with Query Parameters
• User Identity and Access Context
• Security and Privacy Audit Logging
• Accounting of Disclosures and Access Reports
• Privacy Consent
• Provenance
• Digital and Electronic Signatures
• De-Identification, Anonymization, and Pseudonymization
• Security Considerations on Test Data

6.0.3 Security

FHIR is focused on the data access methods and encoding leveraging existing Security solutions. Security in FHIR needs to focus on the set of considerations required to ensure that data can be discovered, accessed, or altered only in accordance with expectations and policies. Implementation should leverage existing security standards and implementations to ensure that:

• All communications can be encrypted to prevent unauthorized access.
• No information leaks when errors occur
• No active script content can be injected into narrative resources
• Full audit trails can be constructed and used to detect anomalous access patterns

For general security considerations and principles, see Security. Please leverage mature Security Frameworks covering device security, cloud security, big-data security, service-to-service security, etc. See NIST Mobile Device Security and OWASP Mobile Security . These security frameworks include prioritized lists of most important concerns.

6.0.4 Privacy

Privacy in FHIR includes the set of considerations required to ensure that individual data are treated according to an individual's Privacy Principles and Privacy-By-Design. FHIR includes implementation guidance to ensure that:

• Individual preferences can be communicated through standards-based protocols (e.g., OAuth, User-Managed Access) or using an explicit FHIR representation (Consent)
• Resources can be tagged to indicate the sensitivity or confidentiality of the data they represent (Security Labels)

• Data access records and audit logs can be shared with individuals, e.g. for accounting of disclosures (Audit Event)

6.0.5 Common Use Cases

6.0.5.1 Authorization and Access Control

Use case: A FHIR server should ensure that API access is allowed for authorized requests and denied for unauthorized requests.

Approach: Authorization details can vary according to local policy, and according to the access scenario (e.g. sharing data among institution-internal subsystems vs. sharing data with trusted partners vs. sharing data with third-party user-facing apps). In general, FHIR enables a separation of concerns between the FHIR REST API and standards-based authorization protocols like OAuth. For the use case of user-facing third-party app authorization, we recommend the OAuth-based SMART protocol see Security: Authentication as an externally-reviewed authorization mechanism with a real-world deployment base - but we note that community efforts are underway to explore a variety of approaches to authorization. For further details, see Security: Authorization and Access Control.

6.0.5.1.1 Query Parameters Considerations

Use-Case: When a user has restricted rights but attempts to do a query they do not have rights to, they should not be given the data. Policy should be used to determine if the user query should result in an error, zero data, or the data one would get after removing the non-authorized parameters.

Approach: Enforcement is by local enforcement methods. Note that community efforts are underway to explore a variety of approaches to enforcement.

Example: Using _include or _revinc to get at resources beyond those authorized. Ignoring (removing) the _include parameter would give some results, just not the _include Resources. This could be silently handled and thus give some results, or it could be returned as error.

6.0.5.2 User Identity and Access Context

Use case: "Access to protected Resources are enabled though user Role-Based, Context-Based, and/or Attribute-Based Access Control."

Approach: Ensure that the level of assurance for identity proofing reflects the appropriate risk, given the issued party's exposure to health information. Users should be identified and should have their Functional and/or Structural role declared when these roles are related to the functionality the user is interacting with. Roles should be conveyed using standard codes from Security Role Vocabulary.

A purpose of use should be asserted for each requested action on a Resource. Purpose of use should be conveyed using standard codes from Purpose of Use Vocabulary.

When using OAuth, the requested action on a Resource for specified one or more purpose of use and the role of the user are managed by the OAuth authorization service (AS) and may be communicated in the security token where JWT tokens are used. For details, see Security: HCS vocabulary.

6.0.5.3 Audit Logging

Use case: "A FHIR server should keep a complete, tamper-proof log of all API access and other security- and privacy-relevant events".

Approach: FHIR provides an AuditEvent resource suitable for use by FHIR clients and servers to record when a security or privacy relevant event has occurred. This form of audit logging records as much detail as reasonable at the time the event happened. The FHIR AuditEvent is aligned and cross-referenced with IHE Audit Trail and Node Authentication (ATNA) Profile. For details, see Security: Audit.

6.0.5.4 Accounting of Disclosures and/or Access Report

Use case: "A Patient should be offered a report that informs about how their data is Collected, Used, and Disclosed."

Approach: The AuditEvent resource can inform this report.

There are many motivations to provide a Patient with some report on how their data was used. There is a very restricted version of this in HIPAA as an "Accounting of Disclosures", there are others that would include more accesses. The result is a human readable report. The raw material used to create this report can be derived from a well recorded 'security audit log', specifically based on AuditEvent. The format of the report delivered to the Patient is not further discussed but might be: printed on paper, PDF, comma separated file, or FHIR Document made up of filtered and crafted AuditEvent Resources. The report would indicate, to the best ability, Who accessed What data from Where at When for Why purpose. The 'best ability' recognizes that some events happen during emergent conditions where some knowledge is not knowable. The report usually does need to be careful not to abuse the Privacy rights of the individual that accessed the data (Who). The report would describe the data that was accessed (What), not duplicate the data.

Some events are known to be subject to the Accounting of Disclosures report when the event happens, thus can be recorded as an Accounting of Disclosures - See example Accounting of Disclosures. Other events must

be pulled from the security audit log. A security audit log will record ALL actions upon data regardless of if they are reportable to the Patient. This is true because the security audit log is used for many other purposes. - See Audit Logging. These recorded AuditEvents may need to be manipulated to protect organization or employee (provider) privacy constraints. Given the large number of AuditEvents, there may be multiple records of the same actual access event, so the reporting will need to de-duplicate.

6.0.5.5 Privacy Consent

Use case: "Documentation of a Patient's Privacy Consent Directive - rules for Collection, Use, and Disclosure of their health data."

Approach: FHIR provides a Consent resource suitable for use by FHIR clients and servers to record current Privacy Consent state. The meaning of a consent or the absence of the consent is a local policy concern. The Privacy Consent may be a pointer to privacy rules documented elsewhere, such as a policy identifier or identifier in XACML. The Privacy Consent has the ability to point at a scanned image of an ink-on-paper signing ceremony, and supports digital signatures through use of Provenance. The Privacy Consent has the ability to include some simple FHIR centric base and exception rules.

All uses of FHIR Resources would be security/privacy relevant and thus should be recorded in an AuditEvent. The data access that qualifies as a Disclosure should additionally be recorded as a Disclosure, see Disclosure Audit Event Example.

For Privacy Consent guidance and examples, see Consent Resource.

6.0.5.6 Provenance

Use case: "All FHIR Resources should be capable of having the Provenance fully described."

Approach: FHIR provides the Provenance resource suitable for use by FHIR clients and servers to record the full provenance details: who, what, where, when, and why. A Provenance resource can record details for Create, Update, and Delete; or any other activity. Generally, Read operations would be recorded using AuditEvent. Many Resources include these elements within; this is done when that provenance element is critical to the use of that Resource. This overlap is expected and cross-referenced on the Five Ws pattern. For details, see Provenance Resource.

Use case: "For any given query, need Provenance records also."

Approach: Given that a system is using Provenance records. When one needs the Provenance records in addition to the results of a query on other records (e.g. Query on MedicationRequest), then one uses reverse include to request that all Provenance records also be returned. That is to add ?_revinclude=Provenance:target. For details, see _revinclude.

6.0.5.7 Signature

Use case: "Digital Signature is needed to prove authenticity, integrity, and non-repudiation."

Approach: FHIR Resources are often parts of Medical Record or are communicated as part of formal Medical Documentation. As such there is a need to cryptographically bind a signature so that the receiving or consuming actor can verify authenticity, integrity, and non-repudiation. This functionality is provided through the signature element in Provenance Resource. Where the signature can be any local policy agreed to signature including Digital Signature methods and Electronic Signature. For details, see Security: Digital Signatures.

Digital Signatures bind cryptographically the exact contents, so that any changes will make the Digital Signature invalid. When a Resource is created, or updated the server is expected to update relevant elements that it manages (id, lastupdated, etc.). These changes, although expected of normal RESTful create/update operations, will break any Digital Signature that has been calculated prior. One solution is to create the Digital Signature after the REST create operation completes, one must first confirm that the resulting created/updated Resource is as expected, then the Digital Signature is formed.

A variation of this happens in Messaging, Documents, and other interaction models. For details, see Ramifications of storage/retrieval variations

6.0.5.8 De-Identification, pseudonymization, anonymization

De-Identification is inclusive of pseudonymization and anonymization; which are the processes of reducing privacy risk by eliminating and modifying data elements to meet a targeted use-case.

Use-Case: "Requesting Client should have access to De-Identified data only."

Trigger: Based on an Access Control decision that results in a permit with an Obligation to De-Identify, the Results delivered to the Requesting Client would be de-identified.

Consideration: This assumes the system knows the type and intensity of the de-identification algorithm, where de-identification is best viewed as a process, not an algorithm - a process that reduces Privacy risk while enabling a targeted and authorized use-case.

Modifying an element: The de-identification process may determine that specific elements need to be modified to lower privacy risk. Some methods of modifying are: eliminating the element, setting to a static value (e.g. "removed"), fuzzing (e.g. adjusting by some random value), masking (e.g. encryption), pseudonym

(e.g. replace with an alias), etc. Narrative and Attachment elements present particularly difficult challenges. See standards below for further details.

Discussion: Obviously the most important elements for de-identification are names and identifiers. FHIR resources have many different types of ids and identifiers that serve different purposes. Some (ids) are the basis for internal links between different resources, while identifiers are mainly - but not exclusively - for correlating with external data sources. Strategies for de-identification need to consider whether re-identification with the source system is a problem, in which case ids will need to be modified - and consistently across the resource set being de-identified. External identifiers will mostly need to be removed, but even then, where they are used for internal references within the resource set, they'll need to be changed consistently.

Then, there is the question of where to make the de-identification changes. For example, the Observation Resource has a subject element that mostly refers to a Patient resource. Should it be removed? Left and the Patient resource it refers to be de-identified? Updated to a new patient resource randomly or consistently? There are many other Reference elements on Observation that can easily be used to navigate back to the Subject; e.g., Observation.context value of Encounter or EpisodeOfCare; or Observation.performer. These also need to be de-identified, and it will depend on the intended use of the data whether these all need to be consistent with each other.

Some identifiers in Observation Resource:

• Direct Identifiers: .identifier, .subject, .performer, .encounter, .focus, .note, .specimen, .basedOn

• Indirect Identifiers: .category, .code, .issued, .effective[x], .method, .bodySite, .interpretation, .value[x], .component

Emphasis: The .specimen is a direct identifier of a particular specimen; and would be a direct identifier of a particular patient. This is a ramification of having the specimen identifier. One solution is to create pseudo specimen resources that will stand-in for the original specimen resource. This pseudo specimen management is supplied by a trusted-third-party that maintains a database of pseudo-identifiers with authorized reversibility.

Care should be taken when modifying an isModifier elements, as the modification will change the meaning of the Resource.

In practice, then, the de-identification process depends on the intended use of the data, the scope of the data being extracted, and the risk associated with the release of the data (e.g. data released into the public domain has a different risk than internal sharing of data within a tightly managed organization with strong information security policies.

Security-label: The resulting Resource should be marked with security-label to indicate that it has been de-identified. This would assure that downstream use doesn't mistake this Resource as representing full fidelity. These security-labels come from the Security Integrity Observation ValueSet. Some useful security-tag vocabulary: ANONYED, MASKED, PSEUDED, REDACTED

Further Standards: Health: ISO Pseudonymization , NIST IR 8053 - De-Identification of Personal Information , IHE De-Identification Handbook , DICOM (Part 15, Chapter E)

6.0.5.9 Using FHIR to publish test data

Use-Case: There are times when test data is needed. Test data are data that is not associated with any real patient. Test data are usually representative of expected data that is published for the purpose of testing. Test data may be fully fabricated, synthetic, or derived from use-cases that had previously caused failures.

Trigger: When test data are published it may be important to identify the data as test data.

Consideration: This identification may be to assure that the test data is not misunderstood as real data, and that the test data is not factored into statistics or reporting. However, there is a risk that identifying test data may inappropriately thwart the intended test that the data are published to test.

Discussion:

Test data could be isolated in a server specific to test data.

Test data could be intermingled with real-patient data using one or both of the following methods:

• Security-label -- All test data Resources could be tagged with PurposeOfUse HTEST

• Test Patient -- The Patient that the data is associated with could be a clear indication of a test patient. (For example, in the USA the SSN starting with "666-" will never be issued so is commonly used as an indicator of a test patient)

Considerations: Note there is a risk when co-mingling test data with real patient data that someone will accidentally use test data without realizing it is test data.

6.0.6 Developmental Roadmap

In the STU3 release, FHIR includes building blocks and principles for creating secure, privacy-oriented health IT systems; FHIR does not mandate a single technical approach to security and privacy.

In future releases, we anticipate including guidance on:

• incorporate the SMART on FHIR authorization specification, for user-authorized apps,

• methods for organization-to-organization authorization,

• more details about how to use digital Signatures for data integrity and non-repudiation, including an approach that supports some level of manipulation of resources (e.g. separating the entries in a bundle, or conversion between XML and JSON during processing),

• more detailed Consent management, including support for specific consent use cases.

• guidance and methods to support GDPR or other emerging Privacy regulations.

6.1 FHIR Security

URL: https://hl7.org/fhir/R4/security.html

The FHIR RESTful API provides several ways that a client may request or create information. When designing a system to authorize access to information, all potential access methods must be considered. They include the following:

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Fast Healthcare Interoperability Resources (FHIR) is not a security protocol, nor does it define any security related functionality. However, FHIR does define exchange protocols and content models that need to be used with various security protocols defined elsewhere. This section gathers all information about security in one section. A summary:

• Time Keeping - all clocks should be synchronized using NTP/SNTP, and the design of the system should be robust against a system clock with the wrong value

• Communications Security - all exchange of production data should be secured using TLS (e.g., https).

• Authentication - Users/Clients must be authenticated. For web-centric, OAuth is recommended. When using OAuth, a profile of OAuth will be needed. Consider use of Smart-On-FHIR where appropriate.

• Authorization/Access Control - FHIR defines a Security Label infrastructure to support access control management. FHIR may also define a set of resources to administer access control management, but does not define any at present

• Audit - FHIR defines provenance and audit event resources suitable for tracking the origins, authorship, history, status, and access of resources

• Digital Signatures - FHIR includes several specifically reserved locations for digital signatures

• Attachments - FHIR allows for binary resources and attachments. These have their own concerns

• Labels - FHIR allows for set of security related tags that affect the way resources are handled

• Data Management Policies - FHIR defines a set of capabilities to support data exchange. Not all the capabilities that FHIR enables may be appropriate or legal for use in some combinations of context and jurisdiction (e.g. HIPAA, GDPR). It is the responsibility of implementers to ensure that relevant regulations and other requirements are met

• Narrative - Care must be taken when displaying the narrative from FHIR resources

• Input Validation - Validate all input received from other actors to assure the data is well formed and does not contain content that would cause unwanted system behaviour. Testing ensures that the input is not susceptible to data input validation errors by using techniques such as fuzzing, invalid input attacks, and injection attacks.

Time critical concerns regarding security flaws in the FHIR specification should be addressed to the FHIR email list for prompt consideration.

6.1.0.1 General Considerations

A production FHIR system will need some kind of security sub-system that administers users, user authentication, and user authorization. Where this subsystem fits into the deployment architecture is a matter for system design:

| | The consumer that is using a healthcare related system The client application the user is using (application, mobile app, website, etc.) The security system (authentication and access control) The clinical/healthcare repository
| The consumer that is using a healthcare related system
| The client application the user is using (application, mobile app, website, etc.)
| The security system (authentication and access control)
| The clinical/healthcare repository

In this diagram, the red lines represent FHIR interfaces. From the perspective of the FHIR API, the client (consumer of FHIR services) may either interact with a security system that manifests as a FHIR server, and which depends on a subsequent FHIR interface to provide the actual storage, or either the client or server interacts with the security system independently. In each of these 3 scenarios, the different components may be assembled into applications or network components differently, but the same logical layout applies. The FHIR specification assumes that a security system exists, and that it may be deployed in front of or behind the FHIR API.

The security system includes the following subsystems:

• Authentication: identifies and authenticates the user

• Access Control decision engine: decides whether FHIR operations are allowed

• Audit Log: records actions to allow for subsequent review and detection of intrusion or inappropriate usage

Because there are a plethora of standards relating to the administration and functionality of the security system, FHIR does not provide user, profile, or other such administration resources. Instead, the FHIR resources are the targets of the policies expressed in these other approaches. What FHIR does specify is a way to apply security labels to resources so that a security system may use these (along with the contents of the resources if appropriate) to determine whether a user is authorized to perform a particular FHIR operation or not.

6.1.0.2 Security and Privacy Considerations

The appropriate protections for Privacy and Security are specific to the risks to Privacy and the risks to Security of that data being protected. This concept of appropriate protections is a very specific thing to the actual data. Any declaration of 'required' or 'optional' requirements that could be mentioned here are only recommendations for that kind of Resource in general for the most common use of that Resource. Where one uses the Resource in a way that is different than this most common use, one will have different risks and thus need different protections.

Most Resources will need some form of Access Control to Create, Update, or Delete. The following general guidance is given only as general guidance for READ and QUERY access:

6.1.0.2.1 Anonymous READ Access Resources

These resources tend to not contain any individual data, or business sensitive data. Most often these Resources will be available for anonymous access, meaning there is no access control based on the user or system requesting. However, these Resources do tend to contain important information that must be authenticated back to the source publishing them, and protected from integrity failures in communication. For this reason, server authenticated https (TLS) is recommended to provide authentication of the server and integrity protection in transit. This is normal web-server use of https.

6.1.0.2.2 Business Sensitive Resources

These Resources tend to not contain any individual data, but do have data that describe business or service sensitive data. The use of the term Business is not intended to only mean an incorporated business, but rather the broader concept of an organization, location, or other group that is not identifiable as individuals. Often these resources will require some sort of client authentication to assure that only authorized access is given. The client access control may be to individuals, or may be to system identity. For this purpose, possible client authentication methods such as: mutual-authenticated-TLS, APIKey, App signed JWT, or App OAuth client-id JWT For example: an App that uses a Business protected Provider Directory to determine other business endpoint details.

6.1.0.2.3 Individual Sensitive

These Resources do NOT contain Patient data, but do contain individual information about other participants. These other individuals are Practitioners, PractionerRole, CareTeam, or other users. These identities are needed to enable the practice of healthcare. These identities are identities under general privacy regulations, and thus must consider Privacy risk. Often access to these other identities are covered by business relationships. For this purpose, access to these Resources will tend to be Role specific using methods such as RBAC or ABAC.

6.1.0.2.4 Patient Sensitive

These Resources make up the bulk of FHIR and therefore are the most commonly understood. These Resources contain highly sensitive health information, or are closely linked to highly sensitive health information. These Resources will often use the security labels to differentiate various confidentiality levels within this broad group of Patient Sensitive data. Access to these Resources often requires a declared Purpose Of Use. Access to these Resources is often controlled by a Privacy Consent. See the section below on Authorization and Access Control.

6.1.0.2.5 Not classified

Some Resources can be used for a wide scope of use-cases that span very sensitive to very non-sensitive. These Resources do not fall into any of the above classifications, as their sensitivity is highly variable. These Resources will need special handling. These Resources often contain metadata that describes the content in a way that can be used for Access Control decisions.

6.1.0.3 Communications

For the RESTful API, normal HTTP security rules apply. Please follow the HTTP specification Security Considerations section 9 . The Service Base URL will specify whether TLS is required. Client authentication may be required by the server, possibly including the requirement for client certificates. When returning

responses to non-authorized clients, ensure that Hypertext Transfer Protocol (HTTP) headers of a web server and API error messages or faults do not disclose detailed information about the underlying web server that could be the source of potential exploitation.

Please follow IETF Best Current Practice (BCP 195) - "Recommendations for Secure Use of Transport Layer Security (TLS) and Datagram Layer Security (DTLS)" . This recommends TLS 1.2 or higher be used for all production data exchange, limits support for lower versions of TLS, and forbids SSL. Versions of TLS prior to TLS 1.2 should be used only where TLS 1.2 or higher is not available between partners. When using TLS, use with strong cipher suites (e.g. AES).

Consider using additional methods of security for an API to help authenticate where Domain Name System (DNS) responses are coming from and ensure that they are valid. For example, the use of Domain Name System Security Extensions (DNSSEC), a suite of extensions that add security to the DNS protocol, can ensure that domains associated with API endpoints that transmit health information or information required for API access are secure. DNSSEC provides origin authority, data integrity, and authenticated denial of existence. With DNSSEC, the DNS protocol is much less susceptible to certain types of attacks, particularly DNS spoofing attacks.

The TLS communications are established prior to any HTTP command/response, so the whole FHIR interaction is protected by the TLS communications. The security of the endpoints of the TLS communications must be risk-managed, so as to prevent inappropriate risks (e.g. audit logging of the GET parameters into an unprotected audit log).

When it is desirable to support browser-based javascript client applications, servers SHOULD consider enabling cross-origin resource sharing (CORS) for the REST operations. Consider advice from sources including Enable-CORS and Moesif blog on Guide to CORS Pitfalls . Experience shows that this is an area where ongoing issues may be expected as security holes are found and closed on an ongoing basis.

6.1.0.4 Authentication

Other than testing systems, FHIR servers should authenticate the clients. The server may choose to authenticate the client system and trust it, or to authenticate the individual user by a variety of techniques. For web-centric environments it is recommended to use OpenID Connect (or other suitable authentication protocol) to verify identity of the end user, where it is necessary that end-users be identified to the application. It is recommended that OAuth be used to authenticate and/or authorize the client and user. The Smart-On-FHIR profile on OAuth is a recommended method for using OAuth.

The HEART Working Group has developed a set of privacy and security specifications that enable an individual to control the authorization of access to RESTful health-related data sharing APIs, and to facilitate the development of interoperable implementations of these specifications by others. IHE IUA Profile constrains OAuth token attributes to support Healthcare.

All systems are shall protect authenticator mechanisms, and select the type of credential/strength of authenticator based on use-case and risk management.

6.1.0.5 Authorization/Access Control

Correctly identifying people, devices, locations and organizations is one of the foundations that any security system is built on. Most applications of security protocols, whether authentication, access control, digital signatures, etc. rely on the correct mapping between the relevant resources and the underlying systems. Note that this isn't necessary. There is nothing in FHIR that requires or relies on any security being in place, or any particular security implementation. However, real world usage will generally require this.

A holder of data should not allow the data to be communicated unless there are enough assurances that the other party is authorized to receive it. This is true for a client creating a resource through a PUT/POST, as much as it is true for a server returning resources on a GET. The presumption is that without proper authorization, to the satisfaction of the data holder, the data does not get communicated.

Two of the classic Access Control models are: Role-Based Access Control (RBAC), and Attribute-Based Access Control (ABAC).

In Role-Based Access Control (RBAC), permissions are operations on an object that a user wishes to access. Permissions are grouped into roles. A role characterizes the functions a user can perform. Roles are assigned to users. If the user's role has the appropriate permissions to access an object, then that user is granted access to the object. FHIR readily enables RBAC, as FHIR Resources are object types and the CRUDE (Create, Read, Update, Delete, Execute) events (the FHIR equivalent to permissions in the RBAC scheme) are operations on those objects.

In Attribute-Based Access Control (ABAC), a user requests to perform operations on objects. That user's access request is granted or denied based on a set of access control policies that are specified in terms of attributes and conditions. FHIR readily enables ABAC, as instances of a Resource in FHIR (again, Resources are object types) can have attributes associated with them. These attributes include security tags, environment conditions, and a host of user and object characteristics, which are the same attributes as those used in ABAC. Attributes help define the access control policies that determine the operations a user may perform on a Resource (in FHIR) or object (in ABAC). For example, a tag (or attribute) may specify that the

identified Resource (object) is not to be further disclosed without explicit consent from the patient.

The rules behind the access control decision are often very complex, and potentially depend on information sourced from:

• Client, such as user identity, user role, location, level of assurance

• Resource, such as confidentiality, sensitivity, type of data, date ranges covered by the data, author of the data

• Patient, such as the patient identity, patient relationship to the user, patient consent policies

• Context of the transaction, system identity, time-of-day, purpose of use, workflow state, and transport security

For one source of further information, see the IHE Access Control white paper

Access control constraints may result in data returned in a read or search being redacted or otherwise restricted. See Variations between Submitted data and Retrieved data.

6.1.0.5.1 Access Control Considerations

• The basic CRUD methods on resources. A security implementation must evaluate whether a client can read, update create or delete a given resource.

• Chained search provides the ability to disclose information on related resources. A security implementation must consider whether a client has the permission to access the resource being searched on, as well as the chained resource(s)

• _include and _revinclude search parameters allow client to request related resources. A security implementation must determine if the client has access to the included resources.

• security labels

• Several resources, including Bundle, Composition, Group and List, are designed to contain other resources. A security implementation should consider whether access to an individual resource, such as a Bundle, should permit access to all resources contained within the resource.

• FHIR defines several operations that may be supported by a server. Security implementations must evaluate whether a client can invoke these operations and what information should be returned from them. Fetch Encounter Record, Evaluate Measure, Observation Statistics, Find Patient Matches using MPI-based Logic, and Fetch Patient Record specifically provide the ability to disclose patient information.

• Batch and transaction processing provide ways for clients to create and update information in bulk. Security implementations should consider whether a client can initiate one of these interactions and make authorization decisions on each action in the batch/transaction.

• Security implementations must be aware of the Break the Glass protocol (e.g. break the glass) (example).

6.1.0.5.2 Access Denied Response Handling

A web-server, especially hosting FHIR, must choose the response carefully when an Access Denied condition exists. Returning too much information may expose details that should not be communicated. The Access Denied condition might be because of missing but required Authentication, the user is not authorized to access the endpoint, the user is not authorized to access specific data, or other policy reasons.

To balance usability of the returned result vs appropriate protection, the actual result method used needs to be controlled by policy and context. Typical methods of handling Access Denied used are:

Return a Success with Bundle containing zero results - This result is indistinguishable from the case where no data is known. When consistently returned on Access Denied, this will not expose which patients exist, or what data might be blinded. This method is also consistent with cases where some results are authorized while other results are blinded. This can only be used when returning a Bundle is a valid result.

Return a 404 "Not Found" - This also protects from data leakage as it is indistinguishable from a query against a resource that doesn't exist. It does however leak that the user authentication is validated.

Return a 403 "Forbidden" - This communicates that the reason for the failure is an Authorization failure. It should only be used when the client and/or user is well enough known to be given this information. Thus this method is most used when the user can know that they are forbidden access. It doesn't explain how the user might change things to become authorized.

Return a 401 "Unauthorized" - This communicates that user authentication was attempted and failed to be authenticated.

Note that if a server allows PUT to a new location, it is not feasible to return 404 Not Found. This means that clients can use this to test whether content exists that they are not able to access, which is a minor, but potentially significant, leak of information.

6.1.0.6 Audit Logging

FHIR provides an AuditEvent resource suitable for use by FHIR clients and servers to record when a security or privacy relevant event has occurred. This form of audit logging records as much detail as reasonable at the time the event happened.

When used to record security and privacy relevant events, the AuditEvent can then be used by properly authorized applications to support audit reporting, alerting, filtering, and forwarding. This model has been developed and used in healthcare for a decade as IHE-ATNA profile . ATNA log events can be automatically converted to AuditEvent resources, and from there, client applications are able to search the audit events, or subscribe to them.

For HTTP logs, implementers need to consider the implications of distributing access to the logs. HTTP logs, including those that only contain the URL itself, should be regarded as being as sensitive as the resources themselves. Even if direct PHI is kept out of the logs by careful avoidance of search parameters (e.g. by using POST), the logs will still contain a rich set of information about the clinical records.

6.1.0.7 Attachments

Several FHIR resources include attachments. Attachments can either be references to content found elsewhere or included inline encoded in base64. Attachments represent security risks in a way that FHIR resources do not, since some attachments contain executable code. Implementers should always use caution when handling resources.

6.1.0.8 Security Labels

See Security Labels.

6.1.0.9 Narrative

FHIR resources include an XHTML narrative, so that applications can display the contents of the resource to users without having to fully and correctly process the data in the resource. However, displaying HTML is associated with several known security issues that have been observed in production systems in other contexts (e.g. with CDA). For this reason, the FHIR narrative can't contain active content. However, care is still needed when displaying the narrative:

• Validate the narrative (the standard FHIR schemas do not allow active content, and the reference implementations won't handle it). Note, though, that external references could still be included in CSS, and removing/preventing these are outside the scope of schemas and reference implementations.

• Ensure that any external references to images or anchors (e.g. outside the resource) do not cause the display software to leak sensitive information in headers

• Do not allow external links to run in a privileged context such as the EHR unless you are sure they can be trusted

• Care should be taken to differentiate HTTP RESTful (API) from browser-based server content. Specifically, one should separate user session cookies, as an attacker could create content that serves up with content-type "text/html" and has content like "<script>send_to_attacker(document.cookie);</script>".

Also note that the inclusion of an external reference to an image can allow the server that hosts the image to track when the resource is displayed. This may be a feature or a problem depending on the context.

In addition to narrative, Documents may also contain stylesheets. Unlike with CDA, the stylesheets are simple CSS stylesheets, not executable XSLT, so the same security risks do not apply. However, CSS stylesheets may still reference external content (e.g. background images), and applications displaying documents should ensure that CSS links are not automatically followed without checking their safety first, and that session/identifying information does not leak with any use of external links.

6.1.1 Security Labels

URL: https://hl7.org/fhir/R4/security-labels.html
FHIR Infrastructure Work Group | Maturity Level: 3 | Standards Status: Trial Use

A security label is a concept attached to a resource or bundle that provides specific security metadata about the information it is fixed to. The Access Control decision engine uses the security label together with any provenance resources associated with the resource and other metadata (e.g. the resource type, resource contents, etc.) to

• approve read, change, and other operations

• determine what resources can be returned

• determine what handling caveats must be conveyed with the data

Security Labels enable more data to flow as they enable policy fragments to accompany the resource data.

The intent of a security label is that the recipient of resources or bundles with security-tags is obligated to enforce the handling caveats of the tags and carry the security labels forward as appropriate.

Security labels are only a device to connect specific resources, bundles, or operations to a wider security framework; a full set of policy and consent statements and their consequent obligations is needed to give the labels meaning. Because of this, security labels are most effective in fully trusted environments - that is, where all trading partners have agreed to abide by them in a Mutual Trust Framework. Note also that security labels support policy, and specific tagging of individual resources is not always required to implement policy

correctly.

In the absence of this kind of pre-agreement, Security Labels may still be used by individual parties to assist with security role checking, but they might not all be recognized and enforced, which in turn limits what information can flow.

Local agreements and implementation profiles for the use security labels should describe how the security labels connect to the relevant consent and policy statements, and in particular:

• Which Security Labels are able to be used

• What to do if a resource has an unrecognized security label on it

• Authoring obligations around security labels

• Operational implications of security labels

This specification defines a basic set of labels for the most common use cases trading partners, and a wider array of security labels that allow much finer grained management of the information.

6.1.1.1 Representing Security Labels

A security label is represented as a Coding, with the following important properties:

system | The coding scheme from which label is taken (see code system URI, and below)
code | a code from the coding scheme that identifies the security label and code is a value from the code system
display | The display form for the code (mostly for use when a system doesn't recognize the code)

An example XML Patient Resource with a "Restricted" tag associated with it, as represented in an HTTP response:

```
<Patient xmlns="http://hl7.org/fhir">
  <meta>
    <security>
      <system value="http://terminology.hl7.org/CodeSystem/v3-Confidentiality"/>
      <code value="R"/>
      <display value="Restricted"/>
    </security>
  </meta>
...  [snip] ...
</Patient>
```

A JSON search result that includes a resource that the receiving application must delete all copies of the resource after using it:

```
{
  "resourceType" : "Bundle",
  "type" : "searchset",
  ... other headers etc.....
  "entry" : [
     ... other entries ....
     {
       "resource": {
         "id" : "1",
         "meta" : {
           "security" : [{
             "system" : "http://terminology.hl7.org/CodeSystem/v3-ActCode",
             "code" : "DELAU",
             "display" : "delete after use"
           }]
         }
         ... other content etc.....
       }
     },
     ... other entries ....
  ]
}
```

Note: the actual terms used in these examples are described below.

The basic framework for security labels is described by the HL7 Healthcare Classification System . This specification identifies how security labels are defined and provides a relatively comprehensive list of labels. All the HCS defined labels (see below for the lists) can be used as security labels on FHIR resources and bundles (e.g. requests and responses).

In addition, other security labels not defined here or in the HCS can be defined by jurisdictions, vendors and/or projects and used as appropriate. However, note that:

• Defining additional security labels will increase costs associated with information and system portability

• Implementation guides and applications SHOULD always use the applicable label defined by the HCS if one exists

Note: The use of security labels and the expression of common shared security policies is a matter of ongoing discussion and development in several communities.

6.1.1.2 Core Security Labels

This specification defines a set of core security labels for all FHIR systems. All conformant FHIR Applications SHOULD use these labels where appropriate. For all these labels, how they are operationalized - their use and interpretation - is subject to the applicable Mutual Trust Framework agreement as described above.

Name/ Tag | Description

Context of Use

Purpose of Use | These Purpose of Use (system = http://terminology.hl7.org/CodeSystem/v3-PurposeOfUse) is an indication of a reason for performing one or more operations on information. which may be permitted by source system's security policy in accordance with one or more privacy policies and consent directives. Such as collecting personal health information for research or public health purposes. Notes may be used as: The rationale or purpose for a request for data The use limitation on a data Bundle See discussion on HCS below

Data Sensitivity

Confidentiality codes | These confidentiality class (system = http://terminology.hl7.org/CodeSystem/v3-Confidentiality) can be applied to any resource or bundle. They are generally assigned by the author of the resource but can be modified subsequently as a matter of operational management. The Confidentiality classifications describe the sensitivity of the information in a resource about whether it should made available or disclosed to unauthorized individuals, entities, or processes. Notes: In the absence of a confidentiality code, the basic confidentiality of a resource may be implied by its definition and content; e.g. a patient's condition is far more likely to be confidential than a practitioner resource, and a Diagnostic Report with an HIV test is always highly confidential, whereas a routine electrolytes report is rarely particularly confidential The confidentiality of a bundle is always as confidential as the most confidential resource in the bundle The additional security labels are more specific to support very specific fine-grained access control and should always be used in association with an appropriate confidentiality label. See discussion on HCS below

Control of Flow

Delete After Use: ActCode.DELAU | An application receiving a resource with this label must delete all copies after the immediate use for which the data was exchanged, is complete. Notes: This may imply a prohibition not storing the resource in any audit trail as well Additional security labels can make exceptions to the blanket restriction this implies. This allows a resource to be exchanged with a blanket rule not to retain copies unless the exact rules for retaining it can be followed

Do Not Re-use: ActCode.NOREUSE | An application receiving a resource with this label may only use it for the immediate purpose of use. In particular, the application is not authorized to re-distribute (i.e. exchange this resource with any other application). Notes: The exact interpretation of "immediate purpose of use" and the boundaries of "the application" are determined by local policy Additional security labels can make exceptions to the blanket restriction this implies. This allows a resource to be exchanged with a blanket rule not to re-use unless the exact rules for doing so can be followed

Test Data: ActCode.HTEST | This marks that a resource has been created to test an application, and is not real production data Notes: Most testing is performed on dedicated test systems where all the data is test data Some testing is performed on production systems, which is where this security label is used This is a security label because different access control rules may apply to test resources

6.1.1.3 Break the Glass

There is a special security label to support the commonly encountered "break-the-glass" protocol, where a clinician (usually in an emergency context) requests emergency unauthorized access to the patient's record.

Break the Glass | http://hl7.org/fhir/security-label#break-the-glass | The requester is asking for emergency access for patient treatment. Typically, this means that the patient is unconscious and not able to provide relevant information or consent.

This purpose of use label is represented as a security label on the request, rather than on a resource, and so is represented in the request as a web category :

```
HTTP/1.1 GET fhir/Patient/482735/condition
Content-Type: text/xml
Access-Control-Allow-Origin: *
Last-Modified: Thu, 19 Nov 2013 07:07:32 +1100
ETag: 24
Category: http://terminology.hl7.org/CodeSystem/v3-ActReason#BTG; scheme="http://hl7.org/fhir/tag/security";
    label="break the glass"
```

While the principle of break-the-glass is well understood, implementing it well has some challenges. This specification defines a method to represent break-the-glass in an HTTP request, but does not define any policy or protocol around such requests. At a minimum, implementations must ensure:

• How, when and why to initiate the break-the-glass is well understood

• Appropriate authorization, consent checking, and access control is used to ensure it is used properly (e.g. if using OAuth, checking that the Authorization Server allows this)

• Any use is well-represented in an AuditEvent (todo: profile for break the glass)

See this paper for discussion of the issues involved in break-the-glass operations.

6.1.1.4 Healthcare Privacy and Security Classification System (HCS)

The security labels described above are a subset of the full set of security labels defined by the HL7 Healthcare Privacy and Security Classification System . The HCS defines 5 categories of security labels that may be applied to a resource:

Security Label | Card. | Values | Description

Confidentiality Classification | 0..1 | ConfidentialityClassification | Security label metadata classifying an IT resource (clinical fact, data, information object, service, or system capability) according to its level of sensitivity, which is based on an analysis of applicable privacy policies and the risk of financial, reputational, or other harm to an individual or entity that could result if made available or disclosed to unauthorized individuals, entities, or processes. Example Uses: Unrestricted, Normal, Very restricted

Sensitivity Category | 0..* | InformationSensitivityPolicy | Security label metadata that "segments" an IT resource by categorizing the value, importance, and vulnerability of an IT resource perceived as undesirable to share. Example Uses: STDs, Psychiatric

care, Celebrity status

Compartment Category | 0..* | Compartment | Security label metadata that "segments" an IT resource by indicating that access and use is restricted to members of a defined community or project Note: this is a different use of "Compartment" to the Patient Compartment use. Example Uses: Research, HR records

Integrity Category | 0..* | SecurityIntegrityObservationValue | Security label metadata that "segments" an IT resource by conveying the completeness, veracity, reliability, trustworthiness, and provenance of an IT resource Example Uses: Anonymized, signed, patient reported

Handling Caveat | 0..* | SecurityControlObservationValue | Security label metadata conveying dissemination controls and information handling instructions such as obligations and retention policies to which an IT resource custodian or receiver must comply. This type of handling caveat SHALL be assigned to a clinical fact if required by jurisdictional or organizational policy, which may be triggered by a patient consent directive Example Uses: do not disclose, various restrictions on use, and policy marks

Each of these security labels identifies a ValueSet that lists a set of possible codes for the security label.

6.1.1.4.1 Jurisdiction Specific Security Labels

The HL7 Healthcare Classification System also allows for Realm-specific privacy law or policy category codes for use in security labels in specific domains. These domains are included with this specification:

Security Label | Card. | Values | Description

US Privacy Law | 0..* | ActUSPrivacyLaw | Security label metadata that "segments" an IT resource by indicating the legal provisions to which the assignment of a Confidentiality Classification complies in the US.

6.1.2 Digital Signatures

URL: https://hl7.org/fhir/R4/signatures.html

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Trial Use

This specification recommends the use of W3C Digital Signatures or JSON Digital Signatures for digital signatures. Resources can be signed using the Provenance resource to carry a detached digital signature . The Signature datatype is available to support various signature types including non-repudiation purposes. Further details on creation and validation of Signatures are defined.

In addition, documents may be signed using an enveloped signature. A specification for enveloped signature is profiled in the IHE DSG profile .

Neither of these definitions prohibits policies that accept the use of other ways of using digital signatures or scanned wet signatures.

Trial-Use Note: The use of signatures with RESTful interfaces is a poorly understood area, and we would welcome reports of implementation experience. See discussion on use of Digital Signature in FHIR

Feedback is welcome here .

6.2 Resource Consent - Content

URL: https://hl7.org/fhir/R4/consent.html

Community Based Collaborative Care Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Patient

A record of a healthcare consumer’s choices, which permits or denies identified recipient(s) or recipient role(s) to perform one or more actions within a given policy context, for specific purposes and periods of time.

6.2.1 Scope and Usage

The purpose of this Resource is to be used to express a Consent regarding Healthcare. There are four anticipated uses for the Consent Resource, all of which are written or verbal agreements by a healthcare consumer [grantor] or a personal representative, made to an authorized entity [grantee] concerning authorized or restricted actions with any limitations on purpose of use, and handling instructions to which the authorized entity must comply:

• Privacy Consent Directive: Agreement to collect, access, use or disclose (share) information.

• Medical Treatment Consent Directive: Consent to undergo a specific treatment (or record of refusal to consent).

• Research Consent Directive: Consent to participate in research protocol and information sharing required.

• Advance Care Directives: Consent to instructions for potentially needed medical treatment (e.g. DNR).

This resource is scoped to cover all four uses, but at this time, only the privacy use case is modeled. The scope of the resource may change when the other possible scopes are investigated, tested, or profiled.

A FHIR Consent Directive instance is considered the encoded legally binding Consent Directive if it meets requirements of a policy domain requirements for an enforceable contract. In some domains, electronic signatures of one or both of the parties to the content of an encoded representation of a Consent Form is deemed to constitute a legally binding Consent Directive. Some domains accept a notary’s electronic signature over the wet or electronic signature of a party to the Consent Directive as the additional identity proofing required to make an encoded Consent Directive legally binding. Other domains may only accept a wet signature or might not require the parties’ signatures at all.

Whatever the criteria are for making an encoded FHIR Consent Directive legally binding, anything less than a legally binding representation of a Consent Directive must be identified as such, i.e., as a derivative of the legally binding Consent Directive, which has specific usage in Consent Directive workflow management.

Definitions:

Consent | The record of a healthcare consumer’s policy choices, which permits or denies identified recipient(s) or recipient role(s) to perform one or more actions within a given policy context, for specific purposes and periods of time

Consent Directive | The legal record of a healthcare consumer's agreement with a party responsible for enforcing the consumer’s choices, which permits or denies identified actors or roles to perform actions affecting the consumer within a given context for specific purposes and periods of time

Consent Form | Human readable consent content describing one or more actions impacting the grantor for which the grantee would be authorized or prohibited from performing. It includes the terms, rules, and conditions pertaining to the authorization or restrictions, such as effective time, applicability or scope, purposes of use, obligations and prohibitions to which the grantee must comply. Once a Consent Form is “executed” by means required by policy, such as verbal agreement, wet signature, or electronic/digital signature, it becomes a legally binding Consent Directive.

Consent Directive Derivative | Consent Content that conveys the minimal set of information needed to manage Consent Directive workflow, including providing Consent Directive content sufficient to: Represent a Consent Directive Register or index a Consent Directive Query and respond about a Consent Directive Retrieve a Consent Directive Notify authorized entities about Consent Directive status changes Determine entities authorized to collect, access, use or disclose information about the Consent Directive or about the information governed by the Consent Directive. Derived Consent content includes the Security Labels encoding the applicable privacy and security policies. Consent Security Labels inform recipients about specific access control measures required for compliance.

Consent Statement | A Consent Directive derivative has less than full fidelity to the legally binding Consent Directive from which it was "transcribed". It provides recipients with the full content representation they may require for compliance purposes, and typically include a reference to or an attached unstructured representation for recipients needing an exact copy of the legal agreement.

Consent Registration | The legal record of a healthcare consumer's agreement with a party responsible for enforcing the consumer’s choices, which permits or denies identified actors or roles to perform actions affecting the consumer within a given context for specific purposes and periods of timeA Consent Directive derivative that conveys the minimal set of information needed to register an active and revoked Consent Directive, or to update Consent status as it changes during its lifecycle.

Consent Query/Response Types | The FHIR Consent Resource specifies multiple Consent Search parameters, which support many types of queries for Consent Resource content. There are several Query/Response patterns that are typically used for obtaining information about consent directive content for the following use cases: Find Active Consent Directive: A query that includes sufficient consent directive content to determine whether a specific party is authorized to share information governed by a consent directive with another specific party. The Response is either: “Yes” meaning that both parties are authorized to share the information with one another. “No” meaning that the authorized querier is not permitted to share with another specific party “No information found” meaning that there is no active Consent Directive in which the querier is authorized to share the governed information. Find Consent Directive Authorized Entities: A query that includes sufficient consent directive content to return a list of entities with which the querier is authorized to share governed information. The response to an authorized querier is the list of any authorized entities with which the querier is permitted to share governed information. The response to an unauthorized querier is that “no information is found”. Find Consent Directive(s): A query that includes sufficient consent directive content to return a list of Consent Directive metadata for an authorized querier to determine what Consent Directives are available, and to locate and retrieve one or more of those Consent Directives as needed.

Policy context | Any organizational or jurisdictional policies, which may limit the consumer’s policy choices, and which includes the named range of actions allowed

Healthcare Consumer | The individual establishing his/her personal consent (i.e. Consenter). In FHIR, this is referred to as the 'Patient' though this word is not used across all contexts of care

6.2.1.1 Privacy Consent Directive (PCD)

Privacy policies define how Individually Identifiable Health Information (IIHI) is to be collected, accessed, used and disclosed. A Privacy Consent Directive as a legal record of a patient's (e.g. a healthcare consumer) agreement with a party responsible for enforcing the patient's choices, which permits or denies identified actors or roles to perform actions affecting the patient within a given context for specific purposes and periods of time. All consent directives have a policy context, which is any set of organizational or jurisdictional policies which may limit the consumer’s policy choices, and which include a named range of actions allowed. In addition, Privacy Consent Directives provide the ability for a healthcare consumer to delegate authority to a Substitute Decision Maker who may act on behalf of that individual. Alternatively, a consumer may author/publish their privacy preferences as a self-declared Privacy Consent Directive.

The Consent resource on FHIR provides support for alternative representations for expressing interoperable health information privacy consent directives in a standard form for the exchange and enforcement by sending, intermediating, or receiving systems of privacy policies that can be enforced by consuming systems (e.g., scanned documents, of computable structured entries elements, FHIR structures with optional attached, or referenced unstructured representations.) It may be used to represent the Privacy Consent Directive itself, a Consent Statement, which electronically represents a Consent Directive, or Consent Metadata, which is the minimum necessary consent content derived from a Consent Directive for use in workflow management.

6.2.2 Boundaries and Relationships

Consent management - particularly privacy consent - is complicated by the fact that consent to share is often itself necessary to protect. The need to protect the privacy of the privacy statement itself competes with the execution of the consent statement. For this reason, it is common to deal with 'consent statements' that are only partial representations of the full consent statement that the patient provided.

For this reason, the consent resource contains two elements that refer back to the source: a master identifier, and a direct reference to content from which this Consent Statement was derived. That reference can be one of several things:

• A reference to another consent resource from which this limited statement was derived

• A reference to a document format for the original source (e.g. PDF or CDA - see the HL7 CDAR2 ConsentDirective Implementation Guide , which incorporated the IHE Basic Patient Privacy Consents (BPPC)), either directly, or in a reference

• The source can be included in the consent as an attachment

The consent statements represent a chain that refers back to the original source consent directive. Applications may be able to follow the chain back to the source but should not generally assume that they are authorized to do this.

Consent Directives are executed by verbal acknowledge or by being signed - either on paper, or digitally. Consent Signatures will be found in the Provenance resource (example consent and signature). Implementation Guides will generally make rules about what signatures are required, and how they are to be shared and used.

6.2.3 Background and Context

Change to "The Consent resource is structured with a base policy (represented as Consent.policy/Consent.policyRule) which is either opt-in or opt-out, followed by a listing of exceptions to that policy (represented as Consent.provision(s)). The exceptions can be additional positive or negative exceptions upon the base policy. The set of exceptions include a list of data objects, list of authors, list of recipients, list of Organizations, list of purposeOfUse, and Date Range.

The enforcement of the Privacy Consent Directive is not included but is expected that enforcement can be done using a mix of the various Access Control enforcement methodologies (e.g. OAuth, UMA, XACML). This enforcement includes the details of the enforcement meaning of the elements of the Privacy Consent Directive, such as the rules in place when there is an opt-in consent would be specific about which organizational roles have access to what kinds of resources (e.g. RBAC, ABAC). The specification of these details is not in scope for the Consent resource.

This resource is referenced by itself and ResearchSubject

6.2.4 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints

Consent | ITU | | DomainResource | A healthcare consumer's choices to permit or deny recipients or roles to perform actions for specific purposes and periods of time + Rule: Either a Policy or PolicyRule + Rule: IF Scope=privacy, there must be a patient + Rule: IF Scope=research, there must be a patient + Rule: IF Scope=adr, there must be a patient + Rule: IF Scope=treatment, there must be a patient Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | Identifier for this record (external references)

status | ?!Σ | 1..1 | code | draft | proposed | active | rejected | inactive | entered-in-error ConsentState (Required)

scope | ?!Σ | 1..1 | CodeableConcept | Which of the four areas this resource covers (extensible) Consent Scope Codes (Extensible)

category | Σ | 1..* | CodeableConcept | Classification of the consent statement - for indexing/retrieval Consent Category Codes (Extensible)

patient | Σ | 0..1 | Reference(Patient) | Who the consent applies to

dateTime | Σ | 0..1 | dateTime | When this Consent was created or indexed

performer | Σ | 0..* | Reference(Organization | Patient | Practitioner | RelatedPerson | PractitionerRole) | Who is agreeing to the policy and rules

organization | Σ | 0..* | Reference(Organization) | Custodian of the consent

source[x] | Σ | 0..1 | | Source from which this consent is taken

sourceAttachment | | | Attachment

sourceReference | | | Reference(Consent | DocumentReference | Contract | QuestionnaireResponse)

policy | | 0..* | BackboneElement | Policies covered by this consent

authority | I | 0..1 | uri | Enforcement source for policy

uri | I | 0..1 | uri | Specific policy covered by this consent

policyRule | ΣI | 0..1 | CodeableConcept | Regulation that this consents to Consent PolicyRule Codes (Extensible)

verification | Σ | 0..* | BackboneElement | Consent Verified by patient or family

verified | Σ | 1..1 | boolean | Has been verified

verifiedWith | | 0..1 | Reference(Patient | RelatedPerson) | Person who verified
verificationDate | | 0..1 | dateTime | When consent verified
provision | Σ | 0..1 | BackboneElement | Constraints to the base Consent.policyRule
type | Σ | 0..1 | code | deny | permit ConsentProvisionType (Required)
period | Σ | 0..1 | Period | Timeframe for this rule
actor | | 0..* | BackboneElement | Who|what controlled by this rule (or group, by role)
role | | 1..1 | CodeableConcept | How the actor is involved SecurityRoleType (Extensible)
reference | | 1..1 | Reference(Device | Group | CareTeam | Organization | Patient | Practitioner | RelatedPerson | PractitionerRole) | Resource for the actor (or group, by role)
action | Σ | 0..* | CodeableConcept | Actions controlled by this rule Consent Action Codes (Example)
securityLabel | Σ | 0..* | Coding | Security Labels that define affected resources SecurityLabels (Extensible)
purpose | Σ | 0..* | Coding | Context of activities covered by this rule V3 Value SetPurposeOfUse (Extensible)
class | Σ | 0..* | Coding | e.g. Resource Type, Profile, CDA, etc. Consent Content Class (Extensible)
code | Σ | 0..* | CodeableConcept | e.g. LOINC or SNOMED CT code, etc. in the content Consent Content Codes (Example)
dataPeriod | Σ | 0..1 | Period | Timeframe for data controlled by this rule
data | Σ | 0..* | BackboneElement | Data controlled by this rule
meaning | Σ | 1..1 | code | instance | related | dependents | authoredby ConsentDataMeaning (Required)
reference | Σ | 1..1 | Reference(Any) | The actual data reference
provision | | 0..* | see provision | Nested Exception Rules
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Consent xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Identifier for this record (external references) --></identifier>
 <status value="[code]"/><!-- 1..1 draft | proposed | active | rejected | inactive | entered-in-error -->
 <scope><!-- 1..1 CodeableConcept Which of the four areas this resource covers (extensible) --></scope>
 <category><!-- 1..* CodeableConcept Classification of the consent statement - for indexing/retrieval --></category>
 <patient><!-- 0..1 Reference(Patient) Who the consent applies to --></patient>
 <dateTime value="[dateTime]"/><!-- 0..1 When this Consent was created or indexed -->
 <performer><!-- 0..* Reference(Organization|Patient|Practitioner|RelatedPerson|
   PractitionerRole) Who is agreeing to the policy and rules --></performer>
 <organization><!-- 0..* Reference(Organization) Custodian of the consent --></organization>
 <source[x]><!-- 0..1 Attachment|Reference(Consent|DocumentReference|Contract|
   QuestionnaireResponse) Source from which this consent is taken --></source[x]>
 <policy>  <!-- 0..* Policies covered by this consent -->
  <authority value="[uri]"/><!--  0..1 Enforcement source for policy -->
  <uri value="[uri]"/><!--  0..1 Specific policy covered by this consent -->
 </policy>
 <policyRule><!--  0..1 CodeableConcept Regulation that this consents to --></policyRule>
 <verification>  <!-- 0..* Consent Verified by patient or family -->
  <verified value="[boolean]"/><!-- 1..1 Has been verified -->
  <verifiedWith><!-- 0..1 Reference(Patient|RelatedPerson) Person who verified --></verifiedWith>
  <verificationDate value="[dateTime]"/><!-- 0..1 When consent verified -->
 </verification>
 <provision>  <!-- 0..1 Constraints to the base Consent.policyRule -->
  <type value="[code]"/><!-- 0..1 deny | permit -->
  <period><!-- 0..1 Period Timeframe for this rule --></period>
  <actor>  <!-- 0..* Who|what controlled by this rule (or group, by role) -->
   <role><!-- 1..1 CodeableConcept How the actor is involved --></role>
   <reference><!-- 1..1 Reference(Device|Group|CareTeam|Organization|Patient|
     Practitioner|RelatedPerson|PractitionerRole) Resource for the actor (or group, by role) --></reference>
  </actor>
  <action><!-- 0..* CodeableConcept Actions controlled by this rule --></action>
  <securityLabel><!-- 0..* Coding Security Labels that define affected resources --></securityLabel>
  <purpose><!-- 0..* Coding Context of activities covered by this rule --></purpose>
  <class><!-- 0..* Coding e.g. Resource Type, Profile, CDA, etc. --></class>
  <code><!-- 0..* CodeableConcept e.g. LOINC or SNOMED CT code, etc. in the content --></code>
  <dataPeriod><!-- 0..1 Period Timeframe for data controlled by this rule --></dataPeriod>
  <data>  <!-- 0..* Data controlled by this rule -->
   <meaning value="[code]"/><!-- 1..1 instance | related | dependents | authoredby -->
   <reference><!-- 1..1 Reference(Any) The actual data reference --></reference>
  </data>
  <provision><!-- 0..* Content as for Consent.provision Nested Exception Rules --></provision>
 </provision>
</Consent>
```

JSON Template

```
{
  "resourceType" : "Consent",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Identifier for this record (external references)
  "status" : "<code>", // R!  draft | proposed | active | rejected | inactive | entered-in-error
  "scope" : { CodeableConcept }, // R!  Which of the four areas this resource covers (extensible)
  "category" : [{ CodeableConcept }], // R!  Classification of the consent statement - for indexing/retrieval
  "patient" : { Reference(Patient) }, // Who the consent applies to
  "dateTime" : "<dateTime>", // When this Consent was created or indexed
  "performer" : [{ Reference(Organization|Patient|Practitioner|RelatedPerson|
   PractitionerRole) }], // Who is agreeing to the policy and rules
  "organization" : [{ Reference(Organization) }], // Custodian of the consent
  // source[x]: Source from which this consent is taken. One of these 2:
```

```
  "sourceAttachment" : { Attachment },
  "sourceReference" : { Reference(Consent|DocumentReference|Contract|
   QuestionnaireResponse) },
  "policy" : [{ // Policies covered by this consent
    "authority" : "<uri>", // C? Enforcement source for policy
    "uri" : "<uri>" // C? Specific policy covered by this consent
  }],
  "policyRule" : { CodeableConcept }, // C? Regulation that this consents to
  "verification" : [{ // Consent Verified by patient or family
    "verified" : <boolean>, // R!  Has been verified
    "verifiedWith" : { Reference(Patient|RelatedPerson) }, // Person who verified
    "verificationDate" : "<dateTime>" // When consent verified
  }],
  "provision" : { // Constraints to the base Consent.policyRule
    "type" : "<code>", // deny | permit
    "period" : { Period }, // Timeframe for this rule
    "actor" : [{ // Who|what controlled by this rule (or group, by role)
      "role" : { CodeableConcept }, // R!  How the actor is involved
      "reference" : { Reference(Device|Group|CareTeam|Organization|Patient|
     Practitioner|RelatedPerson|PractitionerRole) } // R!  Resource for the actor (or group, by role)
    }],
    "action" : [{ CodeableConcept }], // Actions controlled by this rule
    "securityLabel" : [{ Coding }], // Security Labels that define affected resources
    "purpose" : [{ Coding }], // Context of activities covered by this rule
    "class" : [{ Coding }], // e.g. Resource Type, Profile, CDA, etc.
    "code" : [{ CodeableConcept }], // e.g. LOINC or SNOMED CT code, etc. in the content
    "dataPeriod" : { Period }, // Timeframe for data controlled by this rule
    "data" : [{ // Data controlled by this rule
      "meaning" : "<code>", // R!  instance | related | dependents | authoredby
      "reference" : { Reference(Any) } // R!  The actual data reference
    }],
    "provision" : [{ Content as for Consent.provision }] // Nested Exception Rules
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Consent;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Consent.identifier [ Identifier ], ... ; # 0..* Identifier for this record (external references)
  fhir:Consent.status [ code ]; # 1..1 draft | proposed | active | rejected | inactive | entered-in-error
  fhir:Consent.scope [ CodeableConcept ]; # 1..1 Which of the four areas this resource covers (extensible)
  fhir:Consent.category [ CodeableConcept ], ... ; # 1..* Classification of the consent statement - for
    indexing/retrieval
  fhir:Consent.patient [ Reference(Patient) ]; # 0..1 Who the consent applies to
  fhir:Consent.dateTime [ dateTime ]; # 0..1 When this Consent was created or indexed
  fhir:Consent.performer [ Reference(Organization|Patient|Practitioner|RelatedPerson|PractitionerRole) ], ... ; # 0..*
    Who is agreeing to the policy and rules
  fhir:Consent.organization [ Reference(Organization) ], ... ; # 0..* Custodian of the consent
  # Consent.source[x] : 0..1 Source from which this consent is taken. One of these 2
    fhir:Consent.sourceAttachment [ Attachment ]
    fhir:Consent.sourceReference [ Reference(Consent|DocumentReference|Contract|QuestionnaireResponse) ]
  fhir:Consent.policy [ # 0..* Policies covered by this consent
    fhir:Consent.policy.authority [ uri ]; # 0..1 Enforcement source for policy
    fhir:Consent.policy.uri [ uri ]; # 0..1 Specific policy covered by this consent
  ], ...;
  fhir:Consent.policyRule [ CodeableConcept ]; # 0..1 Regulation that this consents to
  fhir:Consent.verification [ # 0..* Consent Verified by patient or family
    fhir:Consent.verification.verified [ boolean ]; # 1..1 Has been verified
    fhir:Consent.verification.verifiedWith [ Reference(Patient|RelatedPerson) ]; # 0..1 Person who verified
    fhir:Consent.verification.verificationDate [ dateTime ]; # 0..1 When consent verified
  ], ...;
  fhir:Consent.provision [ # 0..1 Constraints to the base Consent.policyRule
    fhir:Consent.provision.type [ code ]; # 0..1 deny | permit
    fhir:Consent.provision.period [ Period ]; # 0..1 Timeframe for this rule
    fhir:Consent.provision.actor [ # 0..* Who|what controlled by this rule (or group, by role)
      fhir:Consent.provision.actor.role [ CodeableConcept ]; # 1..1 How the actor is involved
      fhir:Consent.provision.actor.reference [
    Reference(Device|Group|CareTeam|Organization|Patient|Practitioner|RelatedPerson|
  PractitionerRole) ]; # 1..1 Resource for the actor (or group, by role)
    ], ...;
    fhir:Consent.provision.action [ CodeableConcept ], ... ; # 0..* Actions controlled by this rule
    fhir:Consent.provision.securityLabel [ Coding ], ... ; # 0..* Security Labels that define affected resources
    fhir:Consent.provision.purpose [ Coding ], ... ; # 0..* Context of activities covered by this rule
    fhir:Consent.provision.class [ Coding ], ... ; # 0..* e.g. Resource Type, Profile, CDA, etc.
    fhir:Consent.provision.code [ CodeableConcept ], ... ; # 0..* e.g. LOINC or SNOMED CT code, etc. in the content
    fhir:Consent.provision.dataPeriod [ Period ]; # 0..1 Timeframe for data controlled by this rule
    fhir:Consent.provision.data [ # 0..* Data controlled by this rule
      fhir:Consent.provision.data.meaning [ code ]; # 1..1 instance | related | dependents | authoredby
      fhir:Consent.provision.data.reference [ Reference(Any) ]; # 1..1 The actual data reference
    ], ...;
    fhir:Consent.provision.provision [ See Consent.provision ], ... ; # 0..* Nested Exception Rules
  ];
]
```

Changes since R3

Consent
Consent.identifier | Max Cardinality changed from 1 to *
Consent.status | Change value set from http://hl7.org/fhir/ValueSet/consent-state-codes to http://hl7.org/fhir/ValueSet/consent-state-codes|4.0.1
Consent.scope | Added Mandatory Element
Consent.category | Min Cardinality changed from 0 to 1 Add Binding http://hl7.org/fhir/ValueSet/consent-category (extensible)
Consent.patient | Min Cardinality changed from 1 to 0
Consent.performer | Added Element
Consent.source[x] | Remove Type Identifier
Consent.policyRule | Type changed from uri to CodeableConcept Add Binding http://hl7.org/fhir/ValueSet/consent-policy (extensible)
Consent.verification | Added Element
Consent.verification.verified | Added Mandatory Element
Consent.verification.verifiedWith | Added Element
Consent.verification.verificationDate | Added Element
Consent.provision | Added Element
Consent.provision.type | Added Element
Consent.provision.period | Added Element
Consent.provision.actor | Added Element
Consent.provision.actor.role | Added Mandatory Element
Consent.provision.actor.reference | Added Mandatory Element
Consent.provision.action | Added Element
Consent.provision.securityLabel | Added Element
Consent.provision.purpose | Added Element
Consent.provision.class | Added Element
Consent.provision.code | Added Element
Consent.provision.dataPeriod | Added Element
Consent.provision.data | Added Element
Consent.provision.data.meaning | Added Mandatory Element
Consent.provision.data.reference | Added Mandatory Element
Consent.provision.provision | Added Element
Consent.period | deleted
Consent.consentingParty | deleted
Consent.actor | deleted
Consent.action | deleted
Consent.securityLabel | deleted
Consent.purpose | deleted
Consent.dataPeriod | deleted
Consent.data | deleted
Consent.except | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 12 tests that all execute ok. All tests pass round-trip testing and 12 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
Consent | ITU | | DomainResource | A healthcare consumer's choices to permit or deny recipients or roles to perform actions for specific purposes and periods of time + Rule: Either a Policy or PolicyRule + Rule: IF Scope=privacy, there must be a patient + Rule: IF Scope=research, there must be a patient + Rule: IF Scope=adr, there must be a patient + Rule: IF Scope=treatment, there must be a patient Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Identifier for this record (external references)
status | ?!Σ | 1..1 | code | draft | proposed | active | rejected | inactive | entered-in-error ConsentState (Required)
scope | ?!Σ | 1..1 | CodeableConcept | Which of the four areas this resource covers (extensible) Consent Scope Codes (Extensible)
category | Σ | 1..* | CodeableConcept | Classification of the consent statement - for indexing/retrieval Consent Category Codes (Extensible)
patient | Σ | 0..1 | Reference(Patient) | Who the consent applies to
dateTime | Σ | 0..1 | dateTime | When this Consent was created or indexed
performer | Σ | 0..* | Reference(Organization | Patient | Practitioner | RelatedPerson | PractitionerRole) | Who is agreeing to the policy and rules
organization | Σ | 0..* | Reference(Organization) | Custodian of the consent
source[x] | Σ | 0..1 | | Source from which this consent is taken
sourceAttachment | | | Attachment
sourceReference | | | Reference(Consent | DocumentReference | Contract | QuestionnaireResponse)
policy | | 0..* | BackboneElement | Policies covered by this consent
authority | I | 0..1 | uri | Enforcement source for policy
uri | I | 0..1 | uri | Specific policy covered by this consent
policyRule | ΣI | 0..1 | CodeableConcept | Regulation that this consents to Consent PolicyRule Codes (Extensible)
verification | Σ | 0..* | BackboneElement | Consent Verified by patient or family
verified | Σ | 1..1 | boolean | Has been verified

verifiedWith | | 0..1 | Reference(Patient | RelatedPerson) | Person who verified
verificationDate | | 0..1 | dateTime | When consent verified
provision | Σ | 0..1 | BackboneElement | Constraints to the base Consent.policyRule
type | Σ | 0..1 | code | deny | permit ConsentProvisionType (Required)
period | Σ | 0..1 | Period | Timeframe for this rule
actor | | 0..* | BackboneElement | Who|what controlled by this rule (or group, by role)
role | | 1..1 | CodeableConcept | How the actor is involved SecurityRoleType (Extensible)
reference | | 1..1 | Reference(Device | Group | CareTeam | Organization | Patient | Practitioner | RelatedPerson | PractitionerRole) | Resource for the actor (or group, by role)
action | Σ | 0..* | CodeableConcept | Actions controlled by this rule Consent Action Codes (Example)
securityLabel | Σ | 0..* | Coding | Security Labels that define affected resources SecurityLabels (Extensible)
purpose | Σ | 0..* | Coding | Context of activities covered by this rule V3 Value SetPurposeOfUse (Extensible)
class | Σ | 0..* | Coding | e.g. Resource Type, Profile, CDA, etc. Consent Content Class (Extensible)
code | Σ | 0..* | CodeableConcept | e.g. LOINC or SNOMED CT code, etc. in the content Consent Content Codes (Example)
dataPeriod | Σ | 0..1 | Period | Timeframe for data controlled by this rule
data | Σ | 0..* | BackboneElement | Data controlled by this rule
meaning | Σ | 1..1 | code | instance | related | dependents | authoredby ConsentDataMeaning (Required)
reference | Σ | 1..1 | Reference(Any) | The actual data reference
provision | | 0..* | see provision | Nested Exception Rules
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Consent xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Identifier for this record (external references) --></identifier>
 <status value="[code]"/><!-- 1..1 draft | proposed | active | rejected | inactive | entered-in-error -->
 <scope><!-- 1..1 CodeableConcept Which of the four areas this resource covers (extensible) --></scope>
 <category><!-- 1..* CodeableConcept Classification of the consent statement - for indexing/retrieval --></category>
 <patient><!-- 0..1 Reference(Patient) Who the consent applies to --></patient>
 <dateTime value="[dateTime]"/><!-- 0..1 When this Consent was created or indexed -->
 <performer><!-- 0..* Reference(Organization|Patient|Practitioner|RelatedPerson|
   PractitionerRole) Who is agreeing to the policy and rules --></performer>
 <organization><!-- 0..* Reference(Organization) Custodian of the consent --></organization>
 <source[x]><!-- 0..1 Attachment|Reference(Consent|DocumentReference|Contract|
   QuestionnaireResponse) Source from which this consent is taken --></source[x]>
 <policy>  <!-- 0..* Policies covered by this consent -->
  <authority value="[uri]"/><!--  0..1 Enforcement source for policy -->
  <uri value="[uri]"/><!--  0..1 Specific policy covered by this consent -->
 </policy>
 <policyRule><!--  0..1 CodeableConcept Regulation that this consents to --></policyRule>
 <verification>  <!-- 0..* Consent Verified by patient or family -->
  <verified value="[boolean]"/><!-- 1..1 Has been verified -->
  <verifiedWith><!-- 0..1 Reference(Patient|RelatedPerson) Person who verified --></verifiedWith>
  <verificationDate value="[dateTime]"/><!-- 0..1 When consent verified -->
 </verification>
 <provision>  <!-- 0..1 Constraints to the base Consent.policyRule -->
  <type value="[code]"/><!-- 0..1 deny | permit -->
  <period><!-- 0..1 Period Timeframe for this rule --></period>
  <actor>  <!-- 0..* Who|what controlled by this rule (or group, by role) -->
   <role><!-- 1..1 CodeableConcept How the actor is involved --></role>
   <reference><!-- 1..1 Reference(Device|Group|CareTeam|Organization|Patient|
     Practitioner|RelatedPerson|PractitionerRole) Resource for the actor (or group, by role) --></reference>
  </actor>
  <action><!-- 0..* CodeableConcept Actions controlled by this rule --></action>
  <securityLabel><!-- 0..* Coding Security Labels that define affected resources --></securityLabel>
  <purpose><!-- 0..* Coding Context of activities covered by this rule --></purpose>
  <class><!-- 0..* Coding e.g. Resource Type, Profile, CDA, etc. --></class>
  <code><!-- 0..* CodeableConcept e.g. LOINC or SNOMED CT code, etc. in the content --></code>
  <dataPeriod><!-- 0..1 Period Timeframe for data controlled by this rule --></dataPeriod>
  <data>  <!-- 0..* Data controlled by this rule -->
   <meaning value="[code]"/><!-- 1..1 instance | related | dependents | authoredby -->
   <reference><!-- 1..1 Reference(Any) The actual data reference --></reference>
  </data>
  <provision><!-- 0..* Content as for Consent.provision Nested Exception Rules --></provision>
 </provision>
</Consent>
```

JSON Template

```
{
  "resourceType" : "Consent",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Identifier for this record (external references)
  "status" : "<code>", // R!  draft | proposed | active | rejected | inactive | entered-in-error
  "scope" : { CodeableConcept }, // R!  Which of the four areas this resource covers (extensible)
  "category" : [{ CodeableConcept }], // R!  Classification of the consent statement - for indexing/retrieval
  "patient" : { Reference(Patient) }, // Who the consent applies to
  "dateTime" : "<dateTime>", // When this Consent was created or indexed
  "performer" : [{ Reference(Organization|Patient|Practitioner|RelatedPerson|
   PractitionerRole) }], // Who is agreeing to the policy and rules
  "organization" : [{ Reference(Organization) }], // Custodian of the consent
  // source[x]: Source from which this consent is taken. One of these 2:
```

```
  "sourceAttachment" : { Attachment },
  "sourceReference" : { Reference(Consent|DocumentReference|Contract|
   QuestionnaireResponse) },
  "policy" : [{ // Policies covered by this consent
    "authority" : "<uri>", // C? Enforcement source for policy
    "uri" : "<uri>" // C? Specific policy covered by this consent
  }],
  "policyRule" : { CodeableConcept }, // C? Regulation that this consents to
  "verification" : [{ // Consent Verified by patient or family
    "verified" : <boolean>, // R!  Has been verified
    "verifiedWith" : { Reference(Patient|RelatedPerson) }, // Person who verified
    "verificationDate" : "<dateTime>" // When consent verified
  }],
  "provision" : { // Constraints to the base Consent.policyRule
    "type" : "<code>", // deny | permit
    "period" : { Period }, // Timeframe for this rule
    "actor" : [{ // Who|what controlled by this rule (or group, by role)
      "role" : { CodeableConcept }, // R!  How the actor is involved
      "reference" : { Reference(Device|Group|CareTeam|Organization|Patient|
     Practitioner|RelatedPerson|PractitionerRole) } // R!  Resource for the actor (or group, by role)
    }],
    "action" : [{ CodeableConcept }], // Actions controlled by this rule
    "securityLabel" : [{ Coding }], // Security Labels that define affected resources
    "purpose" : [{ Coding }], // Context of activities covered by this rule
    "class" : [{ Coding }], // e.g. Resource Type, Profile, CDA, etc.
    "code" : [{ CodeableConcept }], // e.g. LOINC or SNOMED CT code, etc. in the content
    "dataPeriod" : { Period }, // Timeframe for data controlled by this rule
    "data" : [{ // Data controlled by this rule
      "meaning" : "<code>", // R!  instance | related | dependents | authoredby
      "reference" : { Reference(Any) } // R!  The actual data reference
    }],
    "provision" : [{ Content as for Consent.provision }] // Nested Exception Rules
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Consent;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Consent.identifier [ Identifier ], ... ; # 0..* Identifier for this record (external references)
  fhir:Consent.status [ code ]; # 1..1 draft | proposed | active | rejected | inactive | entered-in-error
  fhir:Consent.scope [ CodeableConcept ]; # 1..1 Which of the four areas this resource covers (extensible)
  fhir:Consent.category [ CodeableConcept ], ... ; # 1..* Classification of the consent statement - for
    indexing/retrieval
  fhir:Consent.patient [ Reference(Patient) ]; # 0..1 Who the consent applies to
  fhir:Consent.dateTime [ dateTime ]; # 0..1 When this Consent was created or indexed
  fhir:Consent.performer [ Reference(Organization|Patient|Practitioner|RelatedPerson|PractitionerRole) ], ... ; # 0..*
    Who is agreeing to the policy and rules
  fhir:Consent.organization [ Reference(Organization) ], ... ; # 0..* Custodian of the consent
  # Consent.source[x] : 0..1 Source from which this consent is taken. One of these 2
    fhir:Consent.sourceAttachment [ Attachment ]
    fhir:Consent.sourceReference [ Reference(Consent|DocumentReference|Contract|QuestionnaireResponse) ]
  fhir:Consent.policy [ # 0..* Policies covered by this consent
    fhir:Consent.policy.authority [ uri ]; # 0..1 Enforcement source for policy
    fhir:Consent.policy.uri [ uri ]; # 0..1 Specific policy covered by this consent
  ], ...;
  fhir:Consent.policyRule [ CodeableConcept ]; # 0..1 Regulation that this consents to
  fhir:Consent.verification [ # 0..* Consent Verified by patient or family
    fhir:Consent.verification.verified [ boolean ]; # 1..1 Has been verified
    fhir:Consent.verification.verifiedWith [ Reference(Patient|RelatedPerson) ]; # 0..1 Person who verified
    fhir:Consent.verification.verificationDate [ dateTime ]; # 0..1 When consent verified
  ], ...;
  fhir:Consent.provision [ # 0..1 Constraints to the base Consent.policyRule
    fhir:Consent.provision.type [ code ]; # 0..1 deny | permit
    fhir:Consent.provision.period [ Period ]; # 0..1 Timeframe for this rule
    fhir:Consent.provision.actor [ # 0..* Who|what controlled by this rule (or group, by role)
      fhir:Consent.provision.actor.role [ CodeableConcept ]; # 1..1 How the actor is involved
      fhir:Consent.provision.actor.reference [
    Reference(Device|Group|CareTeam|Organization|Patient|Practitioner|RelatedPerson|
  PractitionerRole) ]; # 1..1 Resource for the actor (or group, by role)
    ], ...;
    fhir:Consent.provision.action [ CodeableConcept ], ... ; # 0..* Actions controlled by this rule
    fhir:Consent.provision.securityLabel [ Coding ], ... ; # 0..* Security Labels that define affected resources
    fhir:Consent.provision.purpose [ Coding ], ... ; # 0..* Context of activities covered by this rule
    fhir:Consent.provision.class [ Coding ], ... ; # 0..* e.g. Resource Type, Profile, CDA, etc.
    fhir:Consent.provision.code [ CodeableConcept ], ... ; # 0..* e.g. LOINC or SNOMED CT code, etc. in the content
    fhir:Consent.provision.dataPeriod [ Period ]; # 0..1 Timeframe for data controlled by this rule
    fhir:Consent.provision.data [ # 0..* Data controlled by this rule
      fhir:Consent.provision.data.meaning [ code ]; # 1..1 instance | related | dependents | authoredby
      fhir:Consent.provision.data.reference [ Reference(Any) ]; # 1..1 The actual data reference
    ], ...;
    fhir:Consent.provision.provision [ See Consent.provision ], ... ; # 0..* Nested Exception Rules
  ];
]
```

Changes since Release 3

Consent
Consent.identifier | Max Cardinality changed from 1 to *
Consent.status | Change value set from http://hl7.org/fhir/ValueSet/consent-state-codes to http://hl7.org/fhir/ValueSet/consent-state-codes|4.0.1
Consent.scope | Added Mandatory Element
Consent.category | Min Cardinality changed from 0 to 1 Add Binding http://hl7.org/fhir/ValueSet/consent-category (extensible)
Consent.patient | Min Cardinality changed from 1 to 0
Consent.performer | Added Element
Consent.source[x] | Remove Type Identifier
Consent.policyRule | Type changed from uri to CodeableConcept Add Binding http://hl7.org/fhir/ValueSet/consent-policy (extensible)
Consent.verification | Added Element
Consent.verification.verified | Added Mandatory Element
Consent.verification.verifiedWith | Added Element
Consent.verification.verificationDate | Added Element
Consent.provision | Added Element
Consent.provision.type | Added Element
Consent.provision.period | Added Element
Consent.provision.actor | Added Element
Consent.provision.actor.role | Added Mandatory Element
Consent.provision.actor.reference | Added Mandatory Element
Consent.provision.action | Added Element
Consent.provision.securityLabel | Added Element
Consent.provision.purpose | Added Element
Consent.provision.class | Added Element
Consent.provision.code | Added Element
Consent.provision.dataPeriod | Added Element
Consent.provision.data | Added Element
Consent.provision.data.meaning | Added Mandatory Element
Consent.provision.data.reference | Added Mandatory Element
Consent.provision.provision | Added Element
Consent.period | deleted
Consent.consentingParty | deleted
Consent.actor | deleted
Consent.action | deleted
Consent.securityLabel | deleted
Consent.purpose | deleted
Consent.dataPeriod | deleted
Consent.data | deleted
Consent.except | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 12 tests that all execute ok. All tests pass round-trip testing and 12 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

6.2.4.1 Terminology Bindings

Path | Definition | Type | Reference
Consent.status | Indicates the state of the consent. | Required | ConsentState
Consent.scope | The four anticipated uses for the Consent Resource. | Extensible | ConsentScopeCodes
Consent.category | A classification of the type of consents found in a consent statement. | Extensible | ConsentCategoryCodes
Consent.policyRule | Regulatory policy examples. | Extensible | ConsentPolicyRuleCodes
Consent.provision.type | How a rule statement is applied, such as adding additional consent or removing consent. | Required | ConsentProvisionType
Consent.provision.actor.role | How an actor is involved in the consent considerations. | Extensible | SecurityRoleType
Consent.provision.action | Detailed codes for the consent action. | Example | ConsentActionCodes
Consent.provision.securityLabel | Security Labels from the Healthcare Privacy and Security Classification System. | Extensible | All Security Labels
Consent.provision.purpose | What purposes of use are controlled by this exception. If more than one label is specified, operations must have all the specified labels. | Extensible | v3.PurposeOfUse
Consent.provision.class | The class (type) of information a consent rule covers. | Extensible | ConsentContentClass
Consent.provision.code | If this code is found in an instance, then the exception applies. | Example | ConsentContentCodes
Consent.provision.data.meaning | How a resource reference is interpreted when testing consent restrictions. | Required | ConsentDataMeaning

6.2.4.2 Constraints

id | Level | Location | Description | Expression
ppc-1 | Rule | (base) | Either a Policy or PolicyRule | policy.exists() or policyRule.exists()

ppc-2 | Rule | (base) | IF Scope=privacy, there must be a patient | patient.exists() or scope.coding.where(system='something' and code='patient-privacy').exists().not()
ppc-3 | Rule | (base) | IF Scope=research, there must be a patient | patient.exists() or scope.coding.where(system='something' and code='research').exists().not()
ppc-4 | Rule | (base) | IF Scope=adr, there must be a patient | patient.exists() or scope.coding.where(system='something' and code='adr').exists().not()
ppc-5 | Rule | (base) | IF Scope=treatment, there must be a patient | patient.exists() or scope.coding.where(system='something' and code='treatment').exists().not()

6.2.5 Policies

The Consent resource has a reference to a single policyRule. Many organizations will work in a context where multiple different consent regulations and policies apply. In these cases, the single policy rule reference refers to a policy document that resolves and reconciles the various policies and presents a single policy for patient consent. If it is still necessary to track which of the underlying policies an exception is make in regard to, the policy may be used.

6.2.6 General Model

The following is the general model of Privacy Consent Directives.

There are context setting parameters:

• Who - The patient

• What - The data - specific resources are listed, empty list means all data covered by the consent.

• Where - The domain and authority - what is the location boundary and authority boundary of this consent

• When - The issued or captured

• When - The timeframe for which the Consent applies

• How - The actions covered. (such as purposes of use that are covered)

• Whom - The recipient are grantees by the consent.

A Privacy Consent may transition through many states including: that no consent has been sought, consent has been proposed, consent has been rejected, and consent approved.

There are set of patterns.

• No consent: All settings need a policy for when no consent has been captured. Often this allows treatment only.;

• Opt-out: No sharing allowed for the specified domain, location, actions, and purposes;

• Opt-out with exceptions: No sharing allowed, with some exceptions where it is allowed. Example: Withhold Authorization for Treatment except for Emergency Treatment;

• Opt-in: Sharing for some purpose of use is authorized Sharing allowed for Treatment, Payment, and normal Operations; and

• Opt-in with restrictions: Sharing allowed, but the patient may make exceptions (See the Canadian examples).

For each of these patterns (positive or negative pattern), there can be exceptions. These exceptions are explicitly recorded in the except element.

6.2.7 Realm specifics

6.2.7.1 US Realm Sample Use-Cases

Five categories of Privacy Consent Directives are described in the Office of the National Coordinator for Health Information (ONC) Consent Directives Document released March 31, 2010, and include the following US-specific "Core consent options" for electronic exchange:

• No consent: Health information of patients is automatically included—patients cannot opt out;

• Opt-out: Default is for health information of patients to be included automatically, but the patient can opt out completely;

• Opt-out with exceptions: Default is for health information of patients to be included, but the patient can opt out completely or allow only select data to be included;

• Opt-in: Default is that no patient health information is included; patients must actively express consent to be included, but if they do so then their information must be all in or all out; and

• Opt-in with restrictions: Default is that no patient health information is made available, but the patient may allow a subset of select data to be included.

A common exception is to explicitly exclude or explicitly include a period of time.

6.2.7.2 Canada Realm Sample Use-Cases

The following scenarios are based on existing jurisdictional policy and are realized in existing systems in Canada. The default policy is one of implied consent for the provision of care, so these scenarios all deal with

withdrawal or withholding consent for that purpose. In other jurisdictions, where an express consent model is used (Opt-In), these examples would contain the phrase "consent to" rather than "withhold" or "withdraw" consent for.

• Withhold or withdraw consent for disclosure of records related to specific domain (e.g. DI, LAB, etc.)

• Withhold or withdraw consent for disclosure of a specific record (e.g. Lab Order/Result)

• Withhold or withdraw consent for disclosure to a specific provider organization

• Withhold or withdraw consent for disclosure to a specific provider agent (an individual within an organization)

• Withhold or withdraw consent for disclosure of records that were authored by a specific organization (or service delivery location).

• Combinations of the above

6.2.7.3 Non-Treatment Use-Cases

Also shown is an example where a Patient has authorized disclosure to a specific individual for purposes directed by the patient (possibly not a treatment case).

6.2.8 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
action | token | Actions controlled by this rule | Consent.provision.action
actor | reference | Resource for the actor (or group, by role) | Consent.provision.actor.reference (Practitioner, Group, Organization, CareTeam, Device, Patient, PractitionerRole, RelatedPerson)
category | token | Classification of the consent statement - for indexing/retrieval | Consent.category
consentor | reference | Who is agreeing to the policy and rules | Consent.performer (Practitioner, Organization, Patient, PractitionerRole, RelatedPerson)
data | reference | The actual data reference | Consent.provision.data.reference (Any)
date | date | When this Consent was created or indexed | Consent.dateTime | 17 Resources
identifier | token | Identifier for this record (external references) | Consent.identifier | 30 Resources
organization | reference | Custodian of the consent | Consent.organization (Organization)
patient | reference | Who the consent applies to | Consent.patient (Patient) | 33 Resources
period | date | Timeframe for this rule | Consent.provision.period
purpose | token | Context of activities covered by this rule | Consent.provision.purpose
scope | token | Which of the four areas this resource covers (extensible) | Consent.scope
security-label | token | Security Labels that define affected resources | Consent.provision.securityLabel
source-reference | reference | Search by reference to a Consent, DocumentReference, Contract or QuestionnaireResponse | Consent.source (Consent, Contract, QuestionnaireResponse, DocumentReference)
status | token | draft | proposed | active | rejected | inactive | entered-in-error | Consent.status

6.2.9 Resource Consent - Examples

URL: https://hl7.org/fhir/R4/consent-examples.html

These are various examples that have been used to drive the creaion of the Consent resource.

6.2.9 Resource Consent - Examples

Community Based Collaborative Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

Example Name | id | Format
General Consent Example | consent-example-basic | XML | JSON | Turtle
Consent withholding access | consent-example-Out | XML | JSON | Turtle
Consent withholding access to data for Treatment exept for Emergency Treatment | consent-example-Emergency | XML | JSON | Turtle
Withhold or withdraw consent for disclosure of records related to specific domain (e.g. DI, LAB, etc.) | consent-example-notThis | XML | JSON | Turtle
Withhold or withdraw consent for disclosure of a specific timeframe | consent-example-notTime | XML | JSON | Turtle
Withhold or withdraw consent for disclosure to a specific provider organization | consent-example-notOrg | XML | JSON | Turtle
Withhold or withdraw consent for disclosure to a specific provider agent (an individual within an organization) | consent-example-notThem | XML | JSON | Turtle
Withhold or withdraw consent for disclosure of records that were authored by a specific organization (or service delivery location). | consent-example-notAuthor | XML | JSON | Turtle
Patient grants access to a specified individual for read-only access | consent-example-grantor | XML | JSON | Turtle
Example of Patients Know Best Usage | consent-example-pkb | XML | JSON | Turtle
Template for recording a Smart on FHIR Authorization | consent-example-smartonfhir | XML | JSON | Turtle
Consent Example with a signature | consent-example-signature | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

6.2.10 Resource Consent - Detailed Descriptions

URL: https://hl7.org/fhir/R4/consent-definitions.html

6.2.10 Resource Consent - Detailed Descriptions

Community Based Collaborative Care Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Patient

Detailed Descriptions for the elements in the Consent resource.

Consent
Element Id | Consent
Definition | A record of a healthcare consumer’s choices, which permits or denies identified recipient(s) or recipient role(s) to perform one or more actions within a given policy context, for specific purposes and periods of time.
Cardinality | 0..*
Type | DomainResource
Comments | Broadly, there are 3 key areas of consent for patients: Consent around sharing information (aka Privacy Consent Directive - Authorization to Collect, Use, or Disclose information), consent for specific treatment, or kinds of treatment, and general advance care directives.
Invariants | Defined on this element ppc-1 Rule Either a Policy or PolicyRule policy.exists() or policyRule.exists() ppc-2 Rule IF Scope=privacy, there must be a patient patient.exists() or scope.coding.where(system='something' and code='patient-privacy').exists().not() ppc-3 Rule IF Scope=research, there must be a patient patient.exists() or scope.coding.where(system='something' and code='research').exists().not() ppc-4 Rule IF Scope=adr, there must be a patient patient.exists() or scope.coding.where(system='something' and code='adr').exists().not() ppc-5 Rule IF Scope=treatment, there must be a patient patient.exists() or scope.coding.where(system='something' and code='treatment').exists().not()
Defined on this element
ppc-1 | Rule | Either a Policy or PolicyRule | policy.exists() or policyRule.exists()
ppc-2 | Rule | IF Scope=privacy, there must be a patient | patient.exists() or scope.coding.where(system='something' and code='patient-privacy').exists().not()
ppc-3 | Rule | IF Scope=research, there must be a patient | patient.exists() or scope.coding.where(system='something' and code='research').exists().not()
ppc-4 | Rule | IF Scope=adr, there must be a patient | patient.exists() or scope.coding.where(system='something' and code='adr').exists().not()
ppc-5 | Rule | IF Scope=treatment, there must be a patient | patient.exists() or scope.coding.where(system='something' and code='treatment').exists().not()
Consent.identifier
Element Id | Consent.identifier
Definition | Unique identifier for this copy of the Consent Statement.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
Comments | This identifier identifies this copy of the consent. Where this identifier is also used elsewhere as the identifier for a consent record (e.g. a CDA consent document) then the consent details are expected to be the same.
Consent.status
Element Id | Consent.status
Definition | Indicates the current state of this consent.
Cardinality | 1..1
Terminology Binding | ConsentState (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Requirements | The Consent Directive that is pointed to might be in various lifecycle states, e.g., a revoked Consent Directive.
Summary | true
Comments | This element is labeled as a modifier because the status contains the codes rejected and entered-in-error that mark the Consent as not currently valid.
Consent.scope
Element Id | Consent.scope
Definition | A selector of the type of consent being presented: ADR, Privacy, Treatment, Research. This list is now extensible.
Cardinality | 1..1
Terminology Binding | Consent Scope Codes (Extensible)
Type | CodeableConcept
Is Modifier | true (Reason: Allows changes to codes based on scope selection)
Summary | true
Consent.category
Element Id | Consent.category
Definition | A classification of the type of consents found in the statement. This element supports indexing and retrieval of consent statements.
Cardinality | 1..*
Terminology Binding | Consent Category Codes (Extensible)
Type | CodeableConcept
Summary | true
Consent.patient
Element Id | Consent.patient
Definition | The patient/healthcare consumer to whom this consent applies.

Cardinality | 0..1
Type | Reference(Patient)
Summary | true
Comments | Commonly, the patient the consent pertains to is the author, but for young and old people, it may be some other person.
Consent.dateTime
Element Id | Consent.dateTime
Definition | When this Consent was issued / created / indexed.
Cardinality | 0..1
Type | dateTime
Summary | true
Comments | This is not the time of the original consent, but the time that this statement was made or derived.
Consent.performer
Element Id | Consent.performer
Definition | Either the Grantor, which is the entity responsible for granting the rights listed in a Consent Directive or the Grantee, which is the entity responsible for complying with the Consent Directive, including any obligations or limitations on authorizations and enforcement of prohibitions.
Cardinality | 0..*
Type | Reference(Organization | Patient | Practitioner | RelatedPerson | PractitionerRole)
Alternate Names | consentor
Summary | true
Comments | Commonly, the patient the consent pertains to is the consentor, but particularly for young and old people, it may be some other person - e.g. a legal guardian.
Consent.organization
Element Id | Consent.organization
Definition | The organization that manages the consent, and the framework within which it is executed.
Cardinality | 0..*
Type | Reference(Organization)
Alternate Names | custodian
Summary | true
Consent.source[x]
Element Id | Consent.source[x]
Definition | The source on which this consent statement is based. The source might be a scanned original paper form, or a reference to a consent that links back to such a source, a reference to a document repository (e.g. XDS) that stores the original consent document.
Cardinality | 0..1
Type | Attachment|Reference(Consent | DocumentReference | Contract | QuestionnaireResponse)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | The source can be contained inline (Attachment), referenced directly (Consent), referenced in a consent repository (DocumentReference), or simply by an identifier (Identifier), e.g. a CDA document id.
Consent.policy
Element Id | Consent.policy
Definition | The references to the policies that are included in this consent scope. Policies may be organizational, but are often defined jurisdictionally, or in law.
Cardinality | 0..*
Consent.policy.authority
Element Id | Consent.policy.authority
Definition | Entity or Organization having regulatory jurisdiction or accountability for enforcing policies pertaining to Consent Directives.
Cardinality | 0..1
Type | uri
Invariants | Affect this element ppc-1 Rule Either a Policy or PolicyRule policy.exists() or policyRule.exists()
Affect this element
ppc-1 | Rule | Either a Policy or PolicyRule | policy.exists() or policyRule.exists()
Consent.policy.uri
Element Id | Consent.policy.uri
Definition | The references to the policies that are included in this consent scope. Policies may be organizational, but are often defined jurisdictionally, or in law.
Cardinality | 0..1
Type | uri
Comments | This element is for discoverability / documentation and does not modify or qualify the policy rules.
Invariants | Affect this element ppc-1 Rule Either a Policy or PolicyRule policy.exists() or policyRule.exists()
Affect this element
ppc-1 | Rule | Either a Policy or PolicyRule | policy.exists() or policyRule.exists()
Consent.policyRule
Element Id | Consent.policyRule
Definition | A reference to the specific base computable regulation or policy.
Cardinality | 0..1
Terminology Binding | Consent PolicyRule Codes (Extensible)
Type | CodeableConcept
Requirements | Might be a unique identifier of a policy set in XACML, or other rules engine.
Summary | true

Comments | If the policyRule is absent, computable consent would need to be constructed from the elements of the Consent resource.
Invariants | Affect this element ppc-1 Rule Either a Policy or PolicyRule policy.exists() or policyRule.exists()
Affect this element
ppc-1 | Rule | Either a Policy or PolicyRule | policy.exists() or policyRule.exists()
Consent.verification
Element Id | Consent.verification
Definition | Whether a treatment instruction (e.g. artificial respiration yes or no) was verified with the patient, his/her family or another authorized person.
Cardinality | 0..*
Summary | true
Consent.verification.verified
Element Id | Consent.verification.verified
Definition | Has the instruction been verified.
Cardinality | 1..1
Type | boolean
Summary | true
Consent.verification.verifiedWith
Element Id | Consent.verification.verifiedWith
Definition | Who verified the instruction (Patient, Relative or other Authorized Person).
Cardinality | 0..1
Type | Reference(Patient | RelatedPerson)
Consent.verification.verificationDate
Element Id | Consent.verification.verificationDate
Definition | Date verification was collected.
Cardinality | 0..1
Type | dateTime
Consent.provision
Element Id | Consent.provision
Definition | An exception to the base policy of this consent. An exception can be an addition or removal of access permissions.
Cardinality | 0..1
Summary | true
Consent.provision.type
Element Id | Consent.provision.type
Definition | Action to take - permit or deny - when the rule conditions are met. Not permitted in root rule, required in all nested rules.
Cardinality | 0..1
Terminology Binding | ConsentProvisionType (Required)
Type | code
Summary | true
Consent.provision.period
Element Id | Consent.provision.period
Definition | The timeframe in this rule is valid.
Cardinality | 0..1
Type | Period
Summary | true
Consent.provision.actor
Element Id | Consent.provision.actor
Definition | Who or what is controlled by this rule. Use group to identify a set of actors by some property they share (e.g. 'admitting officers').
Cardinality | 0..*
Meaning if Missing | There is no specific actor associated with the exception
Consent.provision.actor.role
Element Id | Consent.provision.actor.role
Definition | How the individual is involved in the resources content that is described in the exception.
Cardinality | 1..1
Terminology Binding | SecurityRoleType (Extensible)
Type | CodeableConcept
Consent.provision.actor.reference
Element Id | Consent.provision.actor.reference
Definition | The resource that identifies the actor. To identify actors by type, use group to identify a set of actors by some property they share (e.g. 'admitting officers').
Cardinality | 1..1
Type | Reference(Device | Group | CareTeam | Organization | Patient | Practitioner | RelatedPerson | PractitionerRole)
Consent.provision.action
Element Id | Consent.provision.action
Definition | Actions controlled by this Rule.
Cardinality | 0..*
Terminology Binding | Consent Action Codes (Example)
Type | CodeableConcept
Meaning if Missing | all actions
Summary | true

Comments | Note that this is the direct action (not the grounds for the action covered in the purpose element). At present, the only action in the understood and tested scope of this resource is 'read'.
Consent.provision.securityLabel
Element Id | Consent.provision.securityLabel
Definition | A security label, comprised of 0..* security label fields (Privacy tags), which define which resources are controlled by this exception.
Cardinality | 0..*
Terminology Binding | SecurityLabels (Extensible)
Type | Coding
Summary | true
Comments | If the consent specifies a security label of "R" then it applies to all resources that are labeled "R" or lower. E.g. for Confidentiality, it's a high water mark. For other kinds of security labels, subsumption logic applies. When the purpose of use tag is on the data, access request purpose of use shall not conflict.
Consent.provision.purpose
Element Id | Consent.provision.purpose
Definition | The context of the activities a user is taking - why the user is accessing the data - that are controlled by this rule.
Cardinality | 0..*
Terminology Binding | V3 Value SetPurposeOfUse (Extensible)
Type | Coding
Summary | true
Comments | When the purpose of use tag is on the data, access request purpose of use shall not conflict.
Consent.provision.class
Element Id | Consent.provision.class
Definition | The class of information covered by this rule. The type can be a FHIR resource type, a profile on a type, or a CDA document, or some other type that indicates what sort of information the consent relates to.
Cardinality | 0..*
Terminology Binding | Consent Content Class (Extensible)
Type | Coding
Summary | true
Comments | Multiple types are or'ed together. The intention of the contentType element is that the codes refer to profiles or document types defined in a standard or an implementation guide somewhere.
Consent.provision.code
Element Id | Consent.provision.code
Definition | If this code is found in an instance, then the rule applies.
Cardinality | 0..*
Terminology Binding | Consent Content Codes (Example)
Type | CodeableConcept
Summary | true
Comments | Typical use of this is a Document code with class = CDA.
Consent.provision.dataPeriod
Element Id | Consent.provision.dataPeriod
Definition | Clinical or Operational Relevant period of time that bounds the data controlled by this rule.
Cardinality | 0..1
Type | Period
Summary | true
Comments | This has a different sense to the Consent.period - that is when the consent agreement holds. This is the time period of the data that is controlled by the agreement.
Consent.provision.data
Element Id | Consent.provision.data
Definition | The resources controlled by this rule if specific resources are referenced.
Cardinality | 0..*
Meaning if Missing | all data
Summary | true
Consent.provision.data.meaning
Element Id | Consent.provision.data.meaning
Definition | How the resource reference is interpreted when testing consent restrictions.
Cardinality | 1..1
Terminology Binding | ConsentDataMeaning (Required)
Type | code
Summary | true
Consent.provision.data.reference
Element Id | Consent.provision.data.reference
Definition | A reference to a specific resource that defines which resources are covered by this consent.
Cardinality | 1..1
Type | Reference(Any)
Summary | true
Consent.provision.provision
Element Id | Consent.provision.provision
Definition | Rules which provide exceptions to the base rule or subrules.
Cardinality | 0..*
Type | See Consent.provision

6.2.11 Resource Consent - Mappings

URL: https://hl7.org/fhir/R4/consent-mappings.html

6.2.11 Resource Consent - Mappings

Community Based Collaborative Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Mappings for the consent resource (see Mappings to Other Standards for further information & status).

6.2.11.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Consent
identifier | FiveWs.identifier
status | FiveWs.status
category | FiveWs.class
patient | FiveWs.subject[x]
dateTime | FiveWs.recorded
performer | FiveWs.actor
organization | FiveWs.witness

6.2.11.2 Workflow Pattern (http://hl7.org/fhir/workflow)

Consent | Event
identifier | Event.identifier
status | Event.status
category | Event.code
patient | Event.subject
dateTime | Event.occurrence[x]
performer | Event.performer
organization | Event.performer

6.2.11.3 HL7 v2 Mapping (http://hl7.org/v2)

Consent | CON
identifier
status | HL7 Table 0498 - Consent Status
scope
category | HL7 Table 0497 - Consent Type
patient
dateTime | Field 13/ Consent Decision Date
performer | Field 24/ ConsenterID
organization
source[x] | Field 19 Informational Material Supplied Indicator
policy
authority
uri
policyRule
verification
verified
verifiedWith
verificationDate
provision
type
period
actor
role
reference
action
securityLabel
purpose
class
code
dataPeriod
data
meaning
reference
provision

6.2.11.4 RIM Mapping (http://hl7.org/v3)

Consent | FinancialConsent
identifier | .id
status | .statusCode
category | CNTRCT
patient | Role
dateTime | FinancialConsent effectiveTime
data | Role

6.2.12 Resource Consent - Extensions & Profiles

URL: https://hl7.org/fhir/R4/consent-profiles.html

6.2.12 Resource Consent - Extensions & Profiles

Community Based Collaborative Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

This table lists profiles and extensions for the Consent resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

6.2.12.1 Profiles

No Profiles defined for this resource

6.2.12.2 Extensions

consent-NotificationEndpoint | Consent | NotificationEndpoint | for Consent HL7 Extensions
consent-Transcriber | Consent | Transcriber | for Consent HL7 Extensions
consent-Witness | Consent | Witness | for Consent HL7 Extensions
consent-location | Consent.provision | location | for Consent HL7 Extensions
workflow-researchStudy | Consent | researchStudy | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

6.2.12.3 Search Extensions

No Search Extensions defined for this resource

6.2.13 Resource Consent - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/consent-version-maps.html

No validation errors - all conversions are clean

6.2.13 Resource Consent - R3/R4 Conversions

Community Based Collaborative Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

R3 : R4 Conversion maps for Consent.

Functional status for this map: 12 tests that all execute ok. All tests pass round-trip testing and 12 r3 resources are invalid (0 errors). (see documentation)

6.2.13.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Consent3to4" = "R3 to R4 Conversions for Consent"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Consent" alias ConsentR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Consent" alias Consent as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Consent(source src : ConsentR3, target tgt : Consent) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
```

```
  src.status -> tgt.status;
  src.category -> tgt.category;
  src.patient -> tgt.patient;
  src.dateTime -> tgt.dateTime;
  src.consentingParty -> tgt.performer;
  src.organization -> tgt.organization;
  src.source -> tgt.source;
  src.policy as vs -> tgt.policy as vt then policy(vs, vt);
  src.policyRule as v ->  tgt.policyRule as cc,  cc.coding as c,  c.system = 'urn:ietf:rfc:3986',  c.code = v;
  src where securityLabel.exists() or period.exists() or actor.exists() or action.exists() or purpose.exists() or
    dataPeriod.exists() or data.exists() or except.exists() -> tgt.provision as vt then base(src, vt) "securityLabel";
}

group base(source src, target tgt) {
  src.period -> tgt.period;
  src.actor as vs -> tgt.actor as vt then actor(vs, vt);
  src.action -> tgt.action;
  src.securityLabel -> tgt.securityLabel;
  src.purpose -> tgt.purpose;
  src.dataPeriod -> tgt.dataPeriod;
  src.data as vs -> tgt.data as vt then data(vs, vt);
  src.except as vs -> tgt.provision as vt then provision(vs, vt);
}

group policy(source src, target tgt) extends BackboneElement {
  src.authority -> tgt.authority;
  src.uri -> tgt.uri;
}

group provision(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.period -> tgt.period;
  src.actor as vs -> tgt.actor as vt then actor(vs, vt);
  src.action -> tgt.action;
  src.securityLabel -> tgt.securityLabel;
  src.purpose -> tgt.purpose;
  src.class -> tgt.class;
  src.code -> tgt.code;
  src.dataPeriod -> tgt.dataPeriod;
  src.data as vs -> tgt.data as vt then data(vs, vt);
}

group actor(source src, target tgt) extends BackboneElement {
  src.role -> tgt.role;
  src.reference -> tgt.reference;
}

group data(source src, target tgt) extends BackboneElement {
  src.meaning -> tgt.meaning;
  src.reference -> tgt.reference;
}
```

6.2.13.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Consent4to3" = "R4 to R3 Conversion for Consent"

uses "http://hl7.org/fhir/StructureDefinition/Consent" alias Consent as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Consent" alias ConsentR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Consent(source src : Consent, target tgt : ConsentR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.category -> tgt.category;
  src.patient -> tgt.patient;
  src.dateTime -> tgt.dateTime;
  src.performer -> tgt.consentingParty;
  src.organization -> tgt.organization;
  src.source -> tgt.source;
  src.policy as vs -> tgt.policy as vt then policy(vs, vt);
  src.policyRule as cc then {
    cc.coding as c where system = 'urn:ietf:rfc:3986' then {
      c.code -> tgt.policyRule;
    };
  };
  src.provision as vs then baseProvision(vs, tgt);
}

group baseProvision(source src, target tgt) {
  src.period -> tgt.period;
  src.actor as vs -> tgt.actor as vt then actor(vs, vt);
  src.action -> tgt.action;
  src.securityLabel -> tgt.securityLabel;
  src.purpose -> tgt.purpose;
  src.dataPeriod -> tgt.dataPeriod;
  src.data as vs -> tgt.data as vt then data(vs, vt);
  src.provision as vs -> tgt.except as vt then provision(vs, vt);
}
```

```
group policy(source src, target tgt) extends BackboneElement {
  src.authority -> tgt.authority;
  src.uri -> tgt.uri;
}

group provision(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.period -> tgt.period;
  src.actor as vs -> tgt.actor as vt then actor(vs, vt);
  src.action -> tgt.action;
  src.securityLabel -> tgt.securityLabel;
  src.purpose -> tgt.purpose;
  src.class -> tgt.class;
  src.code -> tgt.code;
  src.dataPeriod -> tgt.dataPeriod;
  src.data as vs -> tgt.data as vt then data(vs, vt);
}

group actor(source src, target tgt) extends BackboneElement {
  src.role -> tgt.role;
  src.reference -> tgt.reference;
}

group data(source src, target tgt) extends BackboneElement {
  src.meaning -> tgt.meaning;
  src.reference -> tgt.reference;
}
```

6.2.13.3 R4 Validation Errors

6.2.14 Consent HL7 Extensions

URL: https://hl7.org/fhir/R4/consent-extensions.html

Defines common extensions used with or related to the Consent resource

6.2.14 Consent HL7 Extensions

Community Based Collaborative Care Work Group | Maturity Level: N/A | Standards Status: Informative

6.2.14.1 Content

Extensions:

consent-NotificationEndpoint | NotificationEndpoint : Endpoint for sending Disclosure notifications in the form of FHIR AuditEvent records.

consent-Witness | Witness : Any witness to the consent.

consent-location | location : Restricts this exception to only apply a specific location as defined.

consent-Transcriber | Transcriber : Any person/thing who transcribed the consent into the system.

6.2.15 JSON Schema for Consent

URL: https://hl7.org/fhir/R4/consent.schema.json.html

6.2.15 JSON Schema for Consent

Community Based Collaborative Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient

This is a representation of the json schema for Consent, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Consent",
  "$ref": "#/definitions/Consent",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Consent": {
      "description": "A record of a healthcare consumer’s  choices, which permits or denies identified recipient(s) or
    recipient role(s) to perform one or more actions within a given policy context, for specific purposes and periods of
     time.",
      "properties": {
        "resourceType": {
          "description": "This is a Consent resource",
          "const": "Consent"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
```

```
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Unique identifier for this copy of the Consent Statement.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "Indicates the current state of this consent.",
      "enum": [
        "draft",
        "proposed",
        "active",
        "rejected",
        "inactive",
        "entered-in-error"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "scope": {
      "description": "A selector of the type of consent being presented: ADR, Privacy, Treatment, Research.  This
list is now extensible.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "category": {
      "description": "A classification of the type of consents found in the statement. This element supports
indexing and retrieval of consent statements.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
```

```
    },
    "patient": {
      "description": "The patient/healthcare consumer to whom this consent applies.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "dateTime": {
      "description": "When this  Consent was issued / created / indexed.",
      "$ref": "#/definitions/dateTime"
    },
    "_dateTime": {
      "description": "Extensions for dateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "performer": {
      "description": "Either the Grantor, which is the entity responsible for granting the rights listed in a
Consent Directive or the Grantee, which is the entity responsible for complying with the Consent Directive,
including any obligations or limitations on authorizations and enforcement of prohibitions.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "organization": {
      "description": "The organization that manages the consent, and the framework within which it is executed.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "sourceAttachment": {
      "description": "The source on which this consent statement is based. The source might be a scanned original
paper form, or a reference to a consent that links back to such a source, a reference to a document repository (e.g.
 XDS) that stores the original consent document.",
      "$ref": "Attachment.schema.json#/definitions/Attachment"
    },
    "sourceReference": {
      "description": "The source on which this consent statement is based. The source might be a scanned original
paper form, or a reference to a consent that links back to such a source, a reference to a document repository (e.g.
 XDS) that stores the original consent document.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "policy": {
      "description": "The references to the policies that are included in this consent scope. Policies may be
organizational, but are often defined jurisdictionally, or in law.",
      "items": {
        "$ref": "#/definitions/Consent_Policy"
      },
      "type": "array"
    },
    "policyRule": {
      "description": "A reference to the specific base computable regulation or policy.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "verification": {
      "description": "Whether a treatment instruction (e.g. artificial respiration yes or no) was verified with the
patient, his/her family or another authorized person.",
      "items": {
        "$ref": "#/definitions/Consent_Verification"
      },
      "type": "array"
    },
    "provision": {
      "description": "An exception to the base policy of this consent. An exception can be an addition or removal of
 access permissions.",
      "$ref": "#/definitions/Consent_Provision"
    }
  },
  "additionalProperties": false,
  "required": [
    "scope",
    "category",
    "resourceType"
  ]
},
"Consent_Policy": {
  "description": "A record of a healthcare consumer’s  choices, which permits or denies identified recipient(s) or
recipient role(s) to perform one or more actions within a given policy context, for specific purposes and periods of
 time.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
```

```
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "authority": {
      "description": "Entity or Organization having regulatory jurisdiction or accountability for  enforcing
policies pertaining to Consent Directives.",
      "$ref": "#/definitions/uri"
    },
    "_authority": {
      "description": "Extensions for authority",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "uri": {
      "description": "The references to the policies that are included in this consent scope. Policies may be
organizational, but are often defined jurisdictionally, or in law.",
      "$ref": "#/definitions/uri"
    },
    "_uri": {
      "description": "Extensions for uri",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"Consent_Verification": {
  "description": "A record of a healthcare consumer’s  choices, which permits or denies identified recipient(s) or
recipient role(s) to perform one or more actions within a given policy context, for specific purposes and periods of
 time.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "verified": {
      "description": "Has the instruction been verified.",
      "$ref": "#/definitions/boolean"
    },
    "_verified": {
      "description": "Extensions for verified",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "verifiedWith": {
      "description": "Who verified the instruction (Patient, Relative or other Authorized Person).",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "verificationDate": {
      "description": "Date verification was collected.",
      "$ref": "#/definitions/dateTime"
```

```
    },
    "_verificationDate": {
      "description": "Extensions for verificationDate",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"Consent_Provision": {
  "description": "A record of a healthcare consumer’s  choices, which permits or denies identified recipient(s) or
recipient role(s) to perform one or more actions within a given policy context, for specific purposes and periods of
 time.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "Action  to take - permit or deny - when the rule conditions are met.  Not permitted in root
rule, required in all nested rules.",
      "enum": [
        "deny",
        "permit"
      ]
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "period": {
      "description": "The timeframe in this rule is valid.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "actor": {
      "description": "Who or what is controlled by this rule. Use group to identify a set of actors by some property
 they share (e.g. \u0027admitting officers\u0027).",
      "items": {
        "$ref": "#/definitions/Consent_Actor"
      },
      "type": "array"
    },
    "action": {
      "description": "Actions controlled by this Rule.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "securityLabel": {
      "description": "A security label, comprised of 0..* security label fields (Privacy tags), which define which
resources are controlled by this exception.",
      "items": {
        "$ref": "Coding.schema.json#/definitions/Coding"
      },
      "type": "array"
    },
    "purpose": {
      "description": "The context of the activities a user is taking - why the user is accessing the data - that are
 controlled by this rule.",
      "items": {
        "$ref": "Coding.schema.json#/definitions/Coding"
      },
      "type": "array"
    },
```

```
    "class": {
      "description": "The class of information covered by this rule. The type can be a FHIR resource type, a profile
 on a type, or a CDA document, or some other type that indicates what sort of information the consent relates to.",
      "items": {
        "$ref": "Coding.schema.json#/definitions/Coding"
      },
      "type": "array"
    },
    "code": {
      "description": "If this code is found in an instance, then the rule applies.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "dataPeriod": {
      "description": "Clinical or Operational Relevant period of time that bounds the data controlled by this
rule.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "data": {
      "description": "The resources controlled by this rule if specific resources are referenced.",
      "items": {
        "$ref": "#/definitions/Consent_Data"
      },
      "type": "array"
    },
    "provision": {
      "description": "Rules which provide exceptions to the base rule or subrules.",
      "items": {
        "$ref": "#/definitions/Consent_Provision"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"Consent_Actor": {
  "description": "A record of a healthcare consumer’s  choices, which permits or denies identified recipient(s) or
recipient role(s) to perform one or more actions within a given policy context, for specific purposes and periods of
 time.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "role": {
      "description": "How the individual is involved in the resources content that is described in the exception.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "reference": {
      "description": "The resource that identifies the actor. To identify actors by type, use group to identify a
set of actors by some property they share (e.g. \u0027admitting officers\u0027).",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false,
  "required": [
    "reference",
    "role"
  ]
},
"Consent_Data": {
  "description": "A record of a healthcare consumer’s  choices, which permits or denies identified recipient(s) or
```

```
    recipient role(s) to perform one or more actions within a given policy context, for specific purposes and periods of
     time.",
      "properties": {
        "id": {
          "description": "Unique id for the element within a resource (for internal references). This may be any string
    value that does not contain spaces.",
          "$ref": "string.schema.json#/definitions/string"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "meaning": {
          "description": "How the resource reference is interpreted when testing consent restrictions.",
          "enum": [
            "instance",
            "related",
            "dependents",
            "authoredby"
          ]
        },
        "_meaning": {
          "description": "Extensions for meaning",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "reference": {
          "description": "A reference to a specific resource that defines which resources are covered by this consent.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        }
      },
      "additionalProperties": false,
      "required": [
        "reference"
      ]
    }
  }
}
```


6.2.16 Resource Consent - Maturity Dependency View

URL: https://hl7.org/fhir/R4/consent-dependencies.html

6.2.16 Resource Consent - Maturity Dependency View

Community Based Collaborative Care Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient

Dependency Graph for Consent FMM level 2

Name | Card. | Type | Dependency Analysis
Consent | | DomainResource
identifier | 0..* | Identifier
status | 1..1 | code
scope | 1..1 | CodeableConcept
category | 1..* | CodeableConcept
patient | 0..1 | Reference(Patient)
dateTime | 0..1 | dateTime
performer | 0..* | Reference(Organization | Patient | Practitioner | RelatedPerson | PractitionerRole)
organization | 0..* | Reference(Organization)
source[x] | 0..1
sourceAttachment | | Attachment
sourceReference | | Reference(Consent | DocumentReference | Contract | QuestionnaireResponse
policy | 0..* | BackboneElement
authority | 0..1 | uri
uri | 0..1 | uri

policyRule | 0..1 | CodeableConcept
verification | 0..* | BackboneElement
verified | 1..1 | boolean
verifiedWith | 0..1 | Reference(Patient | RelatedPerson)
verificationDate | 0..1 | dateTime
provision | 0..1 | BackboneElement
type | 0..1 | code
period | 0..1 | Period
actor | 0..* | BackboneElement
role | 1..1 | CodeableConcept
reference | 1..1 | Reference(Device | Group | CareTeam | Organization | Patient | Practitioner | RelatedPerson | PractitionerRole) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM2-Trial Use)
action | 0..* | CodeableConcept
securityLabel | 0..* | Coding
purpose | 0..* | Coding
class | 0..* | Coding
code | 0..* | CodeableConcept
dataPeriod | 0..1 | Period
data | 0..* | BackboneElement
meaning | 1..1 | code
reference | 1..1 | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)
provision | 0..* | see provision

6.2.17 Value Set Sc-valueset-consent-state-codes - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-consent-state-codes.html

6.2.17 Value Set Sc-valueset-consent-state-codes - Status Code Mapping

Community Based Collaborative Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient

This is the narrative for the resource. See also the XML, JSON or Turtle format.

6.2.18 ConsentStateCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-consent-state-codes)

Mapping from http://hl7.org/fhir/ValueSet/consent-state-codes to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "Indicates the state of the consent."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
proposed | equivalent | proposed
draft (Pending) | equivalent | draft
rejected | equivalent | declined
active | equivalent | active
inactive | equivalent | inactive

6.3 Resource Provenance - Content

URL: https://hl7.org/fhir/R4/provenance.html
Security Work Group | Maturity Level: 3 | Trial Use | Security Category: Not Classified | Compartments: Device, Patient, Practitioner, RelatedPerson

Provenance of a resource is a record that describes entities and processes involved in producing and delivering or otherwise influencing that resource. Provenance provides a critical foundation for assessing authenticity, enabling trust, and allowing reproducibility. Provenance assertions are a form of contextual metadata and can themselves become important records with their own provenance. Provenance statement indicates clinical significance in terms of confidence in authenticity, reliability, and trustworthiness, integrity, and stage in lifecycle (e.g. Document Completion - has the artifact been legally authenticated), all of which may impact security, privacy, and trust policies.

6.3.1 Scope and Usage

The Provenance resource tracks information about the activity that created, revised, deleted, or signed a version of a resource, describing the entities and agents involved. This information can be used to form assessments about its quality, reliability, trustworthiness, or to provide pointers for where to go to further investigate the origins of the resource and the information in it.

Provenance resources are a record-keeping assertion that gathers information about the context in which the information in a resource was obtained. Provenance resources are prepared by the application that initiates the create/update etc. of the resource. An AuditEvent resource contains overlapping information, but is

created as events occur, to track and audit the events. AuditEvent resources are often (though not exclusively) created by the application responding to the read/query/create/update/etc. event.

6.3.2 Boundaries and Relationships

Many other FHIR resources contain some elements that represent information about how the resource was obtained, and therefore they overlap with the functionality of the Provenance resource. These properties in other resources should always be used in preference to the Provenance resource, and the Provenance resource should be used where additional information is required, or explicit record or provenance is desired.

The relationship between a resource and its provenance is established by a reference from the provenance resource to its target. In this way, provenance may be provided about any resource or version, including past versions. There may be multiple provenance records for a given resource or version of a resource.

6.3.3 Background and Context

The Provenance resource is based on the W3C Provenance specification , and mappings are provided. The Provenance resource is tailored to fit the FHIR use-cases for provenance more directly. In terms of W3C Provenance the FHIR Provenance resource covers "Generation" of "Entity" with respect to FHIR defined resources for creation or updating; whereas AuditEvent covers "Usage" of "Entity" and all other "Activity" as defined in W3C Provenance.

The W3C Provenance Specification has the following fundamental model:

Where:

• Entity - An entity is a physical, digital, conceptual or other kind of thing with some fixed aspects; entities may be real or imaginary.

• Agent - An agent is something that bears some form of responsibility for an activity taking place, for the existence of an entity, or for another agent's activity.

• Activity - An activity is something that occurs over a time period and acts upon or with entities. It may include consuming, processing, transforming, modifying, relocating, using, or generating entities.

The Provenance resource corresponds to a single activity that identifies a set of resources (target) generated by the activity. The activity also references other entities (entity) that were used and the agents (agent) that were associated with the activity. To record multiple activities that resulted in one (target), record each (activity) in independent Provenance records all pointing at that (target).

The Provenance resource depends upon having References to all the resources, entities, and agents involved in the activity. These References need not be resolvable. The references must provide a unique and unambiguous identification. If a resource, entity, or agent can have different versions that must be identified, then the Reference must have versioning information included.

Versioning and unique identification are not mandated for all systems that provide Resources, entities, and agents. But, inclusion of Provenance requirements may introduce requirements for versioning and unique identification on those systems

The Provenance resource is based on leveraging the W3C Provenance specification to represent HL7 support of provenance throughout its standards and explicitly modeled as functional capabilities in ISO/HL7 10781 EHR System Functional Model Release 2 and ISO 21089 Trusted End-to-End Information Flows. Mappings are provided. The Provenance resource is tailored to fit the FHIR use-cases for provenance more directly. In terms of W3C Provenance the FHIR Provenance resources covers "Generation" of "Entity" with respect to FHIR defined resources for creation or updating; whereas AuditEvent covers "Usage" of "Entity" and all other "Activity" as defined in W3C Provenance.

This resource is referenced by Contract, DeviceRequest, MedicationAdministration, MedicationDispense, MedicationRequest, ServiceRequest and Task

6.3.4 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
Provenance | TU | | DomainResource | Who, What, When for a set of resources Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
target | Σ | 1..* | Reference(Any) | Target Reference(s) (usually version specific)
occurred[x] | | 0..1 | | When the activity occurred

occurredPeriod | | | Period
occurredDateTime | | | dateTime
recorded | Σ | 1..1 | instant | When the activity was recorded / updated
policy | | 0..* | uri | Policy or plan the activity was defined by
location | | 0..1 | Reference(Location) | Where the activity occurred, if relevant
reason | | 0..* | CodeableConcept | Reason the activity is occurring V3 Value SetPurposeOfUse (Extensible)
activity | | 0..1 | CodeableConcept | Activity that occurred Provenance activity type (Extensible)
agent | | 1..* | BackboneElement | Actor involved
type | Σ | 0..1 | CodeableConcept | How the agent participated Provenance participant type (Extensible)
role | | 0..* | CodeableConcept | What the agents role was SecurityRoleType (Example)
who | Σ | 1..1 | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization) | Who participated
onBehalfOf | | 0..1 | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization) | Who the agent is representing
entity | | 0..* | BackboneElement | An entity used in this activity
role | Σ | 1..1 | code | derivation | revision | quotation | source | removal ProvenanceEntityRole (Required)
what | Σ | 1..1 | Reference(Any) | Identity of entity
agent | | 0..* | see agent | Entity is attributed to this agent
signature | | 0..* | Signature | Signature on target
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Provenance xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <target><!-- 1..* Reference(Any) Target Reference(s) (usually version specific) --></target>
 <occurred[x]><!-- 0..1 Period|dateTime When the activity occurred --></occurred[x]>
 <recorded value="[instant]"/><!-- 1..1 When the activity was recorded / updated -->
 <policy value="[uri]"/><!-- 0..* Policy or plan the activity was defined by -->
 <location><!-- 0..1 Reference(Location) Where the activity occurred, if relevant --></location>
 <reason><!-- 0..* CodeableConcept Reason the activity is occurring --></reason>
 <activity><!-- 0..1 CodeableConcept Activity that occurred --></activity>
 <agent>  <!-- 1..* Actor involved -->
  <type><!-- 0..1 CodeableConcept How the agent participated --></type>
  <role><!-- 0..* CodeableConcept What the agents role was --></role>
  <who><!-- 1..1 Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|
    Device|Organization) Who participated --></who>
  <onBehalfOf><!-- 0..1 Reference(Practitioner|PractitionerRole|RelatedPerson|
    Patient|Device|Organization) Who the agent is representing --></onBehalfOf>
 </agent>
 <entity>  <!-- 0..* An entity used in this activity -->
  <role value="[code]"/><!-- 1..1 derivation | revision | quotation | source | removal -->
  <what><!-- 1..1 Reference(Any) Identity of entity --></what>
  <agent><!-- 0..* Content as for Provenance.agent Entity is attributed to this agent --></agent>
 </entity>
 <signature><!-- 0..* Signature Signature on target --></signature>
</Provenance>
```

JSON Template

```
{
  "resourceType" : "Provenance",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "target" : [{ Reference(Any) }], // R!  Target Reference(s) (usually version specific)
  // occurred[x]: When the activity occurred. One of these 2:
  "occurredPeriod" : { Period },
  "occurredDateTime" : "<dateTime>",
  "recorded" : "<instant>", // R!  When the activity was recorded / updated
  "policy" : ["<uri>"], // Policy or plan the activity was defined by
  "location" : { Reference(Location) }, // Where the activity occurred, if relevant
  "reason" : [{ CodeableConcept }], // Reason the activity is occurring
  "activity" : { CodeableConcept }, // Activity that occurred
  "agent" : [{ // R!  Actor involved
    "type" : { CodeableConcept }, // How the agent participated
    "role" : [{ CodeableConcept }], // What the agents role was
    "who" : { Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|
    Device|Organization) }, // R!  Who participated
    "onBehalfOf" : { Reference(Practitioner|PractitionerRole|RelatedPerson|
    Patient|Device|Organization) } // Who the agent is representing
  }],
  "entity" : [{ // An entity used in this activity
    "role" : "<code>", // R!  derivation | revision | quotation | source | removal
    "what" : { Reference(Any) }, // R!  Identity of entity
    "agent" : [{ Content as for Provenance.agent }] // Entity is attributed to this agent
  }],
  "signature" : [{ Signature }] // Signature on target
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Provenance;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root
```

```
  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Provenance.target [ Reference(Any) ], ... ; # 1..* Target Reference(s) (usually version specific)
  # Provenance.occurred[x] : 0..1 When the activity occurred. One of these 2
    fhir:Provenance.occurredPeriod [ Period ]
    fhir:Provenance.occurredDateTime [ dateTime ]
  fhir:Provenance.recorded [ instant ]; # 1..1 When the activity was recorded / updated
  fhir:Provenance.policy [ uri ], ... ; # 0..* Policy or plan the activity was defined by
  fhir:Provenance.location [ Reference(Location) ]; # 0..1 Where the activity occurred, if relevant
  fhir:Provenance.reason [ CodeableConcept ], ... ; # 0..* Reason the activity is occurring
  fhir:Provenance.activity [ CodeableConcept ]; # 0..1 Activity that occurred
  fhir:Provenance.agent [ # 1..* Actor involved
    fhir:Provenance.agent.type [ CodeableConcept ]; # 0..1 How the agent participated
    fhir:Provenance.agent.role [ CodeableConcept ], ... ; # 0..* What the agents role was
    fhir:Provenance.agent.who [ Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|Device|Organization) ]; #
    1..1 Who participated
    fhir:Provenance.agent.onBehalfOf [
    Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|Device|Organization) ]; # 0..1 Who the agent is
    representing
  ], ...;
  fhir:Provenance.entity [ # 0..* An entity used in this activity
    fhir:Provenance.entity.role [ code ]; # 1..1 derivation | revision | quotation | source | removal
    fhir:Provenance.entity.what [ Reference(Any) ]; # 1..1 Identity of entity
    fhir:Provenance.entity.agent [ See Provenance.agent ], ... ; # 0..* Entity is attributed to this agent
  ], ...;
  fhir:Provenance.signature [ Signature ], ... ; # 0..* Signature on target
]
```

Changes since R3

Provenance
Provenance.occurred[x] | Renamed from period to occurred[x] Add Type dateTime
Provenance.reason | Type changed from Coding to CodeableConcept
Provenance.activity | Type changed from Coding to CodeableConcept
Provenance.agent.type | Added Element
Provenance.agent.role | Remove Binding http://hl7.org/fhir/ValueSet/security-role-type (extensible)
Provenance.agent.who | Renamed from who[x] to who Remove Type uri
Provenance.agent.onBehalfOf | Renamed from onBehalfOf[x] to onBehalfOf Remove Type uri
Provenance.entity.role | Change value set from http://hl7.org/fhir/ValueSet/provenance-entity-role to http://hl7.org/fhir/ValueSet/provenance-entity-role|4.0.1
Provenance.entity.what | Renamed from what[x] to what Remove Types uri, Identifier
Provenance.agent.relatedAgentType | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 5 tests that all execute ok. All tests pass round-trip testing and 2 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
Provenance | TU | | DomainResource | Who, What, When for a set of resources Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
target | Σ | 1..* | Reference(Any) | Target Reference(s) (usually version specific)
occurred[x] | | 0..1 | | When the activity occurred
occurredPeriod | | | Period
occurredDateTime | | | dateTime
recorded | Σ | 1..1 | instant | When the activity was recorded / updated
policy | | 0..* | uri | Policy or plan the activity was defined by
location | | 0..1 | Reference(Location) | Where the activity occurred, if relevant
reason | | 0..* | CodeableConcept | Reason the activity is occurring V3 Value SetPurposeOfUse (Extensible)
activity | | 0..1 | CodeableConcept | Activity that occurred Provenance activity type (Extensible)
agent | | 1..* | BackboneElement | Actor involved
type | Σ | 0..1 | CodeableConcept | How the agent participated Provenance participant type (Extensible)
role | | 0..* | CodeableConcept | What the agents role was SecurityRoleType (Example)
who | Σ | 1..1 | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization) | Who participated
onBehalfOf | | 0..1 | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization) | Who the agent is representing
entity | | 0..* | BackboneElement | An entity used in this activity
role | Σ | 1..1 | code | derivation | revision | quotation | source | removal ProvenanceEntityRole (Required)
what | Σ | 1..1 | Reference(Any) | Identity of entity
agent | | 0..* | see agent | Entity is attributed to this agent
signature | | 0..* | Signature | Signature on target
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Provenance xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <target><!-- 1..* Reference(Any) Target Reference(s) (usually version specific) --></target>
```

```
 <occurred[x]><!-- 0..1 Period|dateTime When the activity occurred --></occurred[x]>
 <recorded value="[instant]"/><!-- 1..1 When the activity was recorded / updated -->
 <policy value="[uri]"/><!-- 0..* Policy or plan the activity was defined by -->
 <location><!-- 0..1 Reference(Location) Where the activity occurred, if relevant --></location>
 <reason><!-- 0..* CodeableConcept Reason the activity is occurring --></reason>
 <activity><!-- 0..1 CodeableConcept Activity that occurred --></activity>
 <agent>  <!-- 1..* Actor involved -->
  <type><!-- 0..1 CodeableConcept How the agent participated --></type>
  <role><!-- 0..* CodeableConcept What the agents role was --></role>
  <who><!-- 1..1 Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|
    Device|Organization) Who participated --></who>
  <onBehalfOf><!-- 0..1 Reference(Practitioner|PractitionerRole|RelatedPerson|
    Patient|Device|Organization) Who the agent is representing --></onBehalfOf>
 </agent>
 <entity>  <!-- 0..* An entity used in this activity -->
  <role value="[code]"/><!-- 1..1 derivation | revision | quotation | source | removal -->
  <what><!-- 1..1 Reference(Any) Identity of entity --></what>
  <agent><!-- 0..* Content as for Provenance.agent Entity is attributed to this agent --></agent>
 </entity>
 <signature><!-- 0..* Signature Signature on target --></signature>
</Provenance>
```

JSON Template

```
{
  "resourceType" : "Provenance",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "target" : [{ Reference(Any) }], // R!  Target Reference(s) (usually version specific)
  // occurred[x]: When the activity occurred. One of these 2:
  "occurredPeriod" : { Period },
  "occurredDateTime" : "<dateTime>",
  "recorded" : "<instant>", // R!  When the activity was recorded / updated
  "policy" : ["<uri>"], // Policy or plan the activity was defined by
  "location" : { Reference(Location) }, // Where the activity occurred, if relevant
  "reason" : [{ CodeableConcept }], // Reason the activity is occurring
  "activity" : { CodeableConcept }, // Activity that occurred
  "agent" : [{ // R!  Actor involved
    "type" : { CodeableConcept }, // How the agent participated
    "role" : [{ CodeableConcept }], // What the agents role was
    "who" : { Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|
    Device|Organization) }, // R!  Who participated
    "onBehalfOf" : { Reference(Practitioner|PractitionerRole|RelatedPerson|
    Patient|Device|Organization) } // Who the agent is representing
  }],
  "entity" : [{ // An entity used in this activity
    "role" : "<code>", // R!  derivation | revision | quotation | source | removal
    "what" : { Reference(Any) }, // R!  Identity of entity
    "agent" : [{ Content as for Provenance.agent }] // Entity is attributed to this agent
  }],
  "signature" : [{ Signature }] // Signature on target
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Provenance;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Provenance.target [ Reference(Any) ], ... ; # 1..* Target Reference(s) (usually version specific)
  # Provenance.occurred[x] : 0..1 When the activity occurred. One of these 2
    fhir:Provenance.occurredPeriod [ Period ]
    fhir:Provenance.occurredDateTime [ dateTime ]
  fhir:Provenance.recorded [ instant ]; # 1..1 When the activity was recorded / updated
  fhir:Provenance.policy [ uri ], ... ; # 0..* Policy or plan the activity was defined by
  fhir:Provenance.location [ Reference(Location) ]; # 0..1 Where the activity occurred, if relevant
  fhir:Provenance.reason [ CodeableConcept ], ... ; # 0..* Reason the activity is occurring
  fhir:Provenance.activity [ CodeableConcept ]; # 0..1 Activity that occurred
  fhir:Provenance.agent [ # 1..* Actor involved
    fhir:Provenance.agent.type [ CodeableConcept ]; # 0..1 How the agent participated
    fhir:Provenance.agent.role [ CodeableConcept ], ... ; # 0..* What the agents role was
    fhir:Provenance.agent.who [ Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|Device|Organization) ]; #
    1..1 Who participated
    fhir:Provenance.agent.onBehalfOf [
    Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|Device|Organization) ]; # 0..1 Who the agent is
    representing
  ], ...;
  fhir:Provenance.entity [ # 0..* An entity used in this activity
    fhir:Provenance.entity.role [ code ]; # 1..1 derivation | revision | quotation | source | removal
    fhir:Provenance.entity.what [ Reference(Any) ]; # 1..1 Identity of entity
    fhir:Provenance.entity.agent [ See Provenance.agent ], ... ; # 0..* Entity is attributed to this agent
  ], ...;
  fhir:Provenance.signature [ Signature ], ... ; # 0..* Signature on target
]
```

Changes since Release 3

Provenance
Provenance.occurred[x] | Renamed from period to occurred[x] Add Type dateTime
Provenance.reason | Type changed from Coding to CodeableConcept
Provenance.activity | Type changed from Coding to CodeableConcept
Provenance.agent.type | Added Element
Provenance.agent.role | Remove Binding http://hl7.org/fhir/ValueSet/security-role-type (extensible)
Provenance.agent.who | Renamed from who[x] to who Remove Type uri
Provenance.agent.onBehalfOf | Renamed from onBehalfOf[x] to onBehalfOf Remove Type uri
Provenance.entity.role | Change value set from http://hl7.org/fhir/ValueSet/provenance-entity-role to http://hl7.org/fhir/ValueSet/provenance-entity-role|4.0.1
Provenance.entity.what | Renamed from what[x] to what Remove Types uri, Identifier
Provenance.agent.relatedAgentType | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 5 tests that all execute ok. All tests pass round-trip testing and 2 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

6.3.4.1 Terminology Bindings

Path | Definition | Type | Reference
Provenance.reason | The reason the activity took place. | Extensible | v3.PurposeOfUse
Provenance.activity | The activity that took place. | Extensible | ProvenanceActivityType
Provenance.agent.type | The type of participation that a provenance agent played with respect to the activity. | Extensible | ProvenanceParticipantType
Provenance.agent.role | The role that a provenance agent played with respect to the activity. | Example | SecurityRoleType
Provenance.entity.role | How an entity was used in an activity. | Required | ProvenanceEntityRole

6.3.4.2 Using the Provenance Resource

The Provenance resource identifies information about another resource (the reference element). The Provenance resource may be used in several different ways:

• As part of a document bundle where it identifies the provenance of part or all of the document

• On a RESTful system where it keeps track of provenance information relating to resources

When used in a document bundle, the references are often not explicitly versioned, but they always implicitly pertain to the version of the resource found in the document. On a RESTful system, the target resource reference should be version specific, but this requires special care: For new resources that need to have a corresponding Provenance resource, the version-specific reference is often not knowable until after the target resource has been updated. This can create an integrity problem for the system - what if the Provenance resource cannot be created after the target resource has been updated? To avoid any such integrity problems, the target resource and the Provenance resources should be submitted as a pair using a transaction.

6.3.4.2.1 HTTP Header

The custom header X-Provenance to provide a provenance resource when performing PUT or POST operations using the RESTful interface:

```
POST [base]/Observation
Content-Type: application/fhir+?
X-Provenance: { "resourceType": "Provenance", "location": { "reference": "Location/1" }," agent" ... }

[body]
```

The intent is that the server picks up the provenance, fills out the target, and then stores the provenance information as it normally would. Notes:

• Irrespective of the Content-Type or Accept headers, the Provenance resource is in the JSON format

• The provenance SHALL not have a specified Provenance.target. The server will fill the target in as it processes the contents of the POST/PUT and determines the ids of the resource(s) to which the interaction applies

• Servers MAY ignore the header, but SHOULD process it if they support provenance

• Some server frameworks impose a length limit on individual HTTP headers, or on the the headers as a whole. Servers MAY reject a request if the X-Provenance header is too long

• The use of the header is not documented for other HTTP methods (GET etc)

6.3.4.3 Signatures

The Provenance resource includes a signature element (digital signature) which can be used for standards based integrity verification and non-repudiation purposes. The Signature datatype provides details on use of the signature element. The Signature.type coded value of "Source" should be used when the signature is for

simply proving that the resource content is the same as it was when the resource was updated or created.

6.3.4.4 Provenance of Removal

A Provenance record can be recorded to indicate who deleted a Resource. If versioning is supported, the version that was deleted is referenced in Provenance.target; if versioning is not supported then Provenance.target contains the non-version reference. Provenance.entity is not used unless there is a business requirement to do so.

6.3.4.5 Use of Provenance to record Import and Transform

Provenance can be used to record activities of an automaton that transforms input. Such as middleware that extracts information from a HL7 v2 message and creates FHIR resources, or middleware that extracts information from an HL7 CDA document and creates FHIR resources, etc. The Provenance in these cases is recording the activity of the middleware.

The middleware in this case would, in addition to creating the target resources, create a Provenance resource that indicates all the target resources (using Provenance.target). The middleware is identified as one of the Provenance.agent elements, with the Provenance.agent.role of assembler.

The middleware may record the source as another Provenance.agent element.

The original content is optionally saved. This might be as a DocumentReference, or Binary. The Provenance.entity would then point at this original content.

The original source might include some form of 'provenance' to cover the history of the original content prior to the import transformation. This original source 'provenance' should be converted into FHIR Provenance records as appropriate.

6.3.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
agent | reference | Who participated | Provenance.agent.who (Practitioner, Organization, Device, Patient, PractitionerRole, RelatedPerson)
agent-role | token | What the agents role was | Provenance.agent.role
agent-type | token | How the agent participated | Provenance.agent.type
entity | reference | Identity of entity | Provenance.entity.what (Any)
location | reference | Where the activity occurred, if relevant | Provenance.location (Location)
patient | reference | Target Reference(s) (usually version specific) | Provenance.target.where(resolve() is Patient) (Patient)
recorded | date | When the activity was recorded / updated | Provenance.recorded
signature-type | token | Indication of the reason the entity signed the object(s) | Provenance.signature.type
target | reference | Target Reference(s) (usually version specific) | Provenance.target (Any)
when | date | When the activity occurred | (Provenance.occurred as dateTime)

6.3.6 Resource Provenance - Examples

URL: https://hl7.org/fhir/R4/provenance-examples.html

6.3.6 Resource Provenance - Examples

Security Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Patient, Practitioner, RelatedPerson

Example Name | id | Format
General Provenance Example | example | XML | JSON | Turtle
Provenance holding a signature | signature | XML | JSON | Turtle
Provenance with signature for a Consent example | consent-signature | XML | JSON | Turtle
Provenance example for Biocompute objects | example-biocompute-object | XML | JSON | Turtle
Provenance example for cwl workflow project | example-cwl | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

6.3.7 Resource Provenance - Detailed Descriptions

URL: https://hl7.org/fhir/R4/provenance-definitions.html

6.3.7 Resource Provenance - Detailed Descriptions

Security Work Group | Maturity Level: 3 | Trial Use | Security Category: Not Classified | Compartments: Device, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the Provenance resource.

Provenance
Element Id | Provenance

Definition | Provenance of a resource is a record that describes entities and processes involved in producing and delivering or otherwise influencing that resource. Provenance provides a critical foundation for assessing authenticity, enabling trust, and allowing reproducibility. Provenance assertions are a form of contextual metadata and can themselves become important records with their own provenance. Provenance statement indicates clinical significance in terms of confidence in authenticity, reliability, and trustworthiness, integrity, and stage in lifecycle (e.g. Document Completion - has the artifact been legally authenticated), all of which may impact security, privacy, and trust policies.
Cardinality | 0..*
Type | DomainResource
Alternate Names | History; Event; Activity
Comments | Some parties may be duplicated between the target resource and its provenance. For instance, the prescriber is usually (but not always) the author of the prescription resource. This resource is defined with close consideration for W3C Provenance.
Provenance.target
Element Id | Provenance.target
Definition | The Reference(s) that were generated or updated by the activity described in this resource. A provenance can point to more than one target if multiple resources were created/updated by the same activity.
Cardinality | 1..*
Type | Reference(Any)
Summary | true
Comments | Target references are usually version specific, but might not be, if a version has not been assigned or if the provenance information is part of the set of resources being maintained (i.e. a document). When using the RESTful API, the identity of the resource might not be known (especially not the version specific one); the client may either submit the resource first, and then the provenance, or it may submit both using a single transaction. See the notes on transaction for further discussion.
Provenance.occurred[x]
Element Id | Provenance.occurred[x]
Definition | The period during which the activity occurred.
Cardinality | 0..1
Type | Period|dateTime
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | The period can be a little arbitrary; where possible, the time should correspond to human assessment of the activity time.
Provenance.recorded
Element Id | Provenance.recorded
Definition | The instant of time at which the activity was recorded.
Cardinality | 1..1
Type | instant
Summary | true
Comments | This can be a little different from the time stamp on the resource if there is a delay between recording the event and updating the provenance and target resource.
Provenance.policy
Element Id | Provenance.policy
Definition | Policy or plan the activity was defined by. Typically, a single activity may have multiple applicable policy documents, such as patient consent, guarantor funding, etc.
Cardinality | 0..*
Type | uri
Comments | For example: Where an OAuth token authorizes, the unique identifier from the OAuth token is placed into the policy element Where a policy engine (e.g. XACML) holds policy logic, the unique policy identifier is placed into the policy element.
Provenance.location
Element Id | Provenance.location
Definition | Where the activity occurred, if relevant.
Cardinality | 0..1
Type | Reference(Location)
Provenance.reason
Element Id | Provenance.reason
Definition | The reason that the activity was taking place.
Cardinality | 0..*
Terminology Binding | V3 Value SetPurposeOfUse (Extensible)
Type | CodeableConcept
Provenance.activity
Element Id | Provenance.activity
Definition | An activity is something that occurs over a period of time and acts upon or with entities; it may include consuming, processing, transforming, modifying, relocating, using, or generating entities.
Cardinality | 0..1
Terminology Binding | Provenance activity type (Extensible)
Type | CodeableConcept
Provenance.agent
Element Id | Provenance.agent
Definition | An actor taking a role in an activity for which it can be assigned some degree of responsibility for the activity taking place.
Cardinality | 1..*
Requirements | An agent can be a person, an organization, software, device, or other entities that may be ascribed responsibility.
Comments | Several agents may be associated (i.e. has some responsibility for an activity) with an activity and vice-versa.
Provenance.agent.type

Element Id | Provenance.agent.type
Definition | The participation the agent had with respect to the activity.
Cardinality | 0..1
Terminology Binding | Provenance participant type (Extensible)
Type | CodeableConcept
Summary | true
Comments | For example: author, performer, enterer, attester, etc.
Provenance.agent.role
Element Id | Provenance.agent.role
Definition | The function of the agent with respect to the activity. The security role enabling the agent with respect to the activity.
Cardinality | 0..*
Terminology Binding | SecurityRoleType (Example)
Type | CodeableConcept
Comments | For example: doctor, nurse, clerk, etc.
Provenance.agent.who
Element Id | Provenance.agent.who
Definition | The individual, device or organization that participated in the event.
Cardinality | 1..1
Type | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization)
Summary | true
Comments | whoIdentity should be used when the agent is not a Resource type.
Provenance.agent.onBehalfOf
Element Id | Provenance.agent.onBehalfOf
Definition | The individual, device, or organization for whom the change was made.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization)
Comments | onBehalfOfIdentity should be used when the agent is not a Resource type.
Provenance.entity
Element Id | Provenance.entity
Definition | An entity used in this activity.
Cardinality | 0..*
Provenance.entity.role
Element Id | Provenance.entity.role
Definition | How the entity was used during the activity.
Cardinality | 1..1
Terminology Binding | ProvenanceEntityRole (Required)
Type | code
Summary | true
Provenance.entity.what
Element Id | Provenance.entity.what
Definition | Identity of the Entity used. May be a logical or physical uri and maybe absolute or relative.
Cardinality | 1..1
Type | Reference(Any)
Summary | true
Comments | whatIdentity should be used for entities that are not a Resource type.
Provenance.entity.agent
Element Id | Provenance.entity.agent
Definition | The entity is attributed to an agent to express the agent's responsibility for that entity, possibly along with other agents. This description can be understood as shorthand for saying that the agent was responsible for the activity which generated the entity.
Cardinality | 0..*
Type | See Provenance.agent
Comments | A usecase where one Provenance.entity.agent is used where the Entity that was used in the creation/updating of the Target, is not in the context of the same custodianship as the Target, and thus the meaning of Provenance.entity.agent is to say that the entity referenced is managed elsewhere and that this Agent provided access to it. This would be similar to where the Entity being referenced is managed outside FHIR, such as through HL7 v2, v3, or XDS. This might be where the Entity being referenced is managed in another FHIR resource server. Thus it explains the Provenance of that Entity's use in the context of this Provenance activity.
Provenance.signature
Element Id | Provenance.signature
Definition | A digital signature on the target Reference(s). The signer should match a Provenance.agent. The purpose of the signature is indicated.
Cardinality | 0..*
Type | Signature

6.3.8 Resource Provenance - Mappings

URL: https://hl7.org/fhir/R4/provenance-mappings.html

6.3.8 Resource Provenance - Mappings

Security Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

W3C PROV

FHIR AuditEvent Mapping

Mappings for the provenance resource (see Mappings to Other Standards for further information & status).

6.3.8.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Provenance
target | FiveWs.what[x]
occurred[x] | FiveWs.done[x]
recorded | FiveWs.recorded
location | FiveWs.where[x]
reason | FiveWs.why[x]
activity | FiveWs.why[x]
agent | FiveWs.who
who | FiveWs.actor

6.3.8.2 Workflow Pattern (http://hl7.org/fhir/workflow)

Provenance | Event
occurred[x] | Event.occurred[x]
location | Event.location
reason | Event.reasonCode
activity | Event.code
agent | Event.performer
type | Event.performer.function
who | Event.performer.actor

6.3.8.3 RIM Mapping (http://hl7.org/v3)

Provenance | ControlAct[isNormalAct() and subsumes(CACT, classCode) and moodCode=EVN]
target | ./outboundRelationship[isNormalActRelationship() and typeCode=SUBJ]/target OR ./participation[isNormalParticipation() and typeCode=SBJ]/role OR ./participation[isNormalParticipation() and typeCode=SBJ]/role[isNormalRole()]/player
occurred[x] | ./effectiveTime[type=IVL_TS]
recorded | unique(./participation[isNormalParticipation() and typeCode=AUT]/time[type=TS])
policy | ./inboundRelationship[isNormalActRelationship() and typeCode="SUBJ"]/source[isNormalAct and subsumes(POLICY, classCode) and moodCode=EVN]/text[typeCode='ED'/tel
location | unique(./participation[isNormalParticipation() and typeCode=LOC]/role[isNormalRole() and subsumes(SDLOC, classCode)]/player[isNormalEntity and classCode="LOC" and determinerCode="INST"]
reason | unique(./reasonCode)
activity | Act.code
agent | ./participation[isNormalParticipation()] OR ./outboundRelationship[isNormalActRelationship() and typeCode='DRIV']
type | .role
role | .typecode
who | .id
onBehalfOf | Person, Practitioner, Organization, Device :* .role [classCode = RoleClassMutualRelationship; role.code and *.scopes[Role](classCode=IDENT) and *.plays [Role.Code]
entity | ./subjectOf
role | ./typeCode
what | ./text/reference
agent | ./author/role
signature | ./signatureText

6.3.8.4 W3C PROV (http://www.w3.org/ns/prov)

The provenance resource is based on known practices in the HL7 implementation space, particularly those found in the v2 EVN segment, the v3 ControlAct Wrapper, the CDA header, and IHE ATNA. The conceptual model underlying the design is the W3C provenance Specification . Though the content and format of the resource is designed to meet specific requirements for FHIR, all the parts of the resource are formally mapped to the PROV-O specification, and FHIR resources can be transformed to their W3C PROV equivalent.

Provenance | Activity
target | Entity Created/Updated
occurred[x] | Activity.startTime & Activity.endTime
recorded | Activity.when
policy
location | Activity.location
reason | Activity.Activity
activity | Activity.Activity
agent | Agent

type | Agent.Attribution
role
who
onBehalfOf
entity | Entity
role | Entity.role
what | Entity.Identity
agent
signature

6.3.8.5 FHIR AuditEvent Mapping (http://hl7.org/fhir/auditevent)

Provenance
target | AuditEvent.entity.reference
occurred[x]
recorded | AuditEvent.recorded
policy | AuditEvent.agent.policy
location | AuditEvent.agent.location
reason | AuditEvent.purposeOfEvent
activity
agent | AuditEvent.agent
type | AuditEvent.agent.type
role | AuditEvent.agent.role
who
onBehalfOf
entity | AuditEvent.entity
role | AuditEvent.entity.lifecycle
what | AuditEvent.entity.reference
agent
signature

6.3.9 Resource Provenance - Extensions & Profiles

URL: https://hl7.org/fhir/R4/provenance-profiles.html

6.3.9 Resource Provenance - Extensions & Profiles

Security Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the Provenance resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

6.3.9.1 Profiles

Profile | Description | Context
ehrsrle-provenance | Defines the elements to be supported within the Provenance resource in order to conform with the Electronic Health Record System Functional Model Record Lifecycle Event standard | EHRS FM Record Lifecycle Event - Provenance
provenance-relevant-history | Guidance on using Provenance for related history elements | Provenance Relevant History

6.3.9.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

event-eventHistory | eventHistory | for Event Pattern HL7 Extensions
request-relevantHistory | relevantHistory | for Request Pattern HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

6.3.9.3 Search Extensions

No Search Extensions defined for this resource

6.3.10 Resource Provenance - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/provenance-version-maps.html

No validation errors - all conversions are clean

6.3.10 Resource Provenance - R3/R4 Conversions

Security Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for Provenance.

Functional status for this map: 5 tests that all execute ok. All tests pass round-trip testing and 2 r3 resources are invalid (0 errors). (see documentation)

6.3.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Provenance3to4" = "R3 to R4 Conversions for Provenance"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Provenance" alias ProvenanceR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Provenance" alias Provenance as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Provenance(source src : ProvenanceR3, target tgt : Provenance) extends DomainResource <<type+>> {
  src.target -> tgt.target;
  src.period as vs0 -> tgt.occurred = create('Period') as vt0 then Period(vs0, vt0);
  src.recorded -> tgt.recorded;
  src.policy -> tgt.policy;
  src.location -> tgt.location;
  src.reason -> tgt.reason;
  src.activity -> tgt.activity;
  src.agent as vs0 -> tgt.agent as vt0 then agent(vs0, vt0);
  src.entity as vs0 -> tgt.entity as vt0 then entity(vs0, vt0);
  src.signature -> tgt.signature;
}

group agent(source src : ProvenanceR3, target tgt : Provenance) extends BackboneElement {
  src.role -> tgt.role;
  src.who : uri as vs ->  tgt.who as ref,  ref.reference = vs,  ref.extension as ext,  ext.url =
    'http://hl7.org/fhir/3.0/StructureDefinition/BaseType',  ext.value = 'uri';
  src.who : Reference as vs0 -> tgt.who as vt0 then Reference(vs0, vt0);
  src.onBehalfOf : uri as vs ->  tgt.onBehalfOf as ref,  ref.reference = vs,  ref.extension as ext,  ext.url =
    'http://hl7.org/fhir/3.0/StructureDefinition/BaseType',  ext.value = 'uri';
  src.onBehalfOf : Reference as vs0 -> tgt.onBehalfOf as vt0 then reference(vs0, vt0);
  src.relatedAgentType -> tgt.type;
}

group entity(source src : ProvenanceR3, target tgt : Provenance) extends BackboneElement {
  src.role -> tgt.role;
  src.what : uri as vs ->  tgt.what as ref,  ref.reference = vs,  ref.extension as ext,  ext.url =
    'http://hl7.org/fhir/3.0/StructureDefinition/BaseType',  ext.value = 'uri';
  src.what : Reference as vs0 -> tgt.what as vt0 then Reference(vs0, vt0);
  src.what : Identifier as vs0 -> tgt.what as vt0 then Identifier2Reference(vs0, vt0);
  src.agent as vs0 -> tgt.agent as vt0 then agent(vs0, vt0);
}
```

6.3.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Provenance4to3" = "R4 to R3 Conversion for Provenance"

uses "http://hl7.org/fhir/StructureDefinition/Provenance" alias Provenance as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Provenance" alias ProvenanceR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Provenance(source src : Provenance, target tgt : ProvenanceR3) extends DomainResource <<type+>> {
  src.target -> tgt.target;
  src.occurred : Period as vs0 -> tgt.period as vt0 then Period(vs0, vt0) "period";
  src.occurred : dateTime as vs0 ->  tgt.period as vt0,  vt0.start = vs0,  vt0.end = vs0 "period";
  src.recorded -> tgt.recorded;
  src.policy -> tgt.policy;
  src.location -> tgt.location;
  src.reason -> tgt.reason;
  src.activity -> tgt.activity;
  src.agent as vs0 -> tgt.agent as vt0 then agent(vs0, vt0);
  src.entity as vs0 -> tgt.entity as vt0 then entity(vs0, vt0);
  src.signature -> tgt.signature;
}

group agent(source src : ProvenanceR3, target tgt : Provenance) extends BackboneElement {
  src.role -> tgt.role;
  src.who as vs0 where extension('http://hl7.org/fhir/3.0/StructureDefinition/BaseType').empty() -> tgt.who =
    create('Reference') as vt0 then Reference(vs0, vt0);
  src.who as vs0 where extension('http://hl7.org/fhir/3.0/StructureDefinition/BaseType').exists() -> tgt.who =
    create('uri') as vt0 then {
    vs0.reference as vs1 then uri(vs1, vt0);
  };
  src.onBehalfOf as vs0 where extension('http://hl7.org/fhir/3.0/StructureDefinition/BaseType').empty() ->
    tgt.onBehalfOf = create('Reference') as vt0 then Reference(vs0, vt0);
  src.onBehalfOf as vs0 where extension('http://hl7.org/fhir/3.0/StructureDefinition/BaseType').exists() ->
```

```
    tgt.onBehalfOf = create('uri') as vt0 then {
    vs0.reference as vs1 then uri(vs1, vt0);
  };
  src.type -> tgt.relatedAgentType;
}

group entity(source src : ProvenanceR3, target tgt : Provenance) extends BackboneElement {
  src.role -> tgt.role;
  src.what as vs0 where extension('http://hl7.org/fhir/3.0/StructureDefinition/BaseType').empty() and identifier.empty()
     -> tgt.what = create('Reference') as vt0 then Reference(vs0, vt0);
  src.what as vs0 where extension('http://hl7.org/fhir/3.0/StructureDefinition/BaseType').exists() -> tgt.what =
    create('uri') as vt0 then {
    vs0.reference as vs1 then uri(vs1, vt0);
  };
  src.what as vs0 where identifier.exists() -> tgt.what = create('Identifier') as vt0 then Reference2Identifier(vs0,
    vt0);
  src.agent as vs0 -> tgt.agent as vt0 then agent(vs0, vt0);
}
```

6.3.10.3 R4 Validation Errors

6.3.11 Provenance Relevant History

URL: https://hl7.org/fhir/R4/relevant-history.html

Guidance on using Provenance for related history elements

6.3.11 Provenance Relevant History

Security Work Group | Maturity Level: N/A | Standards Status: Informative

6.3.11.1 Content

Profiles:

provenance-relevant-history | Guidance on using Provenance for related history elements

6.3.11.2 StructureDefinition: provenance-relevant-history

URL: https://hl7.org/fhir/R4/provenance-relevant-history.html

6.3.11.2 StructureDefinition: provenance-relevant-history

Security Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/provenance-relevant-history`

Guidance on using Provenance for related history elements

This profile was published on Tue, Dec 3, 2013 00:00+1100 as a draft by Health Level Seven International.

6.3.11.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

• Text Summary
• Differential Table
• Snapshot Table
• XML Template
• JSON Template
• All

This structure is derived from Provenance.

Summary

Mandatory: 3 elements (+2 nested mandatory elements) Must-Support: 9 elements

Slices

This structure defines the following Slices:

• The element Provenance.agent is sliced based on the value of value:type

This structure is derived from Provenance.

Name | Flags | Card. | Type | Description & Constraints
Provenance | | 0..*
target | S | 1..* | Reference(Resource) | Resource version
occurred[x] | S | 1..1 | dateTime
reason | S | 0..* | CodeableConcept
activity | S | 1..1 | CodeableConcept | Record activity Binding: Provenance History Record Activity Codes (extensible)
agent | S | | (Slice Definition) | Who was involved with change Slice: Unordered, Open by value:type
(All Slices) | | | | Content/Rules for all slices
type | S | 1..1 | CodeableConcept | Binding: Provenance Event History Agent Role Codes (extensible)

agent:Author | S | 0..1 | BackboneElement | Author
type | S | 1..1 | CodeableConcept | Required Pattern: At least the following
coding | | 1..* | Coding | Code defined by a terminology system Fixed Value: (complex)
system | | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/v3-ParticipationType
code | | 1..1 | code | Symbol in syntax defined by the system Fixed Value: AUT
who | S | 1..1 | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization) | Author Reference
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Provenance | I | 0..* | | Who, What, When for a set of resources
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
target | SΣI | 1..* | Reference(Resource) | Resource version
occurred[x] | SI | 1..1 | dateTime | When the activity occurred
recorded | ΣI | 1..1 | instant | When the activity was recorded / updated
policy | I | 0..* | uri | Policy or plan the activity was defined by
location | I | 0..1 | Reference(Location) | Where the activity occurred, if relevant
reason | SI | 0..* | CodeableConcept | Reason the activity is occurring Binding: PurposeOfUse (extensible)
activity | SI | 1..1 | CodeableConcept | Record activity Binding: Provenance History Record Activity Codes (extensible)
agent | SI | | (Slice Definition) | Who was involved with change Slice: Unordered, Open by value:type
(All Slices) | | | | Content/Rules for all slices
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
type | SΣI | 1..1 | CodeableConcept | How the agent participated Binding: Provenance Event History Agent Role Codes (extensible)
role | I | 0..* | CodeableConcept | What the agents role was Binding: SecurityRoleType (example)
who | ΣI | 1..1 | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization) | Who participated
onBehalfOf | I | 0..1 | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization) | Who the agent is representing
agent:Author | SI | 0..1 | BackboneElement | Author
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
type | SΣI | 1..1 | CodeableConcept | How the agent participated Binding: Provenance participant type (extensible) Required Pattern: At least the following
id | | 0..1 | string | Unique id for inter-element referencing
extension | | 0..* | Extension | Additional content defined by implementations
coding | | 1..* | Coding | Code defined by a terminology system Fixed Value: (complex)
id | | 0..1 | string | Unique id for inter-element referencing
extension | | 0..* | Extension | Additional content defined by implementations
system | | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/v3-ParticipationType
version | | 0..1 | string | Version of the system - if relevant
code | | 1..1 | code | Symbol in syntax defined by the system Fixed Value: AUT
display | | 0..1 | string | Representation defined by the system
userSelected | | 0..1 | boolean | If this coding was chosen directly by the user
text | | 0..1 | string | Plain text representation of the concept
role | I | 0..* | CodeableConcept | What the agents role was Binding: SecurityRoleType (example)
who | SΣI | 1..1 | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization) | Author Reference
onBehalfOf | I | 0..1 | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization) | Who the agent is representing
entity | I | 0..* | BackboneElement | An entity used in this activity
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
role | ΣI | 1..1 | code | derivation | revision | quotation | source | removal Binding: ProvenanceEntityRole (required)
what | ΣI | 1..1 | Reference(Resource) | Identity of entity
agent | I | 0..* | | Unknown reference to #Provenance.agent:Author Entity is attributed to this agent
signature | I | 0..* | Signature | Signature on target
Documentation for this format

```
<!-- Provenance Relevant History -->

<Provenance xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
```

```
    Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <target><!--  1..* Reference(Resource) Resource version --></target>
 <occurredDateTime value="[dateTime]"/><!--  1..1 When the activity occurred -->
 <recorded value="[instant]"/><!--  1..1 When the activity was recorded / updated -->
 <policy value="[uri]"/><!--  0..* Policy or plan the activity was defined by -->
 <location><!--  0..1 Reference(Location)
     Where the activity occurred, if relevant --></location>
 <reason><!--  0..* CodeableConcept Reason the activity is occurring   --></reason>
 <activity><!--  1..1 CodeableConcept Record activity   --></activity>
 <-- agent sliced by value:type  in the specified orderOpen-->
 <agent>  0..1 BackboneElement  <!--  0..1 Author -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <type><!--  1..1 CodeableConcept How the agent participated   --></type>
  <role><!--  0..* CodeableConcept What the agents role was   --></role>
  <who><!--  1..1 Reference(Practitioner) Author Reference --></who>
  <onBehalfOf><!--  0..1 Reference(Practitioner)
      Who the agent is representing --></onBehalfOf>
 </agent>
 <entity>  0..* BackboneElement  <!--  0..* An entity used in this activity -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <role value="[code]"/><!--  1..1 derivation | revision | quotation | source | removal   -->
  <what><!--  1..1 Reference(Resource) Identity of entity --></what>
  <agent><!-- See #Provenance.agent:Author  Entity is attributed to this agent --></agent>
 </entity>
 <signature><!--  0..* Signature Signature on target --></signature>
</Provenance>
```

```
{ // Provenance Relevant History
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "target" : [{ Reference(Resource) }], // C? R! Resource version
    "occurredDateTime" : "<dateTime>", // C? R! When the activity occurred
    "recorded" : "<instant>", // C? R! When the activity was recorded / updated
    "policy" : ["<uri>"], // C?Policy or plan the activity was defined by
    "location" : { Reference(Location) }, // C?Where the activity occurred, if relevant
    "reason" : [{ CodeableConcept }], // C?Reason the activity is occurring
    "activity" : { CodeableConcept }, // C? R! Record activity
    "agent" : [ //  sliced by value:type  in the specified order, Open
      { // Author // C?
        // from Element: extension
        "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored even if unrecognized
        "type" : { CodeableConcept }, // C? R! How the agent participated
        "role" : [{ CodeableConcept }], // C?What the agents role was
        "who" : { Reference(Practitioner) }, // C? R! Author Reference
        "onBehalfOf" : { Reference(Practitioner) } // C?Who the agent is representing
      }
    ],
    "entity" : [{ BackboneElement }], // C?An entity used in this activity
    "signature" : [{ Signature }], // C?Signature on target
  }
```

This structure is derived from Provenance.

Summary

Mandatory: 3 elements (+2 nested mandatory elements) Must-Support: 9 elements

Slices

This structure defines the following Slices:

• The element Provenance.agent is sliced based on the value of value:type

Differential View

This structure is derived from Provenance.

Name	Flags	Card.	Type	Description & Constraints
Provenance		0..*		
target	S	1..*	Reference(Resource)	Resource version
occurred[x]	S	1..1	dateTime	
reason	S	0..*	CodeableConcept	
activity	S	1..1	CodeableConcept	Record activity Binding: Provenance History Record Activity Codes (extensible)
agent	S		(Slice Definition)	Who was involved with change Slice: Unordered, Open by value:type
(All Slices)				Content/Rules for all slices
type	S	1..1	CodeableConcept	Binding: Provenance Event History Agent Role Codes (extensible)
agent:Author	S	0..1	BackboneElement	Author
type	S	1..1	CodeableConcept	Required Pattern: At least the following
coding		1..*	Coding	Code defined by a terminology system Fixed Value: (complex)
system		1..1	uri	Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/v3-ParticipationType

code | | 1..1 | code | Symbol in syntax defined by the system Fixed Value: AUT
who | S | 1..1 | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization) | Author Reference
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Provenance | I | 0..* | | Who, What, When for a set of resources
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
target | SΣI | 1..* | Reference(Resource) | Resource version
occurred[x] | SI | 1..1 | dateTime | When the activity occurred
recorded | ΣI | 1..1 | instant | When the activity was recorded / updated
policy | I | 0..* | uri | Policy or plan the activity was defined by
location | I | 0..1 | Reference(Location) | Where the activity occurred, if relevant
reason | SI | 0..* | CodeableConcept | Reason the activity is occurring Binding: PurposeOfUse (extensible)
activity | SI | 1..1 | CodeableConcept | Record activity Binding: Provenance History Record Activity Codes (extensible)
agent | SI | | (Slice Definition) | Who was involved with change Slice: Unordered, Open by value:type
(All Slices) | | | | Content/Rules for all slices
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
type | SΣI | 1..1 | CodeableConcept | How the agent participated Binding: Provenance Event History Agent Role Codes (extensible)
role | I | 0..* | CodeableConcept | What the agents role was Binding: SecurityRoleType (example)
who | ΣI | 1..1 | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization) | Who participated
onBehalfOf | I | 0..1 | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization) | Who the agent is representing
agent:Author | SI | 0..1 | BackboneElement | Author
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
type | SΣI | 1..1 | CodeableConcept | How the agent participated Binding: Provenance participant type (extensible) Required Pattern: At least the following
id | | 0..1 | string | Unique id for inter-element referencing
extension | | 0..* | Extension | Additional content defined by implementations
coding | | 1..* | Coding | Code defined by a terminology system Fixed Value: (complex)
id | | 0..1 | string | Unique id for inter-element referencing
extension | | 0..* | Extension | Additional content defined by implementations
system | | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/v3-ParticipationType
version | | 0..1 | string | Version of the system - if relevant
code | | 1..1 | code | Symbol in syntax defined by the system Fixed Value: AUT
display | | 0..1 | string | Representation defined by the system
userSelected | | 0..1 | boolean | If this coding was chosen directly by the user
text | | 0..1 | string | Plain text representation of the concept
role | I | 0..* | CodeableConcept | What the agents role was Binding: SecurityRoleType (example)
who | SΣI | 1..1 | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization) | Author Reference
onBehalfOf | I | 0..1 | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization) | Who the agent is representing
entity | I | 0..* | BackboneElement | An entity used in this activity
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
role | ΣI | 1..1 | code | derivation | revision | quotation | source | removal Binding: ProvenanceEntityRole (required)
what | ΣI | 1..1 | Reference(Resource) | Identity of entity
agent | I | 0..* | | Unknown reference to #Provenance.agent:Author Entity is attributed to this agent
signature | I | 0..* | Signature | Signature on target
Documentation for this format

XML Template

```
<!-- Provenance Relevant History -->

<Provenance xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
```

```
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <target><!--  1..* Reference(Resource) Resource version --></target>
 <occurredDateTime value="[dateTime]"/><!--  1..1 When the activity occurred -->
 <recorded value="[instant]"/><!--  1..1 When the activity was recorded / updated -->
 <policy value="[uri]"/><!--  0..* Policy or plan the activity was defined by -->
 <location><!--  0..1 Reference(Location)
     Where the activity occurred, if relevant --></location>
 <reason><!--  0..* CodeableConcept Reason the activity is occurring   --></reason>
 <activity><!--  1..1 CodeableConcept Record activity   --></activity>
 <-- agent sliced by value:type  in the specified orderOpen-->
 <agent>  0..1 BackboneElement  <!--  0..1 Author -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <type><!--  1..1 CodeableConcept How the agent participated   --></type>
  <role><!--  0..* CodeableConcept What the agents role was   --></role>
  <who><!--  1..1 Reference(Practitioner) Author Reference --></who>
  <onBehalfOf><!--  0..1 Reference(Practitioner)
      Who the agent is representing --></onBehalfOf>
 </agent>
 <entity>  0..* BackboneElement  <!--  0..* An entity used in this activity -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <role value="[code]"/><!--  1..1 derivation | revision | quotation | source | removal   -->
  <what><!--  1..1 Reference(Resource) Identity of entity --></what>
  <agent><!-- See #Provenance.agent:Author  Entity is attributed to this agent --></agent>
 </entity>
 <signature><!--  0..* Signature Signature on target --></signature>
</Provenance>
```

JSON Template

```
{ // Provenance Relevant History
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "target" : [{ Reference(Resource) }], // C? R! Resource version
    "occurredDateTime" : "<dateTime>", // C? R! When the activity occurred
    "recorded" : "<instant>", // C? R! When the activity was recorded / updated
    "policy" : ["<uri>"], // C?Policy or plan the activity was defined by
    "location" : { Reference(Location) }, // C?Where the activity occurred, if relevant
    "reason" : [{ CodeableConcept }], // C?Reason the activity is occurring
    "activity" : { CodeableConcept }, // C? R! Record activity
    "agent" : [ //  sliced by value:type  in the specified order, Open
      { // Author // C?
        // from Element: extension
        "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored even if unrecognized
        "type" : { CodeableConcept }, // C? R! How the agent participated
        "role" : [{ CodeableConcept }], // C?What the agents role was
        "who" : { Reference(Practitioner) }, // C? R! Author Reference
        "onBehalfOf" : { Reference(Practitioner) } // C?Who the agent is representing
      }
    ],
    "entity" : [{ BackboneElement }], // C?An entity used in this activity
    "signature" : [{ Signature }], // C?Signature on target
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

6.3.11.2.2 Terminology Bindings

Path | Name | Conformance | ValueSet
Provenance.language | Common Languages | preferred | Common Languages
Provenance.reason | V3 Value SetPurposeOfUse | extensible | V3 Value SetPurposeOfUse
Provenance.activity | Provenance History Record Activity Codes | extensible | Provenance History Record Activity Codes
Provenance.agent.type | Provenance participant type | extensible | Provenance participant type
Provenance.agent.role | SecurityRoleType | example | SecurityRoleType
Provenance.agent.type | Provenance participant type | extensible | Provenance participant type
Provenance.agent.role | SecurityRoleType | example | SecurityRoleType
Provenance.entity.role | ProvenanceEntityRole | required | ProvenanceEntityRole

6.3.12 JSON Schema for Provenance

URL: https://hl7.org/fhir/R4/provenance.schema.json.html

6.3.12 JSON Schema for Provenance

Security Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

This is a representation of the json schema for Provenance, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
```

```
"id": "http://hl7.org/fhir/json-schema/Provenance",
"$ref": "#/definitions/Provenance",
"description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
"definitions": {
  "Provenance": {
    "description": "Provenance of a resource is a record that describes entities and processes involved in producing
  and delivering or otherwise influencing that resource. Provenance provides a critical foundation for assessing
  authenticity, enabling trust, and allowing reproducibility. Provenance assertions are a form of contextual metadata
  and can themselves become important records with their own provenance. Provenance statement indicates clinical
  significance in terms of confidence in authenticity, reliability, and trustworthiness, integrity, and stage in
  lifecycle (e.g. Document Completion - has the artifact been legally authenticated), all of which may impact
  security, privacy, and trust policies.",
    "properties": {
      "resourceType": {
        "description": "This is a Provenance resource",
        "const": "Provenance"
      },
      "id": {
        "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
   never changes.",
        "$ref": "id.schema.json#/definitions/id"
      },
      "meta": {
        "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
  Changes to the content might not always be associated with version changes to the resource.",
        "$ref": "Meta.schema.json#/definitions/Meta"
      },
      "implicitRules": {
        "description": "A reference to a set of rules that were followed when the resource was constructed, and which
  must be understood when processing the content. Often, this is a reference to an implementation guide that defines
  the special rules along with other profiles etc.",
        "$ref": "#/definitions/uri"
      },
      "_implicitRules": {
        "description": "Extensions for implicitRules",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "language": {
        "description": "The base language in which the resource is written.",
        "$ref": "#/definitions/code"
      },
      "_language": {
        "description": "Extensions for language",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "text": {
        "description": "A human-readable narrative that contains a summary of the resource and can be used to
  represent the content of the resource to a human. The narrative need not encode all the structured data, but is
  required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
  Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
        "$ref": "Narrative.schema.json#/definitions/Narrative"
      },
      "contained": {
        "description": "These resources do not have an independent existence apart from the resource that contains
  them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
        "items": {
          "$ref": "ResourceList.schema.json#/definitions/ResourceList"
        },
        "type": "array"
      },
      "extension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
  the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
   definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
  that SHALL be met as part of the definition of the extension.",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "modifierExtension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
  the resource and that modifies the understanding of the element that contains it and/or the understanding of the
  containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
  of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
  extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
  met as part of the definition of the extension. Applications processing a resource are required to check for
  modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
  DomainResource (including cannot change the meaning of modifierExtension itself).",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "target": {
        "description": "The Reference(s) that were generated or updated by  the activity described in this resource. A
   provenance can point to more than one target if multiple resources were created/updated by the same activity.",
        "items": {
          "$ref": "Reference.schema.json#/definitions/Reference"
```

```
      },
      "type": "array"
    },
    "occurredPeriod": {
      "description": "The period during which the activity occurred.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "occurredDateTime": {
      "description": "The period during which the activity occurred.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_occurredDateTime": {
      "description": "Extensions for occurredDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "recorded": {
      "description": "The instant of time at which the activity was recorded.",
      "$ref": "#/definitions/instant"
    },
    "_recorded": {
      "description": "Extensions for recorded",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "policy": {
      "description": "Policy or plan the activity was defined by. Typically, a single activity may have multiple
applicable policy documents, such as patient consent, guarantor funding, etc.",
      "items": {
        "$ref": "#/definitions/uri"
      },
      "type": "array"
    },
    "_policy": {
      "description": "Extensions for policy",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "location": {
      "description": "Where the activity occurred, if relevant.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "reason": {
      "description": "The reason that the activity was taking place.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "activity": {
      "description": "An activity is something that occurs over a period of time and acts upon or with entities; it
may include consuming, processing, transforming, modifying, relocating, using, or generating entities.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "agent": {
      "description": "An actor taking a role in an activity  for which it can be assigned some degree of
responsibility for the activity taking place.",
      "items": {
        "$ref": "#/definitions/Provenance_Agent"
      },
      "type": "array"
    },
    "entity": {
      "description": "An entity used in this activity.",
      "items": {
        "$ref": "#/definitions/Provenance_Entity"
      },
      "type": "array"
    },
    "signature": {
      "description": "A digital signature on the target Reference(s). The signer should match a Provenance.agent.
The purpose of the signature is indicated.",
      "items": {
        "$ref": "Signature.schema.json#/definitions/Signature"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "agent",
    "resourceType",
    "target"
  ]
},
"Provenance_Agent": {
  "description": "Provenance of a resource is a record that describes entities and processes involved in producing
```

```
and delivering or otherwise influencing that resource. Provenance provides a critical foundation for assessing
authenticity, enabling trust, and allowing reproducibility. Provenance assertions are a form of contextual metadata
and can themselves become important records with their own provenance. Provenance statement indicates clinical
significance in terms of confidence in authenticity, reliability, and trustworthiness, integrity, and stage in
lifecycle (e.g. Document Completion - has the artifact been legally authenticated), all of which may impact
security, privacy, and trust policies.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "The participation the agent had with respect to the activity.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "role": {
      "description": "The function of the agent with respect to the activity. The security role enabling the agent
with respect to the activity.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "who": {
      "description": "The individual, device or organization that participated in the event.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "onBehalfOf": {
      "description": "The individual, device, or organization for whom the change was made.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false,
  "required": [
    "who"
  ]
},
"Provenance_Entity": {
  "description": "Provenance of a resource is a record that describes entities and processes involved in producing
and delivering or otherwise influencing that resource. Provenance provides a critical foundation for assessing
authenticity, enabling trust, and allowing reproducibility. Provenance assertions are a form of contextual metadata
and can themselves become important records with their own provenance. Provenance statement indicates clinical
significance in terms of confidence in authenticity, reliability, and trustworthiness, integrity, and stage in
lifecycle (e.g. Document Completion - has the artifact been legally authenticated), all of which may impact
security, privacy, and trust policies.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
```

```
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "role": {
          "description": "How the entity was used during the activity.",
          "enum": [
            "derivation",
            "revision",
            "quotation",
            "source",
            "removal"
          ]
        },
        "_role": {
          "description": "Extensions for role",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "what": {
          "description": "Identity of the  Entity used. May be a logical or physical uri and maybe absolute or
    relative.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "agent": {
          "description": "The entity is attributed to an agent to express the agent\u0027s responsibility for that
    entity, possibly along with other agents. This description can be understood as shorthand for saying that the agent
    was responsible for the activity which generated the entity.",
          "items": {
            "$ref": "#/definitions/Provenance_Agent"
          },
          "type": "array"
        }
      },
      "additionalProperties": false,
      "required": [
        "what"
      ]
    }
  }
}
```


6.3.13 Resource Provenance - Maturity Dependency View

URL: https://hl7.org/fhir/R4/provenance-dependencies.html

6.3.13 Resource Provenance - Maturity Dependency View

Security Work Group | Maturity Level: 3 | Trial Use | Compartments: Device, Patient, Practitioner, RelatedPerson

Dependency Graph for Provenance FMM level 3

Name | Card. | Type | Dependency Analysis
Provenance | | DomainResource
target | 1..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM3-Trial Use)
occurred[x] | 0..1
occurredPeriod | | Period
occurredDateTime | | dateTime
recorded | 1..1 | instant
policy | 0..* | uri
location | 0..1 | Reference(Location)
reason | 0..* | CodeableConcept
activity | 0..1 | CodeableConcept
agent | 1..* | BackboneElement
type | 0..1 | CodeableConcept
role | 0..* | CodeableConcept
who | 1..1 | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (RelatedPerson = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (Device = FMM2-Trial Use vs. Element = FMM3-Trial Use)
onBehalfOf | 0..1 | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Device | Organization) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (RelatedPerson = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (Device = FMM2-Trial Use vs. Element = FMM3-Trial Use)
entity | 0..* | BackboneElement
role | 1..1 | code
what | 1..1 | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM3-Trial Use)
agent | 0..* | see agent
signature | 0..* | Signature

6.4 Resource AuditEvent - Content

URL: https://hl7.org/fhir/R4/auditevent.html
Security Work Group | Maturity Level: 3 | Trial Use | Security Category: Not Classified | Compartments: Device, Patient, Practitioner

A record of an event made for purposes of maintaining a security log. Typical uses include detection of intrusion attempts and monitoring for inappropriate usage.

6.4.1 Scope and Usage

The audit event is based on the IHE-ATNA Audit record definitions, originally from RFC 3881 , and now managed by DICOM (see DICOM Part 15 Annex A5).

• ASTM E2147 – Setup the concept of security audit logs for healthcare including accounting of disclosures

• IETF RFC 3881 – Defined the Information Model (IETF rule forced this to be informative)

• DICOM Audit Log Message – Made the information model Normative, defined Vocabulary, Transport Binding, and Schema

• IHE ATNA – Defines the grouping with secure transport and access controls; and defined specific audit log records for specific IHE transactions.

• NIST SP800-92 – Shows how to do audit log management and reporting – consistent with our model

• HL7 PASS – Defined an Audit Service with responsibilities and a query interface for reporting use

• ISO 27789 – Defined the subset of audit events that an EHR would need

• ISO/HL7 10781 EHR System Functional Model Release 2

• ISO 21089 Trusted End-to-End Information Flows

This resource is managed collaboratively between HL7, DICOM, and IHE.

The primary purpose of this resource is the maintenance of security audit log information. However, it can also be used for any audit logging needs and simple event-based notification.

6.4.2 Background and Context

All actors - such as applications, processes, and services - involved in an auditable event should record an AuditEvent. This will likely result in multiple AuditEvent entries that show whether privacy and security safeguards, such as access control, are properly functioning across an enterprise's system-of-systems. Thus, it is typical to get an auditable event recorded by both the application in a workflow process and the servers that support them. For this reason, duplicate entries are expected, which is helpful because it may aid in the detection of. For example, fewer than expected actors being recorded in a multi-actor process or attributes related to those records being in conflict, which is an indication of a security problem. There may be non-participating actors, such as trusted intermediary, that also detect a security relevant event and thus would record an AuditEvent, such as a trusted intermediary.

Security relevant events are not limited to communications or RESTful events. They include:

• software start-up and shutdown;

• user login and logout;

• access control decisions;

• configuration events;

• software installation;

• policy rules changes; and

• manipulation of data that exposes the data to users.

See the Audit Event Sub-Type vocabulary for guidance on some security relevant events.

The content of an AuditEvent is intended for use by security system administrators, security and privacy information managers, and records management personnel. This content is not intended to be accessible or used directly by other healthcare users, such as providers or patients, although reports generated from the raw data would be useful. An example is a patient-centric accounting of disclosures or an access report. Servers that provide support for AuditEvent resources would not generally accept update or delete operations on the resources, as this would compromise the integrity of the audit record. Access to the AuditEvent would typically be limited to security, privacy, or other system administration purposes.

Relationship of AuditEvent and Provenance resources are often (though not exclusively) created by the application responding to the create/read/query/update/delete/execute etc. event. A Provenance resource contains overlapping information, but is a record-keeping assertion that gathers information about the context in which the information in a resource "came to be" in its current state, e.g., whether it was created de novo or obtained from another entity in whole, part, or by transformation. Provenance resources are prepared by the application that initiates the create/update of the resource and may be persisted with the AuditEvent target resource.

6.4.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
AuditEvent | TU | | DomainResource | Event record kept for security purposes Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
type | Σ | 1..1 | Coding | Type/identifier of event Audit Event ID (Extensible)
subtype | Σ | 0..* | Coding | More specific type/id for the event Audit Event Sub-Type (Extensible)
action | Σ | 0..1 | code | Type of action performed during the event AuditEventAction (Required)
period | | 0..1 | Period | When the activity occurred
recorded | Σ | 1..1 | instant | Time when the event was recorded
outcome | Σ | 0..1 | code | Whether the event succeeded or failed AuditEventOutcome (Required)
outcomeDesc | Σ | 0..1 | string | Description of the event outcome
purposeOfEvent | Σ | 0..* | CodeableConcept | The purposeOfUse of the event V3 Value SetPurposeOfUse (Extensible)
agent | | 1..* | BackboneElement | Actor involved in the event
type | | 0..1 | CodeableConcept | How agent participated ParticipationRoleType (Extensible)
role | | 0..* | CodeableConcept | Agent role in the event SecurityRoleType (Example)
who | Σ | 0..1 | Reference(PractitionerRole | Practitioner | Organization | Device | Patient | RelatedPerson) | Identifier of who
altId | | 0..1 | string | Alternative User identity
name | | 0..1 | string | Human friendly name for the agent
requestor | Σ | 1..1 | boolean | Whether user is initiator
location | | 0..1 | Reference(Location) | Where
policy | | 0..* | uri | Policy that authorized event
media | | 0..1 | Coding | Type of media Media Type Code (Extensible)
network | | 0..1 | BackboneElement | Logical network location for application activity
address | | 0..1 | string | Identifier for the network access point of the user device
type | | 0..1 | code | The type of network access point AuditEventAgentNetworkType (Required)
purposeOfUse | | 0..* | CodeableConcept | Reason given for this user V3 Value SetPurposeOfUse (Extensible)
source | | 1..1 | BackboneElement | Audit Event Reporter
site | | 0..1 | string | Logical source location within the enterprise
observer | Σ | 1..1 | Reference(PractitionerRole | Practitioner | Organization | Device | Patient | RelatedPerson) | The identity of source detecting the event
type | | 0..* | Coding | The type of source where event originated Audit Event Source Type (Extensible)
entity | I | 0..* | BackboneElement | Data or objects used + Rule: Either a name or a query (NOT both)
what | Σ | 0..1 | Reference(Any) | Specific instance of resource
type | | 0..1 | Coding | Type of entity involved Audit event entity type (Extensible)
role | | 0..1 | Coding | What role the entity played AuditEventEntityRole (Extensible)
lifecycle | | 0..1 | Coding | Life-cycle stage for the entity ObjectLifecycleEvents (Extensible)
securityLabel | | 0..* | Coding | Security labels on the entity SecurityLabels (Extensible)
name | ΣI | 0..1 | string | Descriptor for entity
description | | 0..1 | string | Descriptive text
query | ΣI | 0..1 | base64Binary | Query parameters
detail | | 0..* | BackboneElement | Additional Information about the entity
type | | 1..1 | string | Name of the property
value[x] | | 1..1 | | Property value
valueString | | | string
valueBase64Binary | | | base64Binary
Documentation for this format

UML Diagram (Legend)

XML Template

```
<AuditEvent xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <type><!-- 1..1 Coding Type/identifier of event --></type>
 <subtype><!-- 0..* Coding More specific type/id for the event --></subtype>
 <action value="[code]"/><!-- 0..1 Type of action performed during the event -->
 <period><!-- 0..1 Period When the activity occurred --></period>
 <recorded value="[instant]"/><!-- 1..1 Time when the event was recorded -->
 <outcome value="[code]"/><!-- 0..1 Whether the event succeeded or failed -->
 <outcomeDesc value="[string]"/><!-- 0..1 Description of the event outcome -->
 <purposeOfEvent><!-- 0..* CodeableConcept The purposeOfUse of the event --></purposeOfEvent>
 <agent>  <!-- 1..* Actor involved in the event -->
  <type><!-- 0..1 CodeableConcept How agent participated --></type>
  <role><!-- 0..* CodeableConcept Agent role in the event --></role>
```

```
  <who><!-- 0..1 Reference(PractitionerRole|Practitioner|Organization|Device|
    Patient|RelatedPerson) Identifier of who --></who>
  <altId value="[string]"/><!-- 0..1 Alternative User identity -->
  <name value="[string]"/><!-- 0..1 Human friendly name for the agent -->
  <requestor value="[boolean]"/><!-- 1..1 Whether user is initiator -->
  <location><!-- 0..1 Reference(Location) Where --></location>
  <policy value="[uri]"/><!-- 0..* Policy that authorized event -->
  <media><!-- 0..1 Coding Type of media --></media>
  <network>  <!-- 0..1 Logical network location for application activity -->
   <address value="[string]"/><!-- 0..1 Identifier for the network access point of the user device -->
   <type value="[code]"/><!-- 0..1 The type of network access point -->
  </network>
  <purposeOfUse><!-- 0..* CodeableConcept Reason given for this user --></purposeOfUse>
 </agent>
 <source>  <!-- 1..1 Audit Event Reporter -->
  <site value="[string]"/><!-- 0..1 Logical source location within the enterprise -->
  <observer><!-- 1..1 Reference(PractitionerRole|Practitioner|Organization|Device|
    Patient|RelatedPerson) The identity of source detecting the event --></observer>
  <type><!-- 0..* Coding The type of source where event originated --></type>
 </source>
 <entity>  <!-- 0..* Data or objects used -->
  <what><!-- 0..1 Reference(Any) Specific instance of resource --></what>
  <type><!-- 0..1 Coding Type of entity involved --></type>
  <role><!-- 0..1 Coding What role the entity played --></role>
  <lifecycle><!-- 0..1 Coding Life-cycle stage for the entity --></lifecycle>
  <securityLabel><!-- 0..* Coding Security labels on the entity --></securityLabel>
  <name value="[string]"/><!--  0..1 Descriptor for entity -->
  <description value="[string]"/><!-- 0..1 Descriptive text -->
  <query value="[base64Binary]"/><!--  0..1 Query parameters -->
  <detail>  <!-- 0..* Additional Information about the entity -->
   <type value="[string]"/><!-- 1..1 Name of the property -->
   <value[x]><!-- 1..1 string|base64Binary Property value --></value[x]>
  </detail>
 </entity>
</AuditEvent>
```

JSON Template

```
{
  "resourceType" : "AuditEvent",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "type" : { Coding }, // R!  Type/identifier of event
  "subtype" : [{ Coding }], // More specific type/id for the event
  "action" : "<code>", // Type of action performed during the event
  "period" : { Period }, // When the activity occurred
  "recorded" : "<instant>", // R!  Time when the event was recorded
  "outcome" : "<code>", // Whether the event succeeded or failed
  "outcomeDesc" : "<string>", // Description of the event outcome
  "purposeOfEvent" : [{ CodeableConcept }], // The purposeOfUse of the event
  "agent" : [{ // R!  Actor involved in the event
    "type" : { CodeableConcept }, // How agent participated
    "role" : [{ CodeableConcept }], // Agent role in the event
    "who" : { Reference(PractitionerRole|Practitioner|Organization|Device|
    Patient|RelatedPerson) }, // Identifier of who
    "altId" : "<string>", // Alternative User identity
    "name" : "<string>", // Human friendly name for the agent
    "requestor" : <boolean>, // R!  Whether user is initiator
    "location" : { Reference(Location) }, // Where
    "policy" : ["<uri>"], // Policy that authorized event
    "media" : { Coding }, // Type of media
    "network" : { // Logical network location for application activity
      "address" : "<string>", // Identifier for the network access point of the user device
      "type" : "<code>" // The type of network access point
    },
    "purposeOfUse" : [{ CodeableConcept }] // Reason given for this user
  }],
  "source" : { // R!  Audit Event Reporter
    "site" : "<string>", // Logical source location within the enterprise
    "observer" : { Reference(PractitionerRole|Practitioner|Organization|Device|
    Patient|RelatedPerson) }, // R!  The identity of source detecting the event
    "type" : [{ Coding }] // The type of source where event originated
  },
  "entity" : [{ // Data or objects used
    "what" : { Reference(Any) }, // Specific instance of resource
    "type" : { Coding }, // Type of entity involved
    "role" : { Coding }, // What role the entity played
    "lifecycle" : { Coding }, // Life-cycle stage for the entity
    "securityLabel" : [{ Coding }], // Security labels on the entity
    "name" : "<string>", // C? Descriptor for entity
    "description" : "<string>", // Descriptive text
    "query" : "<base64Binary>", // C? Query parameters
    "detail" : [{ // Additional Information about the entity
      "type" : "<string>", // R!  Name of the property
      // value[x]: Property value. One of these 2:
      "valueString" : "<string>"
      "valueBase64Binary" : "<base64Binary>"
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:AuditEvent;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:AuditEvent.type [ Coding ]; # 1..1 Type/identifier of event
  fhir:AuditEvent.subtype [ Coding ], ... ; # 0..* More specific type/id for the event
  fhir:AuditEvent.action [ code ]; # 0..1 Type of action performed during the event
  fhir:AuditEvent.period [ Period ]; # 0..1 When the activity occurred
  fhir:AuditEvent.recorded [ instant ]; # 1..1 Time when the event was recorded
  fhir:AuditEvent.outcome [ code ]; # 0..1 Whether the event succeeded or failed
  fhir:AuditEvent.outcomeDesc [ string ]; # 0..1 Description of the event outcome
  fhir:AuditEvent.purposeOfEvent [ CodeableConcept ], ... ; # 0..* The purposeOfUse of the event
  fhir:AuditEvent.agent [ # 1..* Actor involved in the event
    fhir:AuditEvent.agent.type [ CodeableConcept ]; # 0..1 How agent participated
    fhir:AuditEvent.agent.role [ CodeableConcept ], ... ; # 0..* Agent role in the event
    fhir:AuditEvent.agent.who [ Reference(PractitionerRole|Practitioner|Organization|Device|Patient|RelatedPerson) ]; #
    0..1 Identifier of who
    fhir:AuditEvent.agent.altId [ string ]; # 0..1 Alternative User identity
    fhir:AuditEvent.agent.name [ string ]; # 0..1 Human friendly name for the agent
    fhir:AuditEvent.agent.requestor [ boolean ]; # 1..1 Whether user is initiator
    fhir:AuditEvent.agent.location [ Reference(Location) ]; # 0..1 Where
    fhir:AuditEvent.agent.policy [ uri ], ... ; # 0..* Policy that authorized event
    fhir:AuditEvent.agent.media [ Coding ]; # 0..1 Type of media
    fhir:AuditEvent.agent.network [ # 0..1 Logical network location for application activity
      fhir:AuditEvent.agent.network.address [ string ]; # 0..1 Identifier for the network access point of the user
    device
      fhir:AuditEvent.agent.network.type [ code ]; # 0..1 The type of network access point
    ];
    fhir:AuditEvent.agent.purposeOfUse [ CodeableConcept ], ... ; # 0..* Reason given for this user
  ], ...;
  fhir:AuditEvent.source [ # 1..1 Audit Event Reporter
    fhir:AuditEvent.source.site [ string ]; # 0..1 Logical source location within the enterprise
    fhir:AuditEvent.source.observer [ Reference(PractitionerRole|Practitioner|Organization|Device|Patient|RelatedPerson)
     ]; # 1..1 The identity of source detecting the event
    fhir:AuditEvent.source.type [ Coding ], ... ; # 0..* The type of source where event originated
  ];
  fhir:AuditEvent.entity [ # 0..* Data or objects used
    fhir:AuditEvent.entity.what [ Reference(Any) ]; # 0..1 Specific instance of resource
    fhir:AuditEvent.entity.type [ Coding ]; # 0..1 Type of entity involved
    fhir:AuditEvent.entity.role [ Coding ]; # 0..1 What role the entity played
    fhir:AuditEvent.entity.lifecycle [ Coding ]; # 0..1 Life-cycle stage for the entity
    fhir:AuditEvent.entity.securityLabel [ Coding ], ... ; # 0..* Security labels on the entity
    fhir:AuditEvent.entity.name [ string ]; # 0..1 Descriptor for entity
    fhir:AuditEvent.entity.description [ string ]; # 0..1 Descriptive text
    fhir:AuditEvent.entity.query [ base64Binary ]; # 0..1 Query parameters
    fhir:AuditEvent.entity.detail [ # 0..* Additional Information about the entity
      fhir:AuditEvent.entity.detail.type [ string ]; # 1..1 Name of the property
      # AuditEvent.entity.detail.value[x] : 1..1 Property value. One of these 2
        fhir:AuditEvent.entity.detail.valueString [ string ]
        fhir:AuditEvent.entity.detail.valueBase64Binary [ base64Binary ]
    ], ...;
  ], ...;
]
```

Changes since R3

AuditEvent
AuditEvent.action | Change value set from http://hl7.org/fhir/ValueSet/audit-event-action to http://hl7.org/fhir/ValueSet/audit-event-action|4.0.1
AuditEvent.period | Added Element
AuditEvent.outcome | Change value set from http://hl7.org/fhir/ValueSet/audit-event-outcome to http://hl7.org/fhir/ValueSet/audit-event-outcome|4.0.1
AuditEvent.agent.type | Added Element
AuditEvent.agent.role | Remove Binding http://hl7.org/fhir/ValueSet/security-role-type (extensible)
AuditEvent.agent.who | Added Element
AuditEvent.agent.network.type | Change value set from http://hl7.org/fhir/ValueSet/network-type to http://hl7.org/fhir/ValueSet/network-type|4.0.1
AuditEvent.source.observer | Added Mandatory Element
AuditEvent.source.type | Change code system for extensibly bound codes from "http://hl7.org/fhir/security-source-type" to "http://terminology.hl7.org/CodeSystem/security-source-type"
AuditEvent.entity.what | Added Element
AuditEvent.entity.role | Change code system for extensibly bound codes from "http://hl7.org/fhir/object-role" to "http://terminology.hl7.org/CodeSystem/object-role"
AuditEvent.entity.detail.value[x] | Renamed from value to value[x] Add Type string
AuditEvent.agent.reference | deleted
AuditEvent.agent.userId | deleted
AuditEvent.source.identifier | deleted
AuditEvent.entity.identifier | deleted
AuditEvent.entity.reference | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 8 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints
AuditEvent | TU | | DomainResource | Event record kept for security purposes Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
type | Σ | 1..1 | Coding | Type/identifier of event Audit Event ID (Extensible)
subtype | Σ | 0..* | Coding | More specific type/id for the event Audit Event Sub-Type (Extensible)
action | Σ | 0..1 | code | Type of action performed during the event AuditEventAction (Required)
period | | 0..1 | Period | When the activity occurred
recorded | Σ | 1..1 | instant | Time when the event was recorded
outcome | Σ | 0..1 | code | Whether the event succeeded or failed AuditEventOutcome (Required)
outcomeDesc | Σ | 0..1 | string | Description of the event outcome
purposeOfEvent | Σ | 0..* | CodeableConcept | The purposeOfUse of the event V3 Value SetPurposeOfUse (Extensible)
agent | | 1..* | BackboneElement | Actor involved in the event
type | | 0..1 | CodeableConcept | How agent participated ParticipationRoleType (Extensible)
role | | 0..* | CodeableConcept | Agent role in the event SecurityRoleType (Example)
who | Σ | 0..1 | Reference(PractitionerRole | Practitioner | Organization | Device | Patient | RelatedPerson) | Identifier of who
altId | | 0..1 | string | Alternative User identity
name | | 0..1 | string | Human friendly name for the agent
requestor | Σ | 1..1 | boolean | Whether user is initiator
location | | 0..1 | Reference(Location) | Where
policy | | 0..* | uri | Policy that authorized event
media | | 0..1 | Coding | Type of media Media Type Code (Extensible)
network | | 0..1 | BackboneElement | Logical network location for application activity
address | | 0..1 | string | Identifier for the network access point of the user device
type | | 0..1 | code | The type of network access point AuditEventAgentNetworkType (Required)
purposeOfUse | | 0..* | CodeableConcept | Reason given for this user V3 Value SetPurposeOfUse (Extensible)
source | | 1..1 | BackboneElement | Audit Event Reporter
site | | 0..1 | string | Logical source location within the enterprise
observer | Σ | 1..1 | Reference(PractitionerRole | Practitioner | Organization | Device | Patient | RelatedPerson) | The identity of source detecting the event
type | | 0..* | Coding | The type of source where event originated Audit Event Source Type (Extensible)
entity | I | 0..* | BackboneElement | Data or objects used + Rule: Either a name or a query (NOT both)
what | Σ | 0..1 | Reference(Any) | Specific instance of resource
type | | 0..1 | Coding | Type of entity involved Audit event entity type (Extensible)
role | | 0..1 | Coding | What role the entity played AuditEventEntityRole (Extensible)
lifecycle | | 0..1 | Coding | Life-cycle stage for the entity ObjectLifecycleEvents (Extensible)
securityLabel | | 0..* | Coding | Security labels on the entity SecurityLabels (Extensible)
name | ΣI | 0..1 | string | Descriptor for entity
description | | 0..1 | string | Descriptive text
query | ΣI | 0..1 | base64Binary | Query parameters
detail | | 0..* | BackboneElement | Additional Information about the entity
type | | 1..1 | string | Name of the property
value[x] | | 1..1 | | Property value
valueString | | | string
valueBase64Binary | | | base64Binary
Documentation for this format

UML Diagram (Legend)

XML Template

```
<AuditEvent xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <type><!-- 1..1 Coding Type/identifier of event --></type>
 <subtype><!-- 0..* Coding More specific type/id for the event --></subtype>
 <action value="[code]"/><!-- 0..1 Type of action performed during the event -->
 <period><!-- 0..1 Period When the activity occurred --></period>
 <recorded value="[instant]"/><!-- 1..1 Time when the event was recorded -->
 <outcome value="[code]"/><!-- 0..1 Whether the event succeeded or failed -->
 <outcomeDesc value="[string]"/><!-- 0..1 Description of the event outcome -->
 <purposeOfEvent><!-- 0..* CodeableConcept The purposeOfUse of the event --></purposeOfEvent>
 <agent>  <!-- 1..* Actor involved in the event -->
  <type><!-- 0..1 CodeableConcept How agent participated --></type>
  <role><!-- 0..* CodeableConcept Agent role in the event --></role>
  <who><!-- 0..1 Reference(PractitionerRole|Practitioner|Organization|Device|
    Patient|RelatedPerson) Identifier of who --></who>
  <altId value="[string]"/><!-- 0..1 Alternative User identity -->
  <name value="[string]"/><!-- 0..1 Human friendly name for the agent -->
  <requestor value="[boolean]"/><!-- 1..1 Whether user is initiator -->
  <location><!-- 0..1 Reference(Location) Where --></location>
  <policy value="[uri]"/><!-- 0..* Policy that authorized event -->
  <media><!-- 0..1 Coding Type of media --></media>
```

```
  <network>  <!-- 0..1 Logical network location for application activity -->
   <address value="[string]"/><!-- 0..1 Identifier for the network access point of the user device -->
   <type value="[code]"/><!-- 0..1 The type of network access point -->
  </network>
  <purposeOfUse><!-- 0..* CodeableConcept Reason given for this user --></purposeOfUse>
 </agent>
 <source>  <!-- 1..1 Audit Event Reporter -->
  <site value="[string]"/><!-- 0..1 Logical source location within the enterprise -->
  <observer><!-- 1..1 Reference(PractitionerRole|Practitioner|Organization|Device|
    Patient|RelatedPerson) The identity of source detecting the event --></observer>
  <type><!-- 0..* Coding The type of source where event originated --></type>
 </source>
 <entity>  <!-- 0..* Data or objects used -->
  <what><!-- 0..1 Reference(Any) Specific instance of resource --></what>
  <type><!-- 0..1 Coding Type of entity involved --></type>
  <role><!-- 0..1 Coding What role the entity played --></role>
  <lifecycle><!-- 0..1 Coding Life-cycle stage for the entity --></lifecycle>
  <securityLabel><!-- 0..* Coding Security labels on the entity --></securityLabel>
  <name value="[string]"/><!--  0..1 Descriptor for entity -->
  <description value="[string]"/><!-- 0..1 Descriptive text -->
  <query value="[base64Binary]"/><!--  0..1 Query parameters -->
  <detail>  <!-- 0..* Additional Information about the entity -->
   <type value="[string]"/><!-- 1..1 Name of the property -->
   <value[x]><!-- 1..1 string|base64Binary Property value --></value[x]>
  </detail>
 </entity>
</AuditEvent>
```

JSON Template

```
{
  "resourceType" : "AuditEvent",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "type" : { Coding }, // R!  Type/identifier of event
  "subtype" : [{ Coding }], // More specific type/id for the event
  "action" : "<code>", // Type of action performed during the event
  "period" : { Period }, // When the activity occurred
  "recorded" : "<instant>", // R!  Time when the event was recorded
  "outcome" : "<code>", // Whether the event succeeded or failed
  "outcomeDesc" : "<string>", // Description of the event outcome
  "purposeOfEvent" : [{ CodeableConcept }], // The purposeOfUse of the event
  "agent" : [{ // R!  Actor involved in the event
    "type" : { CodeableConcept }, // How agent participated
    "role" : [{ CodeableConcept }], // Agent role in the event
    "who" : { Reference(PractitionerRole|Practitioner|Organization|Device|
    Patient|RelatedPerson) }, // Identifier of who
    "altId" : "<string>", // Alternative User identity
    "name" : "<string>", // Human friendly name for the agent
    "requestor" : <boolean>, // R!  Whether user is initiator
    "location" : { Reference(Location) }, // Where
    "policy" : ["<uri>"], // Policy that authorized event
    "media" : { Coding }, // Type of media
    "network" : { // Logical network location for application activity
      "address" : "<string>", // Identifier for the network access point of the user device
      "type" : "<code>" // The type of network access point
    },
    "purposeOfUse" : [{ CodeableConcept }] // Reason given for this user
  }],
  "source" : { // R!  Audit Event Reporter
    "site" : "<string>", // Logical source location within the enterprise
    "observer" : { Reference(PractitionerRole|Practitioner|Organization|Device|
    Patient|RelatedPerson) }, // R!  The identity of source detecting the event
    "type" : [{ Coding }] // The type of source where event originated
  },
  "entity" : [{ // Data or objects used
    "what" : { Reference(Any) }, // Specific instance of resource
    "type" : { Coding }, // Type of entity involved
    "role" : { Coding }, // What role the entity played
    "lifecycle" : { Coding }, // Life-cycle stage for the entity
    "securityLabel" : [{ Coding }], // Security labels on the entity
    "name" : "<string>", // C? Descriptor for entity
    "description" : "<string>", // Descriptive text
    "query" : "<base64Binary>", // C? Query parameters
    "detail" : [{ // Additional Information about the entity
      "type" : "<string>", // R!  Name of the property
      // value[x]: Property value. One of these 2:
      "valueString" : "<string>"
      "valueBase64Binary" : "<base64Binary>"
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:AuditEvent;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root
```

```
  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:AuditEvent.type [ Coding ]; # 1..1 Type/identifier of event
  fhir:AuditEvent.subtype [ Coding ], ... ; # 0..* More specific type/id for the event
  fhir:AuditEvent.action [ code ]; # 0..1 Type of action performed during the event
  fhir:AuditEvent.period [ Period ]; # 0..1 When the activity occurred
  fhir:AuditEvent.recorded [ instant ]; # 1..1 Time when the event was recorded
  fhir:AuditEvent.outcome [ code ]; # 0..1 Whether the event succeeded or failed
  fhir:AuditEvent.outcomeDesc [ string ]; # 0..1 Description of the event outcome
  fhir:AuditEvent.purposeOfEvent [ CodeableConcept ], ... ; # 0..* The purposeOfUse of the event
  fhir:AuditEvent.agent [ # 1..* Actor involved in the event
    fhir:AuditEvent.agent.type [ CodeableConcept ]; # 0..1 How agent participated
    fhir:AuditEvent.agent.role [ CodeableConcept ], ... ; # 0..* Agent role in the event
    fhir:AuditEvent.agent.who [ Reference(PractitionerRole|Practitioner|Organization|Device|Patient|RelatedPerson) ]; #
    0..1 Identifier of who
    fhir:AuditEvent.agent.altId [ string ]; # 0..1 Alternative User identity
    fhir:AuditEvent.agent.name [ string ]; # 0..1 Human friendly name for the agent
    fhir:AuditEvent.agent.requestor [ boolean ]; # 1..1 Whether user is initiator
    fhir:AuditEvent.agent.location [ Reference(Location) ]; # 0..1 Where
    fhir:AuditEvent.agent.policy [ uri ], ... ; # 0..* Policy that authorized event
    fhir:AuditEvent.agent.media [ Coding ]; # 0..1 Type of media
    fhir:AuditEvent.agent.network [ # 0..1 Logical network location for application activity
      fhir:AuditEvent.agent.network.address [ string ]; # 0..1 Identifier for the network access point of the user
    device
      fhir:AuditEvent.agent.network.type [ code ]; # 0..1 The type of network access point
    ];
    fhir:AuditEvent.agent.purposeOfUse [ CodeableConcept ], ... ; # 0..* Reason given for this user
  ], ...;
  fhir:AuditEvent.source [ # 1..1 Audit Event Reporter
    fhir:AuditEvent.source.site [ string ]; # 0..1 Logical source location within the enterprise
    fhir:AuditEvent.source.observer [ Reference(PractitionerRole|Practitioner|Organization|Device|Patient|RelatedPerson)
     ]; # 1..1 The identity of source detecting the event
    fhir:AuditEvent.source.type [ Coding ], ... ; # 0..* The type of source where event originated
  ];
  fhir:AuditEvent.entity [ # 0..* Data or objects used
    fhir:AuditEvent.entity.what [ Reference(Any) ]; # 0..1 Specific instance of resource
    fhir:AuditEvent.entity.type [ Coding ]; # 0..1 Type of entity involved
    fhir:AuditEvent.entity.role [ Coding ]; # 0..1 What role the entity played
    fhir:AuditEvent.entity.lifecycle [ Coding ]; # 0..1 Life-cycle stage for the entity
    fhir:AuditEvent.entity.securityLabel [ Coding ], ... ; # 0..* Security labels on the entity
    fhir:AuditEvent.entity.name [ string ]; # 0..1 Descriptor for entity
    fhir:AuditEvent.entity.description [ string ]; # 0..1 Descriptive text
    fhir:AuditEvent.entity.query [ base64Binary ]; # 0..1 Query parameters
    fhir:AuditEvent.entity.detail [ # 0..* Additional Information about the entity
      fhir:AuditEvent.entity.detail.type [ string ]; # 1..1 Name of the property
      # AuditEvent.entity.detail.value[x] : 1..1 Property value. One of these 2
        fhir:AuditEvent.entity.detail.valueString [ string ]
        fhir:AuditEvent.entity.detail.valueBase64Binary [ base64Binary ]
    ], ...;
  ], ...;
]
```

Changes since Release 3

AuditEvent

AuditEvent.action | Change value set from http://hl7.org/fhir/ValueSet/audit-event-action to http://hl7.org/fhir/ValueSet/audit-event-action|4.0.1

AuditEvent.period | Added Element

AuditEvent.outcome | Change value set from http://hl7.org/fhir/ValueSet/audit-event-outcome to http://hl7.org/fhir/ValueSet/audit-event-outcome|4.0.1

AuditEvent.agent.type | Added Element

AuditEvent.agent.role | Remove Binding http://hl7.org/fhir/ValueSet/security-role-type (extensible)

AuditEvent.agent.who | Added Element

AuditEvent.agent.network.type | Change value set from http://hl7.org/fhir/ValueSet/network-type to http://hl7.org/fhir/ValueSet/network-type|4.0.1

AuditEvent.source.observer | Added Mandatory Element

AuditEvent.source.type | Change code system for extensibly bound codes from "http://hl7.org/fhir/security-source-type" to "http://terminology.hl7.org/CodeSystem/security-source-type"

AuditEvent.entity.what | Added Element

AuditEvent.entity.role | Change code system for extensibly bound codes from "http://hl7.org/fhir/object-role" to "http://terminology.hl7.org/CodeSystem/object-role"

AuditEvent.entity.detail.value[x] | Renamed from value to value[x] Add Type string

AuditEvent.agent.reference | deleted

AuditEvent.agent.userId | deleted

AuditEvent.source.identifier | deleted

AuditEvent.entity.identifier | deleted

AuditEvent.entity.reference | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 8 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

6.4.3.1 Terminology Bindings

Path | Definition | Type | Reference
AuditEvent.type | Type of event. | Extensible | AuditEventID
AuditEvent.subtype | Sub-type of event. | Extensible | AuditEventSub-Type
AuditEvent.action | Indicator for type of action performed during the event that generated the event. | Required | AuditEventAction
AuditEvent.outcome | Indicates whether the event succeeded or failed. | Required | AuditEventOutcome
AuditEvent.purposeOfEvent AuditEvent.agent.purposeOfUse | The reason the activity took place. | Extensible | v3.PurposeOfUse
AuditEvent.agent.type | The Participation type of the agent to the event. | Extensible | ParticipationRoleType
AuditEvent.agent.role | What security role enabled the agent to participate in the event. | Example | SecurityRoleType
AuditEvent.agent.media | Used when the event is about exporting/importing onto media. | Extensible | MediaTypeCode
AuditEvent.agent.network.type | The type of network access point of this agent in the audit event. | Required | AuditEventAgentNetworkType
AuditEvent.source.type | Code specifying the type of system that detected and recorded the event. | Extensible | AuditEventSourceType
AuditEvent.entity.type | Code for the entity type involved in the audit event. | Extensible | AuditEventEntityType
AuditEvent.entity.role | Code representing the role the entity played in the audit event. | Extensible | AuditEventEntityRole
AuditEvent.entity.lifecycle | Identifier for the data life-cycle stage for the entity. | Extensible | ObjectLifecycleEvents
AuditEvent.entity.securityLabel | Security Labels from the Healthcare Privacy and Security Classification System. | Extensible | All Security Labels

6.4.3.2 Constraints

id | Level | Location | Description | Expression
sev-1 | Rule | AuditEvent.entity | Either a name or a query (NOT both) | name.empty() or query.empty()

6.4.3.3 Using Coded Values

The AuditEvent resource and the ATNA Audit record are used in many contexts throughout healthcare. The coded values defined in the "extensible" bindings above are those widely used and/or defined by DICOM, IHE or ISO, who defined these codes to meet very specific use cases. These codes should be used when they are suitable. When needed, other codes can be defined.

Note: When using codes from a vocabulary, the display element for the code can be left off to keep the AuditEvent size small and minimize impact of a large audit log of similar entries.

The set of codes defined for this resource is expected to grow over time, and additional codes may be proposed / requested using the "Propose a change" link above below.

6.4.3.4 Event codes for Common Scenarios

This table summarizes common event scenarios, and the codes that should be used for each case.

Scenario | type | subtype | action | Other
User Login (example) | 110114 User Authentication | 110122 User Authentication | E Execute | One agent which contains the details of the logged-in user.
User Logout (example) | 110114 User Authentication | 110123 User Logout | E Execute | One agent which contains the details of the logged-out user.
REST operation logged on server (example) | rest RESTful Operation | [code] defined for operation | * (see below) | Agent for logged in user, if available.
Search operation logged on server (example) | rest RESTful Operation | [code] defined for operation | E Execute | Agent for logged in user, if available, and one object with a query element.

Audit Event Actions for RESTful operations:

Operation | Action
create | C
read, vread, history-instance, history-type, history-system | R
update | U
delete | D
transaction, operation, conformance, validate, search, search-type, search-system | E

6.4.3.5 Encoding a FHIR operation outcome

FHIR interactions can result in a rich description of the outcome using the OperationOutcome. The OperationOutcome Resource is a collection of error, warning or information messages that result from a system action. This describes in detail the outcome of some operation, such as when a RESTful operation fails.

When recording into an AuditEvent that some FHIR interaction has happened, the AuditEvent should include the OperationOutcome from that FHIR interaction. This is done by placing the OperationOutcome into an AuditEvent.entity. Likely as a contained resource, given that OperationOutcome resources often are not persisted.

entity.who is the OperationOutcome -- Likely contained

entity.type is code OperationOutcome

entity.description explains why this OperationOutcome was included.

See transaction failure example: When a client attempts to post (create) an Observation Resource, using a server Patient endpoint; this would result in an error with an OperationOutcome.

6.4.3.6 PurposeOfEvent and PurposeOfUse

The AuditEvent provides the element purposeOfEvent to convey the purpose of the event and purposeOfUse to convey the reason that a particular actor (machine, person, software) was involved in the event.

purposeOfEvent is an element at the level of AuditEvent and can convey the purpose of the activity that resulted in the event. This will occur when the system that is reporting the event is aware of the purpose of the event. A specific example would be a radiology reporting system where a radiologist has created and is sending a finished report. This system likely knows the purpose, e.g., "treatment". It is multi-valued because the one event may be related to multiple purposes.

It is also commonplace that the reporting system does not have information about the purpose of the event. In these cases, the event report would not have a purposeOfEvent.

It is also likely that the same event will be reported from different perspectives, e.g., by both the sender and recipient of a communication. These two different perspectives can have different knowledge regarding the purposeOfEvent.

purposeOfUse is an element at the level of agent within AuditEvent. This describes the reason that this person, machine, or software is participating in the activity that resulted in the event. For example, an individual person participating in the event may assert a purpose of use from their perspective. It is also possible that they are participating for multiple reasons and report multiple purposeOfUse.

The reporting system might not have knowledge regarding why a particular machine or person was involved and would omit this element in those cases.

When the same event is reported from multiple perspectives, the reports can have different knowledge regarding the purpose.

6.4.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
action | token | Type of action performed during the event | AuditEvent.action
address | string | Identifier for the network access point of the user device | AuditEvent.agent.network.address
agent | reference | Identifier of who | AuditEvent.agent.who (Practitioner, Organization, Device, Patient, PractitionerRole, RelatedPerson)
agent-name | string | Human friendly name for the agent | AuditEvent.agent.name
agent-role | token | Agent role in the event | AuditEvent.agent.role
altid | token | Alternative User identity | AuditEvent.agent.altId
date | date | Time when the event was recorded | AuditEvent.recorded
entity | reference | Specific instance of resource | AuditEvent.entity.what (Any)
entity-name | string | Descriptor for entity | AuditEvent.entity.name
entity-role | token | What role the entity played | AuditEvent.entity.role
entity-type | token | Type of entity involved | AuditEvent.entity.type
outcome | token | Whether the event succeeded or failed | AuditEvent.outcome
patient | reference | Identifier of who | AuditEvent.agent.who.where(resolve() is Patient) | AuditEvent.entity.what.where(resolve() is Patient) (Patient)
policy | uri | Policy that authorized event | AuditEvent.agent.policy
site | token | Logical source location within the enterprise | AuditEvent.source.site
source | reference | The identity of source detecting the event | AuditEvent.source.observer (Practitioner, Organization, Device, Patient, PractitionerRole, RelatedPerson)
subtype | token | More specific type/id for the event | AuditEvent.subtype
type | token | Type/identifier of event | AuditEvent.type

6.4.5 Resource AuditEvent - Examples

URL: https://hl7.org/fhir/R4/auditevent-examples.html

6.4.5 Resource AuditEvent - Examples

Security Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Patient, Practitioner

Example Name | id | Format
General AuditEvent Example | example | XML | JSON | Turtle
Login example | example-login | XML | JSON | Turtle
Logout example | example-logout | XML | JSON | Turtle
RESTful vread Operation | example-rest | XML | JSON | Turtle
Accounting of a Disclosure | example-disclosure | XML | JSON | Turtle
RESTful search operation | example-search | XML | JSON | Turtle
Converted ATNA message from a PIX query | example-pixQuery | XML | JSON | Turtle
Export of DocumentManifest to Media | example-media | XML | JSON | Turtle

Audit of a transaction that was failed resulting in OperationOutcome | example-error | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

6.4.6 Resource AuditEvent - Detailed Descriptions

URL: https://hl7.org/fhir/R4/auditevent-definitions.html

6.4.6 Resource AuditEvent - Detailed Descriptions

Security Work Group | Maturity Level: 3 | Trial Use | Security Category: Not Classified | Compartments: Device, Patient, Practitioner

Detailed Descriptions for the elements in the AuditEvent resource.

AuditEvent
Element Id | AuditEvent
Definition | A record of an event made for purposes of maintaining a security log. Typical uses include detection of intrusion attempts and monitoring for inappropriate usage.
Cardinality | 0..*
Type | DomainResource
Comments | Based on IHE-ATNA.
AuditEvent.type
Element Id | AuditEvent.type
Definition | Identifier for a family of the event. For example, a menu item, program, rule, policy, function code, application name or URL. It identifies the performed function.
Cardinality | 1..1
Terminology Binding | Audit Event ID (Extensible)
Type | Coding
Requirements | This identifies the performed function. For "Execute" Event Action Code audit records, this identifies the application function performed.
Summary | true
AuditEvent.subtype
Element Id | AuditEvent.subtype
Definition | Identifier for the category of event.
Cardinality | 0..*
Terminology Binding | Audit Event Sub-Type (Extensible)
Type | Coding
Requirements | This field enables queries of messages by implementation-defined event categories.
Summary | true
AuditEvent.action
Element Id | AuditEvent.action
Definition | Indicator for type of action performed during the event that generated the audit.
Cardinality | 0..1
Terminology Binding | AuditEventAction (Required)
Type | code
Requirements | This broadly indicates what kind of action was done on the AuditEvent.entity by the AuditEvent.agent.
Summary | true
AuditEvent.period
Element Id | AuditEvent.period
Definition | The period during which the activity occurred.
Cardinality | 0..1
Type | Period
Comments | The period can be a little arbitrary; where possible, the time should correspond to human assessment of the activity time.
AuditEvent.recorded
Element Id | AuditEvent.recorded
Definition | The time when the event was recorded.
Cardinality | 1..1
Type | instant
Requirements | This ties an event to a specific date and time. Security audits typically require a consistent time base (e.g. UTC), to eliminate time-zone issues arising from geographical distribution.
Summary | true
Comments | In a distributed system, some sort of common time base (e.g. an NTP [RFC1305] server) is a good implementation tactic.
AuditEvent.outcome
Element Id | AuditEvent.outcome
Definition | Indicates whether the event succeeded or failed.
Cardinality | 0..1
Terminology Binding | AuditEventOutcome (Required)
Type | code
Summary | true
Comments | In some cases a "success" may be partial, for example, an incomplete or interrupted transfer of a radiological study. For the purpose of establishing accountability, these distinctions are not relevant.
AuditEvent.outcomeDesc

Element Id | AuditEvent.outcomeDesc
Definition | A free text description of the outcome of the event.
Cardinality | 0..1
Type | string
Summary | true
AuditEvent.purposeOfEvent
Element Id | AuditEvent.purposeOfEvent
Definition | The purposeOfUse (reason) that was used during the event being recorded.
Cardinality | 0..*
Terminology Binding | V3 Value SetPurposeOfUse (Extensible)
Type | CodeableConcept
Summary | true
Comments | Use AuditEvent.agent.purposeOfUse when you know that it is specific to the agent, otherwise use AuditEvent.purposeOfEvent. For example, during a machine-to-machine transfer it might not be obvious to the audit system who caused the event, but it does know why.
AuditEvent.agent
Element Id | AuditEvent.agent
Definition | An actor taking an active role in the event or activity that is logged.
Cardinality | 1..*
Requirements | An agent can be a person, an organization, software, device, or other actors that may be ascribed responsibility.
Alternate Names | ActiveParticipant
Comments | Several agents may be associated (i.e. have some responsibility for an activity) with an event or activity. For example, an activity may be initiated by one user for other users or involve more than one user. However, only one user may be the initiator/requestor for the activity.
AuditEvent.agent.type
Element Id | AuditEvent.agent.type
Definition | Specification of the participation type the user plays when performing the event.
Cardinality | 0..1
Terminology Binding | ParticipationRoleType (Extensible)
Type | CodeableConcept
AuditEvent.agent.role
Element Id | AuditEvent.agent.role
Definition | The security role that the user was acting under, that come from local codes defined by the access control security system (e.g. RBAC, ABAC) used in the local context.
Cardinality | 0..*
Terminology Binding | SecurityRoleType (Example)
Type | CodeableConcept
Requirements | This value ties an audited event to a user's role(s). It is an optional value that might be used to group events for analysis by user functional role categories.
Comments | Should be roles relevant to the event. Should not be an exhaustive list of roles.
AuditEvent.agent.who
Element Id | AuditEvent.agent.who
Definition | Reference to who this agent is that was involved in the event.
Cardinality | 0..1
Type | Reference(PractitionerRole | Practitioner | Organization | Device | Patient | RelatedPerson)
Requirements | This field ties an audit event to a specific resource or identifier.
Alternate Names | userId
Summary | true
Comments | Where a User ID is available it will go into who.identifier.
AuditEvent.agent.altId
Element Id | AuditEvent.agent.altId
Definition | Alternative agent Identifier. For a human, this should be a user identifier text string from authentication system. This identifier would be one known to a common authentication system (e.g. single sign-on), if available.
Cardinality | 0..1
Type | string
Requirements | In some situations, a human user may authenticate with one identity but, to access a specific application system, may use a synonymous identify. For example, some "single sign on" implementations will do this. The alternative identifier would then be the original identify used for authentication, and the User ID is the one known to and used by the application.
AuditEvent.agent.name
Element Id | AuditEvent.agent.name
Definition | Human-meaningful name for the agent.
Cardinality | 0..1
Type | string
Requirements | The User ID and Authorization User ID may be internal or otherwise obscure values. This field assists the auditor in identifying the actual user.
AuditEvent.agent.requestor
Element Id | AuditEvent.agent.requestor
Definition | Indicator that the user is or is not the requestor, or initiator, for the event being audited.
Cardinality | 1..1
Type | boolean
Requirements | This value is used to distinguish between requestor-users and recipient-users. For example, one person may initiate a report-output to be sent to another user.
Summary | true

Comments | There can only be one initiator. If the initiator is not clear, then do not choose any one agent as the initiator.
AuditEvent.agent.location
Element Id | AuditEvent.agent.location
Definition | Where the event occurred.
Cardinality | 0..1
Type | Reference(Location)
AuditEvent.agent.policy
Element Id | AuditEvent.agent.policy
Definition | The policy or plan that authorized the activity being recorded. Typically, a single activity may have multiple applicable policies, such as patient consent, guarantor funding, etc. The policy would also indicate the security token used.
Cardinality | 0..*
Type | uri
Requirements | This value is used retrospectively to determine the authorization policies.
Comments | For example: Where an OAuth token authorizes, the unique identifier from the OAuth token is placed into the policy element Where a policy engine (e.g. XACML) holds policy logic, the unique policy identifier is placed into the policy element.
AuditEvent.agent.media
Element Id | AuditEvent.agent.media
Definition | Type of media involved. Used when the event is about exporting/importing onto media.
Cardinality | 0..1
Terminology Binding | Media Type Code (Extensible)
Type | Coding
Requirements | Usually, this is used instead of specifying a network address. This field is not used for Media Id (i.e. the serial number of a CD).
AuditEvent.agent.network
Element Id | AuditEvent.agent.network
Definition | Logical network location for application activity, if the activity has a network location.
Cardinality | 0..1
AuditEvent.agent.network.address
Element Id | AuditEvent.agent.network.address
Definition | An identifier for the network access point of the user device for the audit event.
Cardinality | 0..1
Type | string
Requirements | This datum identifies the user's network access point, which may be distinct from the server that performed the action. It is an optional value that may be used to group events recorded on separate servers for analysis of a specific network access point's data access across all servers.
Comments | This could be a device id, IP address or some other identifier associated with a device.
AuditEvent.agent.network.type
Element Id | AuditEvent.agent.network.type
Definition | An identifier for the type of network access point that originated the audit event.
Cardinality | 0..1
Terminology Binding | AuditEventAgentNetworkType (Required)
Type | code
Requirements | This datum identifies the type of network access point identifier of the user device for the audit event. It is an optional value that may be used to group events recorded on separate servers for analysis of access according to a network access point's type.
AuditEvent.agent.purposeOfUse
Element Id | AuditEvent.agent.purposeOfUse
Definition | The reason (purpose of use), specific to this agent, that was used during the event being recorded.
Cardinality | 0..*
Terminology Binding | V3 Value SetPurposeOfUse (Extensible)
Type | CodeableConcept
Comments | Use AuditEvent.agent.purposeOfUse when you know that is specific to the agent, otherwise use AuditEvent.purposeOfEvent. For example, during a machine-to-machine transfer it might not be obvious to the audit system who caused the event, but it does know why.
AuditEvent.source
Element Id | AuditEvent.source
Definition | The system that is reporting the event.
Cardinality | 1..1
Requirements | The event is reported by one source.
Comments | Since multi-tier, distributed, or composite applications make source identification ambiguous, this collection of fields may repeat for each application or process actively involved in the event. For example, multiple value-sets can identify participating web servers, application processes, and database server threads in an n-tier distributed application. Passive event participants (e.g. low-level network transports) need not be identified.
AuditEvent.source.site
Element Id | AuditEvent.source.site
Definition | Logical source location within the healthcare enterprise network. For example, a hospital or other provider location within a multi-entity provider group.
Cardinality | 0..1
Type | string
Requirements | This value differentiates among the sites in a multi-site enterprise health information system.
AuditEvent.source.observer
Element Id | AuditEvent.source.observer
Definition | Identifier of the source where the event was detected.

Cardinality | 1..1
Type | Reference(PractitionerRole | Practitioner | Organization | Device | Patient | RelatedPerson)
Requirements | This field ties the event to a specific source system. It may be used to group events for analysis according to where the event was detected.
Alternate Names | SourceId
Summary | true
AuditEvent.source.type
Element Id | AuditEvent.source.type
Definition | Code specifying the type of source where event originated.
Cardinality | 0..*
Terminology Binding | Audit Event Source Type (Extensible)
Type | Coding
Requirements | This field indicates which type of source is identified by the Audit Source ID. It is an optional value that may be used to group events for analysis according to the type of source where the event occurred.
AuditEvent.entity
Element Id | AuditEvent.entity
Definition | Specific instances of data or objects that have been accessed.
Cardinality | 0..*
Requirements | The event may have other entities involved.
Alternate Names | ParticipantObject
Comments | Required unless the values for event identification, agent identification, and audit source identification are sufficient to document the entire auditable event. Because events may have more than one entity, this group can be a repeating set of values.
Invariants | Defined on this element sev-1 Rule Either a name or a query (NOT both) name.empty() or query.empty()
Defined on this element
sev-1 | Rule | Either a name or a query (NOT both) | name.empty() or query.empty()
AuditEvent.entity.what
Element Id | AuditEvent.entity.what
Definition | Identifies a specific instance of the entity. The reference should be version specific.
Cardinality | 0..1
Type | Reference(Any)
Summary | true
AuditEvent.entity.type
Element Id | AuditEvent.entity.type
Definition | The type of the object that was involved in this audit event.
Cardinality | 0..1
Terminology Binding | Audit event entity type (Extensible)
Type | Coding
Requirements | To describe the object being acted upon. In addition to queries on the subject of the action in an auditable event, it is also important to be able to query on the object type for the action.
Comments | This value is distinct from the user's role or any user relationship to the entity.
AuditEvent.entity.role
Element Id | AuditEvent.entity.role
Definition | Code representing the role the entity played in the event being audited.
Cardinality | 0..1
Terminology Binding | AuditEventEntityRole (Extensible)
Type | Coding
Requirements | For some detailed audit analysis it may be necessary to indicate a more granular type of entity, based on the application role it serves.
AuditEvent.entity.lifecycle
Element Id | AuditEvent.entity.lifecycle
Definition | Identifier for the data life-cycle stage for the entity.
Cardinality | 0..1
Terminology Binding | ObjectLifecycleEvents (Extensible)
Type | Coding
Requirements | Institutional policies for privacy and security may optionally fall under different accountability rules based on data life cycle. This provides a differentiating value for those cases.
Comments | This can be used to provide an audit trail for data, over time, as it passes through the system.
AuditEvent.entity.securityLabel
Element Id | AuditEvent.entity.securityLabel
Definition | Security labels for the identified entity.
Cardinality | 0..*
Terminology Binding | SecurityLabels (Extensible)
Type | Coding
Requirements | This field identifies the security labels for a specific instance of an object, such as a patient, to detect/track privacy and security issues.
Comments | Copied from entity meta security tags.
AuditEvent.entity.name
Element Id | AuditEvent.entity.name
Definition | A name of the entity in the audit event.
Cardinality | 0..1
Type | string
Requirements | Use only where entity can't be identified with an identifier.

Summary | true
Comments | This field may be used in a query/report to identify audit events for a specific person. For example, where multiple synonymous entity identifiers (patient number, medical record number, encounter number, etc.) have been used.
Invariants | Affect this element sev-1 Rule Either a name or a query (NOT both) name.empty() or query.empty()
Affect this element
sev-1 | Rule | Either a name or a query (NOT both) | name.empty() or query.empty()
AuditEvent.entity.description
Element Id | AuditEvent.entity.description
Definition | Text that describes the entity in more detail.
Cardinality | 0..1
Type | string
Requirements | Use only where entity can't be identified with an identifier.
AuditEvent.entity.query
Element Id | AuditEvent.entity.query
Definition | The query parameters for a query-type entities.
Cardinality | 0..1
Type | base64Binary
Requirements | For query events, it may be necessary to capture the actual query input to the query process in order to identify the specific event. Because of differences among query implementations and data encoding for them, this is a base 64 encoded data blob. It may be subsequently decoded or interpreted by downstream audit analysis processing.
Summary | true
Comments | The meaning and secondary-encoding of the content of base64 encoded blob is specific to the AuditEvent.type, AuditEvent.subtype, AuditEvent.entity.type, and AuditEvent.entity.role. The base64 is a general-use and safe container for event specific data blobs regardless of the encoding used by the transaction being recorded. An AuditEvent consuming application must understand the event it is consuming and the formats used by the event. For example, if auditing an Oracle network database access, the Oracle formats must be understood as they will be simply encoded in the base64binary blob.
Invariants | Affect this element sev-1 Rule Either a name or a query (NOT both) name.empty() or query.empty()
Affect this element
sev-1 | Rule | Either a name or a query (NOT both) | name.empty() or query.empty()
AuditEvent.entity.detail
Element Id | AuditEvent.entity.detail
Definition | Tagged value pairs for conveying additional information about the entity.
Cardinality | 0..*
Requirements | Implementation-defined data about specific details of the object accessed or used.
AuditEvent.entity.detail.type
Element Id | AuditEvent.entity.detail.type
Definition | The type of extra detail provided in the value.
Cardinality | 1..1
Type | string
AuditEvent.entity.detail.value[x]
Element Id | AuditEvent.entity.detail.value[x]
Definition | The value of the extra detail.
Cardinality | 1..1
Type | string|base64Binary
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Should not duplicate the entity value unless absolutely necessary.
Comments | The value can be string when known to be a string, else base64 encoding should be used to protect binary or undefined content. The meaning and secondary-encoding of the content of base64 encoded blob is specific to the AuditEvent.type, AuditEvent.subtype, AuditEvent.entity.type, and AuditEvent.entity.role. The base64 is a general-use and safe container for event specific data blobs regardless of the encoding used by the transaction being recorded. An AuditEvent consuming application must understand the event it is consuming and the formats used by the event. For example if auditing an Oracle network database access, the Oracle formats must be understood as they will be simply encoded in the base64binary blob.

6.4.7 Resource AuditEvent - Mappings

URL: https://hl7.org/fhir/R4/auditevent-mappings.html

6.4.7 Resource AuditEvent - Mappings

Security Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Patient, Practitioner

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

DICOM Tag Mapping

W3C PROV

FHIR Provenance Mapping

Mappings for the auditevent resource (see Mappings to Other Standards for further information & status).

6.4.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

AuditEvent
type | FiveWs.what[x]
subtype | FiveWs.what[x]
action | FiveWs.what[x]
period | FiveWs.done[x]
recorded | FiveWs.recorded
outcome | FiveWs.what[x]
outcomeDesc | FiveWs.what[x]
purposeOfEvent | FiveWs.why[x]
agent | FiveWs.who
type | FiveWs.who
role | FiveWs.who
who | FiveWs.who
altId | FiveWs.who
name | FiveWs.who
requestor | FiveWs.who
location | FiveWs.where[x]
policy | FiveWs.why[x]
media | FiveWs.where[x]
network | FiveWs.where[x]
address | FiveWs.where[x]
type | FiveWs.where[x]
purposeOfUse | FiveWs.why[x]
source | FiveWs.witness
site | FiveWs.witness
observer | FiveWs.witness
type | FiveWs.witness
entity | FiveWs.what[x]
what | FiveWs.what[x]
type | FiveWs.what[x]
role | FiveWs.context
lifecycle | FiveWs.context
securityLabel | FiveWs.context
name | FiveWs.context
description | FiveWs.context
query | FiveWs.context
detail | FiveWs.context
type | FiveWs.context
value[x] | FiveWs.context

6.4.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

AuditEvent | Event
type | Event.code
period | Event.occurred[x]
purposeOfEvent | Event.reasonCode
agent | Event.performer
type | Event.performer.function
who | Event.performer.actor
location | Event.location

6.4.7.3 RIM Mapping (http://hl7.org/v3)

AuditEvent | ControlAct[moodCode=EVN]
type | .code (type, subtype and action are pre-coordinated or sent as translations)
subtype | .code (type, subtype and action are pre-coordinated or sent as translations)
action | .code (type, subtype and action are pre-coordinated or sent as translations)
period | ./effectiveTime[type=IVL_TS]
recorded | .effectiveTime
outcome | .actionNegationInd
outcomeDesc | .outboundRelationship[typeCode=OUT].target.text
purposeOfEvent | * .reasonCode [ControlActReason when Act.class = CACT Control Act]
*.outboundRelationship[typeCode=RSON].target
agent | .participation
type | .typeCode and/or .functionCode
role | .role
who | .id
altId | .id (distinguish id type by root)
name | .name
requestor | If participation.typeCode was author, then true
location | * Role.Class =SDLOC *Role.Code = ServiceDeliveryLocationRoleType *Entity.Code = PlaceEntityType = df.Types of places for Entity.Class = PLC *EntityClass = PLC = df.A physical place or site with its containing structure. May be natural or man-made. The geographic position of a place might or might not be constant.
policy | ActPolicyType

media | .player.description.mediaType
network | .player.description.reference
address | pre-coordinated into URL
type | pre-coordinated into URL
purposeOfUse | *.reasonCode [ActHealthInformationPurposeOfUseReason codes/v:PurposeOfUse (2.16.840.1.113883.1.11.20448)
* .outboundRelationship[typeCode=RSON or SUBJ].target
source | .participation[typeCode=INF].role[classCode=ASSIGN].player[classCode=DEV, determinerCode=INSTANCE]
site | .scopedRole[classCode=LOCE].player.desc
observer | .id
type | .code
entity | .outboundRelationship[typeCode=SUBJ].target or .participation[typeCode=SBJ].role
what | .id
type | [self::Act].code or role.player.code
role | role.code (not sure what this would mean for an Act)
lifecycle | target of ObservationEvent[code="lifecycle"].value
securityLabel | .confidentialityCode
name | .title
description | .text
query | No mapping
detail | .inboundRelationship[typeCode=SUBJ].target[classCode=OBS, moodCode=EVN]
type | .code
value[x] | .value

6.4.7.4 DICOM Tag Mapping (http://nema.org/dicom)

AuditEvent | Message
type | EventId
subtype | EventTypeCode
action | EventActionCode
period | EventDateTime
recorded
outcome | EventOutcomeIndicator
outcomeDesc | EventOutcomeDescription
purposeOfEvent | EventPurposeOfUse
agent | ActiveParticipant
type | RoleIdCode
role | RoleIdCode
who | UserId
altId | AlternativeUserId
name | UserName
requestor | UserIsRequestor
location
policy | ParticipantRoleIDCode
media | MediaType
network
address | NetworkAccessPointID
type | NetworkAccessPointTypeCode
purposeOfUse
source | AuditSourceIdentification
site | AuditEnterpriseSiteId
observer | AuditSourceId
type | AuditSourceTypeCode
entity | ParticipantObjectIdentification
what | ParticipantObjectID and ParticipantObjectIDTypeCode
type | ParticipantObjectTypeCode
role | ParticipantObjectTypeCodeRole
lifecycle | ParticipantObjectDataLifeCycle
securityLabel | ParticipantObjectSensitivity
name | ParticipantObjectName
description | ParticipantObjectDescription
query | ParticipantObjectQuery
detail | ParticipantObjectDetail
type | ParticipantObjectDetail.type
value[x] | ParticipantObjectDetail.value

6.4.7.5 W3C PROV (http://www.w3.org/ns/prov)

The provenance resource is based on known practices in the HL7 implementation space, particularly those found in the v2 EVN segment, the v3 ControlAct Wrapper, the CDA header, and IHE ATNA. The conceptual model underlying the design is the W3C provenance Specification . Though the content and format of the resource is designed to meet specific requirements for FHIR, all the parts of the resource are formally mapped to the PROV-O specification, and FHIR resources can be transformed to their W3C PROV equivalent.

AuditEvent

type | Activity
subtype
action
period | Activity.startTime & Activity.endTime
recorded | Activity.when
outcome
outcomeDesc
purposeOfEvent | Activity.Activity
agent | Agent
type | Agent.Attribution
role | Agent.Attribution
who | Agent.Identity
altId | Agent.Identity
name | Agent.Identity
requestor
location | Activity.location
policy
media
network
address | Agent.Location
type
purposeOfUse | Agent.Activity
source
site
observer
type
entity | Entity
what
type | Entity.type
role | Entity.role
lifecycle | Entity.role
securityLabel
name | Entity.Label
description
query
detail
type
value[x]

6.4.7.6 FHIR Provenance Mapping (http://hl7.org/fhir/provenance)

AuditEvent
type
subtype
action
period | Provenance.occurred[x]
recorded | Provenance.recorded
outcome
outcomeDesc
purposeOfEvent | Provenance.reason, Provenance.activity
agent | Provenance.agent
type | Provenance.agent.type
role | Provenance.agent.role
who | Provenance.agent.who
altId
name
requestor
location | Provenance.location
policy | Provenance.policy
media
network
address
type
purposeOfUse
source
site
observer
type
entity | Provenance.target, Provenance.entity
what | Provenance.target, Provenance.entity.what
type | Provenance.entity.type
role
lifecycle | Provenance.entity.role

securityLabel
name
description
query
detail
type
value[x]

6.4.8 Resource AuditEvent - Extensions & Profiles

URL: https://hl7.org/fhir/R4/auditevent-profiles.html

6.4.8 Resource AuditEvent - Extensions & Profiles

Security Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Patient, Practitioner

This table lists profiles and extensions for the AuditEvent resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

6.4.8.1 Profiles

Profile | Description | Context
ehrsrle-auditevent | Defines the elements to be supported within the AuditEvent resource in order to conform with the Electronic Health Record System Functional Model Record Lifecycle Event standard | EHRS FM Record Lifecycle Event - Audit Event

6.4.8.2 Extensions

auditevent-Accession | AuditEvent.entity | Accession | for AuditEvent HL7 Extensions
auditevent-Anonymized | AuditEvent.entity | Anonymized | for AuditEvent HL7 Extensions
auditevent-Encrypted | AuditEvent.entity | Encrypted | for AuditEvent HL7 Extensions
auditevent-Instance | AuditEvent.entity | Instance | for AuditEvent HL7 Extensions
auditevent-MPPS | AuditEvent.entity | MPPS | for AuditEvent HL7 Extensions
auditevent-NumberOfInstances | AuditEvent.entity | NumberOfInstances | for AuditEvent HL7 Extensions
auditevent-ParticipantObjectContainsStudy | AuditEvent.entity | ParticipantObjectContainsStudy | for AuditEvent HL7 Extensions
auditevent-SOPClass | AuditEvent.entity | SOPClass | for AuditEvent HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

6.4.8.3 Search Extensions

No Search Extensions defined for this resource

6.4.9 Resource AuditEvent - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/auditevent-version-maps.html

No validation errors - all conversions are clean

6.4.9 Resource AuditEvent - R3/R4 Conversions

Security Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Device, Patient, Practitioner

R3 : R4 Conversion maps for AuditEvent.

Functional status for this map: 8 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

6.4.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/AuditEvent3to4" = "R3 to R4 Conversions for AuditEvent"

uses "http://hl7.org/fhir/3.0/StructureDefinition/AuditEvent" alias AuditEventR3 as source
uses "http://hl7.org/fhir/StructureDefinition/AuditEvent" alias AuditEvent as target

imports "http://hl7.org/fhir/StructureMap/*3to4"
```

```
group AuditEvent(source src : AuditEventR3, target tgt : AuditEvent) extends DomainResource <<type+>> {
  src.type -> tgt.type;
  src.subtype -> tgt.subtype;
  src.action -> tgt.action;
  src.recorded -> tgt.recorded;
  src.outcome -> tgt.outcome;
  src.outcomeDesc -> tgt.outcomeDesc;
  src.purposeOfEvent -> tgt.purposeOfEvent;
  src.agent as s -> tgt.agent as t then AuditEventAgent(s, t);
  src.source as s -> tgt.source as t then AuditEventSource(s, t);
  src.entity as s -> tgt.entity as t then AuditEventEntity(s, t);
}

group AuditEventAgent(source src, target tgt) extends BackboneElement {
  src.role -> tgt.role;
  src.reference as vs0 -> tgt.who as vt0 then Reference(vs0, vt0);
  src.userId as vs0 -> tgt.who as vt0 then Identifier2Reference(vs0, vt0);
  src.altId -> tgt.altId;
  src.name -> tgt.name;
  src.requestor -> tgt.requestor;
  src.location -> tgt.location;
  src.policy -> tgt.policy;
  src.media -> tgt.media;
  src.network as s -> tgt.network as t then AuditEventAgentNetwork(s, t);
  src.purposeOfUse -> tgt.purposeOfUse;
}

group AuditEventAgentNetwork(source src, target tgt) extends BackboneElement {
  src.address -> tgt.address;
  src.type -> tgt.type;
}

group AuditEventSource(source src, target tgt) extends BackboneElement {
  src.site -> tgt.site;
  src.identifier as vs0 -> tgt.observer as vt0 then Identifier2Reference(vs0, vt0);
  src.type -> tgt.type;
}

group AuditEventEntity(source src, target tgt) extends BackboneElement {
  src.identifier as vs0 -> tgt.what as vt0 then Identifier2Reference(vs0, vt0);
  src.reference as vs0 -> tgt.what as vt0 then Reference(vs0, vt0);
  src.type -> tgt.type;
  src.role -> tgt.role;
  src.lifecycle -> tgt.lifecycle;
  src.securityLabel -> tgt.securityLabel;
  src.name -> tgt.name;
  src.description -> tgt.description;
  src.query -> tgt.query;
  src.detail as s -> tgt.detail as t then AuditEventEntityDetail(s, t);
}

group AuditEventEntityDetail(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.value as vs0 -> tgt.value = create('base64Binary') as vt0 then base64Binary(vs0, vt0);
}
```

6.4.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/AuditEvent4to3" = "R4 to R3 Conversion for AuditEvent"

uses "http://hl7.org/fhir/StructureDefinition/AuditEvent" alias AuditEvent as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/AuditEvent" alias AuditEventR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group AuditEvent(source src : AuditEventR3, target tgt : AuditEvent) extends DomainResource <<type+>> {
  src.type -> tgt.type;
  src.subtype -> tgt.subtype;
  src.action -> tgt.action;
  src.recorded -> tgt.recorded;
  src.outcome -> tgt.outcome;
  src.outcomeDesc -> tgt.outcomeDesc;
  src.purposeOfEvent -> tgt.purposeOfEvent;
  src.agent as s -> tgt.agent as t then AuditEventAgent(s, t);
  src.source as s -> tgt.source as t then AuditEventSource(s, t);
  src.entity as s -> tgt.entity as t then AuditEventEntity(s, t);
}

group AuditEventAgent(source src, target tgt) extends BackboneElement {
  src.role -> tgt.role;
  src.who as vs0 where reference.exists() -> tgt.reference as vt0 then ReferenceNoIdentifier(vs0, vt0);
  src.who as vs0 where identifier.exists() -> tgt.userId as vt0 then Reference2Identifier(vs0, vt0);
  src.altId -> tgt.altId;
  src.name -> tgt.name;
  src.requestor -> tgt.requestor;
  src.location -> tgt.location;
  src.policy -> tgt.policy;
  src.media -> tgt.media;
  src.network as s -> tgt.network as t then AuditEventAgentNetwork(s, t);
  src.purposeOfUse -> tgt.purposeOfUse;
```

```
}

group AuditEventAgentNetwork(source src, target tgt) extends BackboneElement {
  src.address -> tgt.address;
  src.type -> tgt.type;
}

group AuditEventSource(source src, target tgt) extends BackboneElement {
  src.site -> tgt.site;
  src.observer as vs0 -> tgt.identifier as vt0 then Reference2Identifier(vs0, vt0);
  src.type -> tgt.type;
}

group AuditEventEntity(source src, target tgt) extends BackboneElement {
  src.what as vs0 where identifier.exists() -> tgt.identifier as vt0 then Reference2Identifier(vs0, vt0);
  src.what as vs0 where reference.exists() -> tgt.reference as vt0 then ReferenceNoIdentifier(vs0, vt0);
  src.type -> tgt.type;
  src.role -> tgt.role;
  src.lifecycle -> tgt.lifecycle;
  src.securityLabel -> tgt.securityLabel;
  src.name -> tgt.name;
  src.description -> tgt.description;
  src.query -> tgt.query;
  src.detail as s -> tgt.detail as t then AuditEventEntityDetail(s, t);
}

group AuditEventEntityDetail(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.value where $this is base64Binary -> tgt.value;
}
```

6.4.9.3 R4 Validation Errors

6.4.10 AuditEvent HL7 Extensions

URL: https://hl7.org/fhir/R4/auditevent-extensions.html

Defines common extensions used with or related to the AuditEvent resource

6.4.10 AuditEvent HL7 Extensions

Security Work Group | Maturity Level: N/A | Standards Status: Informative

6.4.10.1 Content

Extensions:

auditevent-SOPClass | SOPClass : Required if ParticipantObjectIDTypeCode is (110180, DCM, "Study Instance UID") and any of the optional fields (AccessionNumber, ContainsMPPS, NumberOfInstances, ContainsSOPInstances,Encrypted,Anonymized) are present in this Participant Object. May be present if ParticipantObjectIDTypeCode is (110180, DCM, "Study Instance UID") even though none of the optional fields are present.

auditevent-Accession | Accession : An Accession Number associated with this participant object.

auditevent-MPPS | MPPS : An MPPS Instance UID associated with this entity.

auditevent-NumberOfInstances | NumberOfInstances : The Number of SOP Instances referred to by this entity.

auditevent-Instance | Instance : Th SOP Instance UID values.

auditevent-Encrypted | Encrypted : True or False indicating whether the data was encrypted.

auditevent-Anonymized | Anonymized : True or False indicating whether all patient identifying information was removed from the data.

auditevent-ParticipantObjectContainsStudy | ParticipantObjectContainsStudy : A Study Instance ID, which may be used when the Entity type is not (110180, DCM, "Study Instance UID").

6.4.11 JSON Schema for AuditEvent

URL: https://hl7.org/fhir/R4/auditevent.schema.json.html

6.4.11 JSON Schema for AuditEvent

Security Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient, Practitioner

This is a representation of the json schema for AuditEvent, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/AuditEvent",
  "$ref": "#/definitions/AuditEvent",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "AuditEvent": {
      "description": "A record of an event made for purposes of maintaining a security log. Typical uses include
    detection of intrusion attempts and monitoring for inappropriate usage.",
      "properties": {
        "resourceType": {
          "description": "This is a AuditEvent resource",
          "const": "AuditEvent"
        },
        "id": {
```

```
      "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
 never changes.",
      "$ref": "id.schema.json#/definitions/id"
    },
    "meta": {
      "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
Changes to the content might not always be associated with version changes to the resource.",
      "$ref": "Meta.schema.json#/definitions/Meta"
    },
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "Identifier for a family of the event.  For example, a menu item, program, rule, policy,
function code, application name or URL. It identifies the performed function.",
      "$ref": "Coding.schema.json#/definitions/Coding"
    },
    "subtype": {
      "description": "Identifier for the category of event.",
      "items": {
        "$ref": "Coding.schema.json#/definitions/Coding"
      },
      "type": "array"
    },
    "action": {
      "description": "Indicator for type of action performed during the event that generated the audit.",
      "enum": [
        "C",
        "R",
        "U",
        "D",
        "E"
      ]
    },
    "_action": {
```

```
      "description": "Extensions for action",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "period": {
      "description": "The period during which the activity occurred.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "recorded": {
      "description": "The time when the event was recorded.",
      "$ref": "#/definitions/instant"
    },
    "_recorded": {
      "description": "Extensions for recorded",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "outcome": {
      "description": "Indicates whether the event succeeded or failed.",
      "enum": [
        "0",
        "4",
        "8",
        "12"
      ]
    },
    "_outcome": {
      "description": "Extensions for outcome",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "outcomeDesc": {
      "description": "A free text description of the outcome of the event.",
      "$ref": "#/definitions/string"
    },
    "_outcomeDesc": {
      "description": "Extensions for outcomeDesc",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "purposeOfEvent": {
      "description": "The purposeOfUse (reason) that was used during the event being recorded.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "agent": {
      "description": "An actor taking an active role in the event or activity that is logged.",
      "items": {
        "$ref": "#/definitions/AuditEvent_Agent"
      },
      "type": "array"
    },
    "source": {
      "description": "The system that is reporting the event.",
      "$ref": "#/definitions/AuditEvent_Source"
    },
    "entity": {
      "description": "Specific instances of data or objects that have been accessed.",
      "items": {
        "$ref": "#/definitions/AuditEvent_Entity"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "agent",
    "source",
    "type",
    "resourceType"
  ]
},
"AuditEvent_Agent": {
  "description": "A record of an event made for purposes of maintaining a security log. Typical uses include
detection of intrusion attempts and monitoring for inappropriate usage.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
```

```
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "Specification of the participation type the user plays when performing the event.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "role": {
      "description": "The security role that the user was acting under, that come from local codes defined by the
access control security system (e.g. RBAC, ABAC) used in the local context.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "who": {
      "description": "Reference to who this agent is that was involved in the event.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "altId": {
      "description": "Alternative agent Identifier. For a human, this should be a user identifier text string from
authentication system. This identifier would be one known to a common authentication system (e.g. single sign-on),
if available.",
      "$ref": "#/definitions/string"
    },
    "_altId": {
      "description": "Extensions for altId",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "Human-meaningful name for the agent.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "requestor": {
      "description": "Indicator that the user is or is not the requestor, or initiator, for the event being
audited.",
      "$ref": "#/definitions/boolean"
    },
    "_requestor": {
      "description": "Extensions for requestor",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "location": {
      "description": "Where the event occurred.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "policy": {
      "description": "The policy or plan that authorized the activity being recorded. Typically, a single activity
may have multiple applicable policies, such as patient consent, guarantor funding, etc. The policy would also
indicate the security token used.",
      "items": {
        "$ref": "#/definitions/uri"
      },
      "type": "array"
    },
    "_policy": {
      "description": "Extensions for policy",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "media": {
      "description": "Type of media involved. Used when the event is about exporting/importing onto media.",
      "$ref": "Coding.schema.json#/definitions/Coding"
    },
    "network": {
      "description": "Logical network location for application activity, if the activity has a network location.",
      "$ref": "#/definitions/AuditEvent_Network"
    },
    "purposeOfUse": {
      "description": "The reason (purpose of use), specific to this agent, that was used during the event being
recorded.",
      "items": {
```

```
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"AuditEvent_Network": {
  "description": "A record of an event made for purposes of maintaining a security log. Typical uses include
detection of intrusion attempts and monitoring for inappropriate usage.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "address": {
      "description": "An identifier for the network access point of the user device for the audit event.",
      "$ref": "#/definitions/string"
    },
    "_address": {
      "description": "Extensions for address",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "An identifier for the type of network access point that originated the audit event.",
      "enum": [
        "1",
        "2",
        "3",
        "4",
        "5"
      ]
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"AuditEvent_Source": {
  "description": "A record of an event made for purposes of maintaining a security log. Typical uses include
detection of intrusion attempts and monitoring for inappropriate usage.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
```

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extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "site": {
      "description": "Logical source location within the healthcare enterprise network.  For example, a hospital or
other provider location within a multi-entity provider group.",
      "$ref": "#/definitions/string"
    },
    "_site": {
      "description": "Extensions for site",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "observer": {
      "description": "Identifier of the source where the event was detected.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "type": {
      "description": "Code specifying the type of source where event originated.",
      "items": {
        "$ref": "Coding.schema.json#/definitions/Coding"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "observer"
  ]
},
"AuditEvent_Entity": {
  "description": "A record of an event made for purposes of maintaining a security log. Typical uses include
detection of intrusion attempts and monitoring for inappropriate usage.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "what": {
      "description": "Identifies a specific instance of the entity. The reference should be version specific.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "type": {
      "description": "The type of the object that was involved in this audit event.",
      "$ref": "Coding.schema.json#/definitions/Coding"
    },
    "role": {
      "description": "Code representing the role the entity played in the event being audited.",
      "$ref": "Coding.schema.json#/definitions/Coding"
    },
    "lifecycle": {
      "description": "Identifier for the data life-cycle stage for the entity.",
      "$ref": "Coding.schema.json#/definitions/Coding"
    },
    "securityLabel": {
      "description": "Security labels for the identified entity.",
      "items": {
        "$ref": "Coding.schema.json#/definitions/Coding"
      },
```

```
      "type": "array"
    },
    "name": {
      "description": "A name of the entity in the audit event.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "description": {
      "description": "Text that describes the entity in more detail.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "query": {
      "description": "The query parameters for a query-type entities.",
      "$ref": "#/definitions/base64Binary"
    },
    "_query": {
      "description": "Extensions for query",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "detail": {
      "description": "Tagged value pairs for conveying additional information about the entity.",
      "items": {
        "$ref": "#/definitions/AuditEvent_Detail"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"AuditEvent_Detail": {
  "description": "A record of an event made for purposes of maintaining a security log. Typical uses include
detection of intrusion attempts and monitoring for inappropriate usage.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "The type of extra detail provided in the value.",
      "$ref": "#/definitions/string"
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueString": {
      "description": "The  value of the extra detail.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_valueString": {
      "description": "Extensions for valueString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueBase64Binary": {
      "description": "The  value of the extra detail.",
      "pattern": "^(\\s*([0-9a-zA-Z\\+/\u003d]){4}\\s*)+$",
```

```
          "type": "string"
        },
        "_valueBase64Binary": {
          "description": "Extensions for valueBase64Binary",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false
    }
  }
}
```

6.4.12 Resource AuditEvent - Maturity Dependency View

URL: https://hl7.org/fhir/R4/auditevent-dependencies.html

6.4.12 Resource AuditEvent - Maturity Dependency View

Security Work Group | Maturity Level: 3 | Trial Use | Compartments: Device, Patient, Practitioner

Dependency Graph for AuditEvent FMM level 3

Name | Card. | Type | Dependency Analysis
AuditEvent | | DomainResource
type | 1..1 | Coding
subtype | 0..* | Coding
action | 0..1 | code
period | 0..1 | Period
recorded | 1..1 | instant
outcome | 0..1 | code
outcomeDesc | 0..1 | string
purposeOfEvent | 0..* | CodeableConcept
agent | 1..* | BackboneElement
type | 0..1 | CodeableConcept
role | 0..* | CodeableConcept
who | 0..1 | Reference(PractitionerRole | Practitioner | Organization | Device | Patient | RelatedPerson) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (Device = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (RelatedPerson = FMM2-Trial Use vs. Element = FMM3-Trial Use)
altId | 0..1 | string
name | 0..1 | string
requestor | 1..1 | boolean
location | 0..1 | Reference(Location)
policy | 0..* | uri
media | 0..1 | Coding
network | 0..1 | BackboneElement
address | 0..1 | string
type | 0..1 | code
purposeOfUse | 0..* | CodeableConcept
source | 1..1 | BackboneElement
site | 0..1 | string
observer | 1..1 | Reference(PractitionerRole | Practitioner | Organization | Device | Patient | RelatedPerson) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (Device = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (RelatedPerson = FMM2-Trial Use vs. Element = FMM3-Trial Use)
type | 0..* | Coding
entity | 0..* | BackboneElement
what | 0..1 | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM3-Trial Use)
type | 0..1 | Coding
role | 0..1 | Coding
lifecycle | 0..1 | Coding
securityLabel | 0..* | Coding
name | 0..1 | string
description | 0..1 | string
query | 0..1 | base64Binary
detail | 0..* | BackboneElement
type | 1..1 | string
value[x] | 1..1
valueString | | string
valueBase64Binary | | base64Binary